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A Co-Reference Latent Model for Dependency Analysis
of Paired Data Sets in Bioinformatics

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서울대학교 대학원
전기컴퓨터공학부
장 정 호

Abstract

An important issue in data mining and machine learning is to identify association patterns between paired data sets, i.e., data sets coming from two different sources or aspects. Except for trivial cases, however, the patterns are often obscured by the complexity of the raw data originating from high-dimensionality of the data.

This dissertation proposes an effective model for simultaneous analysis of paired data sets that solves this problem with dimensionality reduction, on the basis of the idea that the observed association patterns are not random but rather systematic. A key concept is to use two separate models for the respective data sets and to pursue explicitly the inter-relation or co-structure inherent in the paired sets by co-referencing them alternately in an expectation-maximization (EM) algorithm style. Another feature of the model is to extract the underlying regularity in the complex association patterns by mapping them onto the lower-dimensional latent space. This “co-reference latent model” can provide a concise and summarized description of the data sets which can be visualized by the latent factors, which enhances the intuitive understanding of the data.

We evaluate the co-reference latent model on two real-life problems in bioinformatics. In the first application of relating DNA sequence elements to gene expression patterns, we could identify most of the known regulatory sequence motifs, along with a few putative elements, in relation with specific experimental conditions by the use of a linear co-reference latent model. In particular, the dependency structure between gene expression patterns and hundreds of sequence elements could be summarized by a much smaller number of latent features. The reduction also enabled the low-dimensional display of the two entities and it was shown to be of great help in the intuitive understanding of the effects of specific sequence motifs on

transcription activities of the genes which have the motifs in their upstream regions. The other task was the gene-drug dependency mining from the paired set of gene expression and drug activity data. Under the framework of the co-reference latent model, a clustering-based dimensionality reduction was first employed to facilitate the learning and inference for the dependency analysis. Then, a Bayesian network-based dependency model was learned on the condensed data set. By the combined approach, a compact and manageable dependency structure was acquired with a moderate learning cost. More importantly, through the probabilistic inference on the dependency model, which is usually intractable for high-dimensional data, we could detect some meaningful relationships among several genes and drugs in cancer cell lines.

Keywords: Co-Reference Latent Model, Paired Data Sets, Dimensionality Reduction, Dependency Analysis, Bio-Data

Student Number: 97419–816

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Chapter 1

Introduction

1.1 Background

With the great advances in computer technology and automated sensors or systems, we are now more often facing with large and complex data than ever before. In this information age, the ability to extract useful information from those data automatically and organize them in an interpretable way is becoming increasingly important in almost every field of business, science, and engineering. The scientific discipline of digging out valuable information or knowledge from (often large) data sets is well known as *data mining*, where one attempts to find unsuspected relationships and to summarize the data in both understandable and useful ways (Hand et al., 2001).

Machine learning is one of the most practical approaches to data mining. The area of machine learning is concerned with the problem of how to build computer program that automatically improves with *experience* (Mitchell, 1997). In the view point of data mining, the experience is given in the form of *data* available in specific tasks and machine learning can assist human by building the automated analysis tools for the data set. According to learning situation and tasks, the research field of

machine learning can be generally categorized into three classes: *supervised learning*, *reinforcement learning*, and *unsupervised learning*.

In supervised learning, the learning tasks are concerned with the construction of predictive models from data. The main tasks include regression and classification. Training samples are provided in the form of pairs of input values and target (output) values, where target values are provided by a *teacher* (Cherkassky and Mulier, 1998). Reinforcement learning mainly deals with the problem of how an autonomous agent can learn to choose optimal actions for the accomplishment of its goals in its environment (Mitchell, 1997). While there is a teacher as in supervised learning, the teacher does not provide the actual desired responses directly to the learning system. The learner receives only evaluative feedbacks for its actions according to the performance quality.

Unlike previous two learning paradigms, it is often the case that there is no explicit teacher or feedback provider and furthermore the labeling task of a large set of data samples can be surprisingly expensive and time consuming. In this situation where there are not enough prior knowledge such as class information of data samples, the important task is to extract salient patterns underlying the data set itself by an explorative analysis. Unsupervised learning is concerned with the learning paradigm of this case. In the field of bioinformatics, the computational analysis of various massive biological data sets can produce useful information that provides one a proper understanding of the data set. Furthermore, it can also generate meaningful hypotheses that are worthy of experimental test.

An important learning subject in the explorative data analysis is to build a descriptive model of the data set. The descriptive mining is mainly concerned with the presentation of a concise and summarized description of observed data, as opposed to the predictive mining where the main objective is to predict a target value for

unseen data on the basis of a model learned from some observed data (Hand et al., 2001; Han and Kamber, 2001). The summary of a data set is mainly concerned with identifying salient association patterns or dependency structure inherent in a data set, including rules describing data properties, patterns of the frequent occurrence, and grouping of data. Except for a trivial case, however, a data set to be analyzed is of high-volume and high-dimensionality. In this case, the patterns underlying a data set can be obscured by the complexity of the raw data and a simple descriptive statistics are not sufficient. Hence, a sophisticated simplification process is highly needed for clarification of the buried patterns and effective understanding of such data set.

1.2 Focus of Research

The main concern in this dissertation is the explorative analysis of *paired data sets* on the basis of dimensionality reduction, especially the dependency analysis among different attributes of two biological data sets.

Dimensionality Reduction

In most cases, the observed association patterns are not random but rather systematic. By seeking to extract such regularity in the surface associations, one may be able to acquire an efficient explanation of the data set. On the basis of dimensionality reduction, this is achieved by devising a small number of latent factors, with the emphasis being on the revelation of meaningful features for human understanding. The complex association patterns are now represented in terms of the latent factors, by which we can acquire a concise, summarized description of the data set. Another advantage is that the associations can be visualized in a low-dimensional space defined by the latent factors, which highly contributes to the intuitive understanding

of the data set.

One simple but important approach to dimensionality reduction is the *variable subset selection*, where one picks out, among the set of original variables describing a data set, only those most relevant (presumably) for the task at hand (Guyon and Elisseeff, 2003). The other valuable dimensionality reduction technique is *feature extraction* or *feature construction*. Contrary to the subset selection method, the approach reduces the dimensionality by constructing new feature sets from original variables. The generic methods for accomplishing this kind of dimensionality reduction includes clustering, linear or non-linear transformation of original variables, and so on (Guyon and Elisseeff, 2003). For example, the latent semantic analysis (LSA) in text mining field is a well-known technique of this type. It is used to extract and represent the contextual-usage meaning of words and can enhance the accompanying main tasks such as text categorization or retrieval (Landauer et al., 1998).

While the feature extraction method can be used for building new input representation for other main tasks, it takes its own important position as a valuable tool in the explorative analysis of complex data sets. For instance, by employing the feature construction-based dimensionality reduction methods such as singular value decomposition (SVD) or principal component analysis (PCA) in the analysis of a massive microarray data of thousands of genes, one can gain a summarized view on the ways in which gene responses vary under different conditions (Alter et al., 2000; Raychaudhuri et al., 2000). We are concerned with this type of dimensionality reduction throughout this dissertation and the “dimensionality reduction” hereafter means only this case unless otherwise stated.

Low-Dimensional Analysis from Paired Data Sets

By the paired data set, we mean the case where each observation of the data set is given as an instantiation of two or more different but related variable sets. Let $\mathcal{S} = \{s_i\}$ ($1 \leq i \leq N$) be a set of data samples and $\mathcal{X} = \{X_j\}$ and $\mathcal{Y} = \{Y_l\}$ ($1 \leq j \leq M, 1 \leq l \leq L$) be two variable sets. Here a sample $s_i \in \mathcal{S}$ is represented by two vectors, one is the observation in terms of $\mathcal{X}$ and the other is obtained in terms of $\mathcal{Y}$. Finally, the entire data set is given as two matrices, one for each variable set.

For the low-dimensional analysis of the paired data set, one common approach is to first merge the two different data sets into one data matrix such that a sample is represented in a hyperspace of the combined variable set $\mathcal{X} \cup \mathcal{Y}$. Then, a dimensionality technique is applied to the combined data set to construct a lower-dimensional space where two different entities of $\mathcal{X}$ and $\mathcal{Y}$ is mutually related. For example, the association between two words in different languages can be established by the singular value decomposition-based analysis of the merged table of multiple-language documents (Rehder et al., 1997; Littman et al., 1998).

However, this juxtaposition-based approach has some drawbacks in the dependency analysis of two separate data sets. First, the one model scheme in this approach may not appropriate to handle the cases where the two data sets are described by different types of variables, one by the real-valued variables and the other by the count-valued variables, for example. Second, even though the two data sets have the same type of variables and thus can be learned by the same type of model, the model can be biased into one of the data set due to the difference in data-inherent scale which implies the number of variables, the data variance, and so on. In this case, we may have some difficulty in identifying the relation underlying between the two data sets.

In this dissertation, we propose a more effective model for the simultaneous analysis of the paired data sets, called the *co-reference latent model*. The key component of the co-reference latent model is to use the separate models for the respective data sets and to pursue explicitly the inter-relation or co-structure inherent in the two data sets. The association pattern between the attributes of different data sets are efficiently grasped by correlating them on latent spaces produced by the two dimensionality reduction models. The learning of the model can be achieved by an iterative alternation of two learning steps for two separate models while consistently relating the latent variables of the models in the learning. Alternatively, the dependency analysis model is independently introduced after the separate dimensionality reduction of two data sets. The latter learning scheme may be very useful especially in such cases when the ability of the model for dependency analysis can not be fully exploited due to the high complexity of data sets.

Problems to be Explored

We apply the proposed model to the dependency analysis of two kinds of paired biological data sets: one for the paired set of DNA sequence data and gene expression data, and the other for gene expression and drug activity data.

Dependency analysis of DNA sequence elements and gene expression

DNA microarrays from cDNA or oligonucleotide chips provide a global and parallel view on the expression patterns of hundreds or thousands of genes in a cell under various specific experimental conditions or cellular processes. An important issue in computational biology in the analysis of the massive gene expression data is to understand the mechanism underlying the transcriptional regulation of genes. Especially, the transcriptional behaviors of genes are mainly coordinated by the DNA

sequence elements of rather short length, that act as binding sites for transcription factors. We apply our co-reference latent model to the effective identification of such DNA sequence regulatory elements in relation with specific expression patterns and experimental conditions. The meaningful correlations between gene expression and sequence motif profiles are detected by the successive extraction of latent feature pairs from the two data sets.

Dependency analysis of gene expression and drug activity

Analysis of therapeutic response in relation with genomic features is of much interest in the field of pharmacogenomics. As a second application, we deal with the gene-drug dependency mining by the combined analysis of the paired set of microarray and drug activity data from various human tumor cell lines. Basically, Bayesian networks are employed for the dependency modeling. To alleviate several problems due to high-dimensionality of the data set, we take an approach of the dimensionality reduction of gene expression data and drug activity data. Then, the dependency mining proceeds on the two reduced feature spaces, one for genes and the other for drugs.

1.3 Structure of the Dissertation

The remainder of the dissertation is organized as follows. Chapter 2 describes the concept of dimensionality reduction and several representative techniques. Chapter 3 presents the detailed explanation of the co-reference latent model for the analysis of paired data sets on the basis of simultaneous dimensionality reduction of two data sets. In Chapter 4, we present the application of the co-reference latent model to the co-analysis of DNA sequence motif data and gene expression data for an efficient mining of gene regulatory elements. Chapter 5 discusses the dependency analysis

between gene expression and drug activity under the framework of co-reference latent model. Finally, Chapter 6 draws conclusions.

Chapter 2

Dimensionality Reduction

For a data set given by a table or matrix $\mathbf{D}$, let each observation be denoted by $v(x_i, y_j)$ where $x_i \in \mathcal{X}$ and $y_j \in \mathcal{Y}$. Here, the key element for the exploration and description of the data is the association pattern among the elements of $\mathcal{X}$ and $\mathcal{Y}$. One can explain an object of $\mathcal{Y}$ by its specific property in relation with those of $\mathcal{X}$. Furthermore, a relationship (similarity or dissimilarity, for example) between two objects y_i and y_j ($\in \mathcal{Y}$) can be established on the basis of their association patterns across the objects of $\mathcal{X}$, and vice versa. For instance, a text document ($y \in \mathcal{Y}$) is understood by identifying the coordinated usage of words ($x \in \mathcal{X}$) in the document. The similarity of two documents is usually estimated by matching the words in a document with those in the other.

In this aspect, the association pattern is much related to the dependency of the two objects. However, when we are given a large data set, which is usual in practical application, a naive description of every individual in $\mathcal{X}$ or $\mathcal{Y}$ is obscured by the sheer size of the data set. The largeness applies to both object sets which are paired: $|\mathcal{X}|$ may be large or $|\mathcal{Y}|$ may be so. Then, we may need a more sophisticated strategy for an efficient and effective data description.

One most important approach to the explorative learning in this case is *data/dime-*

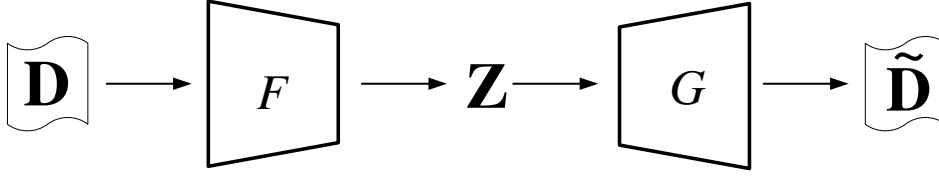


Figure 2.1: A schematic view of dimensionality reduction

nsionality reduction. This is based on a very plausible idea that there underlie some regularities in the data set, which well applies to most real-world data sets. In the generative point of view for a paired data set, it can also be restated that the association pattern between $\mathcal{X}$ and $\mathcal{Y}$ is the result of realization of a set of latent factors. Data reduction is concerned with reducing the number of samples and then representing the data by some representatives. By dimensionality reduction, on the contrary, one reduces the number of attributes describing data samples for the compact description of data. The distinction between samples and attributes depends on which abstract class ($\mathcal{X}$ or $\mathcal{Y}$) we are concerned with. For the time being, we assume that $\mathcal{Y}$ is the set of samples, and thus a sample y_i ($1 \leq i \leq N$) is represented by a vector form $y_i = \mathbf{x}_i = (x_1, x_2, \dots, x_M)^T$.

In the fields of machine learning and data mining, clustering or vector quantization is a main tool for data reduction. Given a data set, one of the main interest here is to describe the data set using a small number of prototypes $\{\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_K\}$ which serve as the representatives for K homogeneous groups of data samples respectively, where $K \ll N$.

Dimensionality reduction is generally defined as a data transformation by which a data set is represented by a reduced number of effective features while retaining most of the intrinsic information of the data (Haykin, 1999). Figure 2.1 shows a schematic view of dimensionality reduction in this sense. By a dimensionality reduction process

F , an M -dimensional input space $\mathbb{R}^M$ is mapped to a lower-dimensional space $\mathbb{R}^K$ ($K \ll M$): $\mathbf{z} = F(\mathbf{x})$, ($\mathbf{x} \in \mathbb{R}^M$, $\mathbf{z} \in \mathbb{R}^K$). This should be done in such a way that its inverse process G could achieve sufficient approximation to the original data, that is, $\tilde{\mathbf{D}} \approx \mathbf{D}$: $\tilde{\mathbf{x}} = G(\mathbf{z})$, ($\tilde{\mathbf{x}} \in \mathbb{R}^M$, $\mathbf{z} \in \mathbb{R}^K$). We are mainly concerned with this type of reduction approach and thus we hereafter explain the descriptive learning in terms of only dimensionality reduction. As a special case, when clustering is applied to the objects in $\mathcal{X}$, rather than $\mathcal{Y}$, it can also serve as dimensionality reduction in terms of $\mathcal{Y}$.¹

2.1 Clustering-Based Analysis

As previously stated, clustering is a typical unsupervised learning task which is concerned with finding groups in data. Besides its own role to provide natural groupings of one's relevant objects or samples, clustering can also be utilized for dimensionality reduction when it is applied to the counter-part objects (features) of samples. Let an element $v_{ij} = v(x_i, y_j)$ of a data matrix $\mathbf{D}$ be real-valued, and $\mathcal{Y}$ and $\mathcal{X}$ be the set of primary objects and the set of objects describing the primary objects, respectively. We assume that $\mathcal{X}$ is partitioned into $K (= |\mathcal{Z}|)$ clusters, $\mathcal{C} = \{\mathcal{C}_1, \mathcal{C}_2, \dots, \mathcal{C}_K\}$. Then, a new feature set $\mathcal{Z}$ is defined by $\mathcal{C}$ and the most intuitive decomposition of $\mathbf{D}$ can be given simply by

$$\begin{aligned} w_{ik} &= \begin{cases} 1 & : \text{ if } x_i \in \mathcal{C}_k \\ 0 & : \text{ otherwise } \end{cases}, \\ z_{kj} &= \frac{1}{|\mathcal{C}_k|} \sum_{x_i \in \mathcal{C}_k} v_{ij}, \end{aligned}$$

where w_{ik} is the element of $\mathbf{W}$ in the i -th row and k -th column and z_{kj} is the element of $\mathbf{Z}$ in the k -th row and j -th column. Consequently, the j -th column vector of $\mathbf{Z}$ gives

¹The use of clustering for dimensionality reduction is described in the application of Chapter 5.

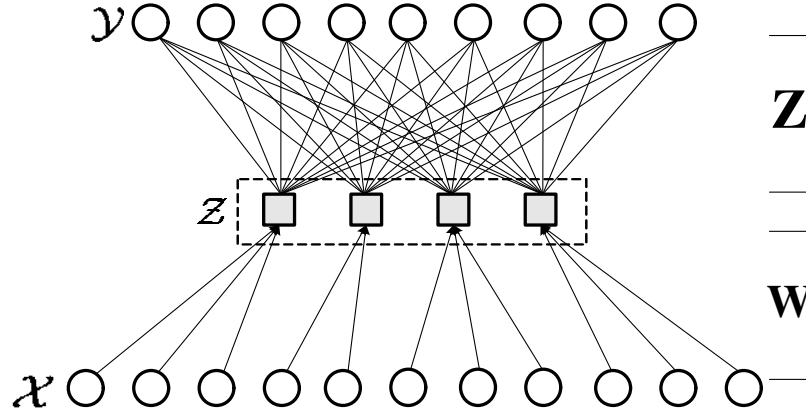


Figure 2.2: A graph-based representation of data dimensionality reduction by the feature clustering

the reduced K -dimensional representation of the j -th object $y_j \in \mathcal{Y}$. The precise settings of $\mathbf{W}$ and $\mathbf{Z}$ depend on clustering algorithms and applications. Figure 2.2 shows a graph-based representation of the process.

This approach is usually intended to make ease of or achieve the performance improvement of the accompanying learning tasks. In the inference of genetic networks on the basis of gene expression data, for example, some researchers have taken the approach to cope with the high dimensionality of data sets which can make the network inference process unstable or even intractable (Wahde and Hertz, 2000; Mjolsness et al., 2000; van Someren et al., 2000; D’haeseleer et al., 2000; Toh and Horimoto, 2002; Herrero et al., 2003). Typically, genes ($\mathcal{X}$) are first partitioned into several groups on the basis of their expression patterns across all experiment conditions ($\mathcal{Y}$) and the value of a gene cluster $\mathcal{C}_k$ in a sample $y_i \in \mathcal{Y}$ is set to a representative value (mean expression value, for instance) of the member genes in $\mathcal{C}_k$ for y_i . Gene clusters can be acquired by various conventional clustering algorithms, including hierarchical clustering (Wahde and Hertz, 2000; van Someren

et al., 2000; Toh and Horimoto, 2002), k -means algorithm (Chang et al., 2005), and mixture model-based clustering (Mjolsness et al., 2000). Then, genetic networks are constructed on the gene clusters rather than individual genes.

In the applications of text mining, for another example, words ($\mathcal{X}$) can be clustered before the accompanying main tasks of document ($\mathcal{Y}$) clustering or classification. Slonim and Tishby (2000) presented a double clustering procedure based on an information-theoretic framework, called the information bottleneck method, and showed that the word clusters can enhance the performance of document clustering compared to other common document distributional clustering algorithms. Baker and McCallum (1998) described the application of distributional word clustering to the task of text classification. Their approach is different from that of Slonim and Tishby (2000) in that words are clustered in a *supervised* way in a sense that each word is associated with class labels, rather than documents. They argue that one can benefit from word clusters since they provide useful groups of semantically-related words, higher classification accuracy, and smaller classification models.

Unlike the previous asymmetric approaches, the two objects $\mathcal{X}$ and $\mathcal{Y}$ can be simultaneously clustered in a symmetric view, where $\mathcal{X}$ serves for the clustering $\mathcal{Y}$ and again $\mathcal{Y}$ serves for the clustering of $\mathcal{X}$. Figure 2.3 shows this approach with a graph-based representation. Cho et al. (2004) presents the co-clustering approach to gene expression data, where a k -means-like algorithm is employed to the simultaneous clustering of genes and measurement conditions in a non-overlapping way.

2.2 Latent Variable Models

Latent variable models are a powerful tool for discovering latent structure underlying data objects. With the ability of encoding a multidimensional data set by a set

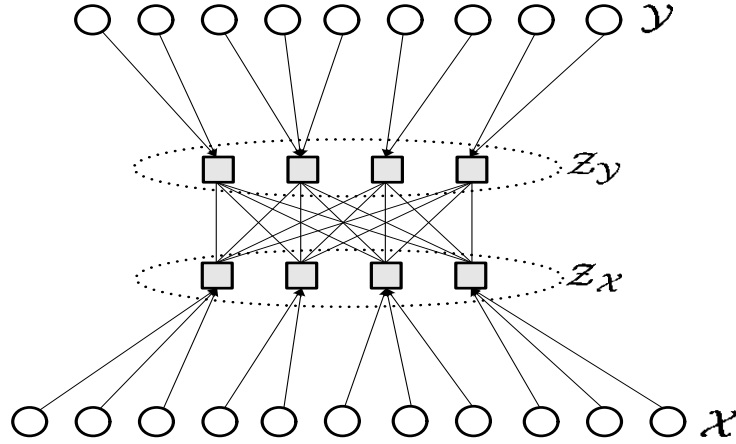


Figure 2.3: A graph-based representation of co-clustering where the objects of $\mathcal{X}$ and $\mathcal{Y}$ are clustered in the same framework. The two cluster sizes, that is, the cardinalities of $\mathcal{Z}_\mathcal{X}$ and $\mathcal{Z}_\mathcal{Y}$ may be same or different.

of latent factors that cannot be directly observed, they are well suited to extract correlational patterns of variables. Latent variable models also provide the meaningful aspects of the data and thus can explain the data in a compact and comprehensible way. Here, what is meant by the *meaningfulness* is the key element. One may be interested in the covariance or correlation structure in the data, and others may be concerned with just the reconstruction error. According to the choice, various latent variable models can be defined.

Figure 2.4 shows a simple example of dimensionality reduction by a latent variable model for a 2-dimensional data set ($|\mathcal{X}| = 2$, $|\mathcal{Y}| = 900$). In the plot of data samples, one can see that the data set is well characterized by its 3-group structure. By the learning of a latent variable model, the regularity of the data set is captured by a line on which data projections are maximally variant. The projected samples reside in a 1-dimensional space defined by the line, rather than the original 2-dimensional space, and are still well reflecting the group structure. Thus, the dimensionality reduction

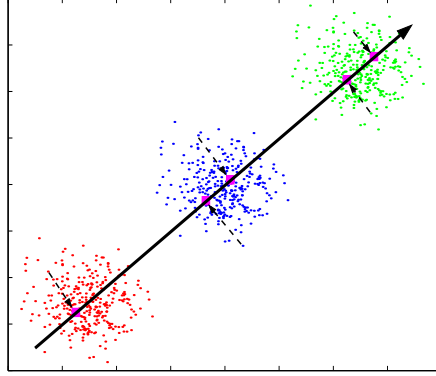


Figure 2.4: An example of dimensionality reduction by a latent variable model for one-table data set

by the latent variable model can be said to give us a concise summarization of the data set.

From the viewpoint of association patterns between $\mathcal{X}$ and $\mathcal{Y}$, such characteristics of latent variable models can be understood in terms of the transformation of the space where the associations take place. While the entities $y \in \mathcal{Y}$ in the original space are related with each other by their properties associated with $\mathcal{X}$, the reduction strategies by latent variable models have the relation associated with newly devised entities. Figure 2.5 depicts this explanation in the framework of network connection and matrix decomposition. Each $z_k \in \mathcal{Z}$ corresponds to a newly created dimension on which samples ($\mathcal{Y}$) are projected. The objects $y_i \in \mathcal{Y}$ are related with each other by their association with $z_k \in \mathcal{Z}$, instead of the elements in $\mathcal{X}$. Unlike the cluster-based approach, a dimensionality reduction by a latent variable model is accomplished in such a way that the sets of original variables defining latent factors are non-exclusive each other, and very often, each factor may involve entire variables.

At the same time, the reduction strategies can also provide new associations among the entities in $\mathcal{X}$ through $\mathcal{Z}$ (see the network connection between $\mathcal{Z}$ and $\mathcal{X}$,

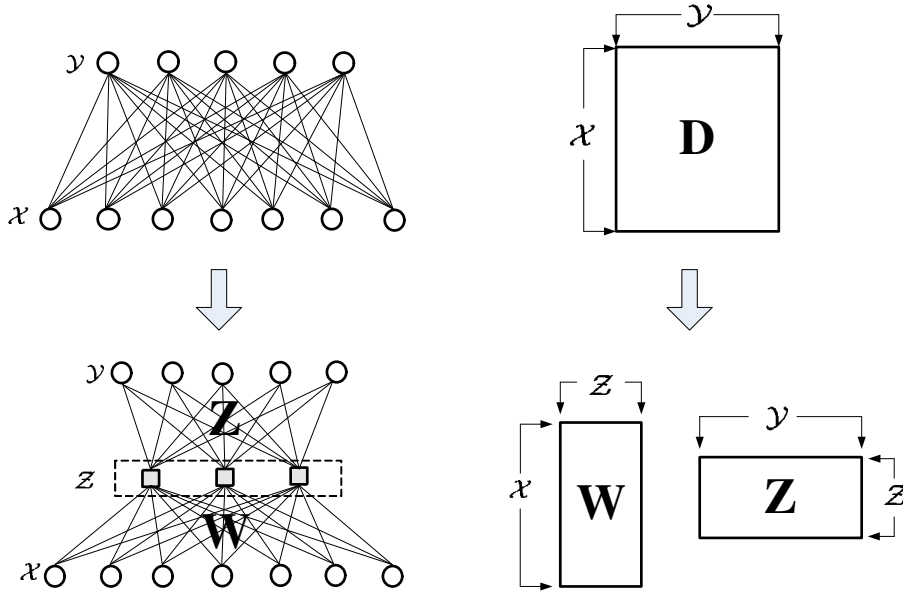


Figure 2.5: Latent variable model-based dimensionality reduction in a network and a matrix form. Objects in $\mathcal{X}$ and $\mathcal{Y}$ are respectively associated via the reduced number of latent factors. As a dimensionality reduction approach, the number of latent factors are usually set to be $|\mathcal{Z}| \ll \min(|\mathcal{X}|, |\mathcal{Y}|)$.

and the matrix $\mathbf{W}$ in Figure 2.5). In particular, the inspection of the association pattern among the objects $\mathcal{X}$ in the definition of a $z_k \in \mathcal{Z}$ may be of a great help in understanding the latent structure revealed by the reduction analysis. There are various type of latent variable models according to their assumption for latent factors and input data set. Among those, the following sections describe four representative models in the framework of dimensionality reduction: principal component analysis (PCA), independent component analysis (ICA), singular value decomposition (SVD), and non-negative matrix factorization (NMF).

2.3 Principal Component Analysis

Principal component analysis (PCA) (Hotelling, 1933) is a classical statistical method for data analysis and is also called as the Karhunen-Loève transform in communication theory. The central aim of PCA is to find a set of orthogonal vectors or principal component axes that account for as much as possible the variation in the data set.

Given a data matrix $\mathbf{D}$, let $\mathbf{x} = (x_1, x_2, \dots, x_M)^T$ be a random vector which describes data samples $\mathcal{Y} = \{y_1, y_2, \dots, y_N\}$, where $y_i = \mathbf{x}_i$. The random vector $\mathbf{x}$ is assumed to have zero mean. Then, the first principal axis is extracted such that the variance of the projections of data samples onto the direction is maximal:

$$\mathbf{w}_1 = \arg \max_{\|\mathbf{w}\|=1} \langle \|\mathbf{w}^T \mathbf{x}\|^2 \rangle, \quad (2.1)$$

where $\langle A \rangle$ means the expectation of A . Subsequent principal component axes can be extracted by the successive applications of Equation (2.1) under the constraint that the i -th axis is orthonormal to the previous $(i - 1)$ axes.

In the viewpoint of matrix decomposition, this can be achieved by the eigen-decomposition of the sample covariance matrix $\mathbf{C} = \frac{1}{N} \mathbf{D} \mathbf{D}^T$:

$$\mathbf{C} = \mathbf{W} \mathbf{\Lambda} \mathbf{W}^T,$$

where $\mathbf{W}$ is the orthogonal matrix of which column vectors are called the *eigenvectors* of $\mathbf{C}$ and $\mathbf{\Lambda}$ is an diagonal matrix of which diagonal elements are the *eigenvalues* of $\mathbf{C}$. The i -th eigenvalue λ_i is non-negative and denotes the variance of the samples projected onto its corresponding eigenvector $\mathbf{w}_i$, that is, $\lambda_i = \langle \|\mathbf{w}_i^T \mathbf{x}\|^2 \rangle$. The sample score on principal axes are given by $\mathbf{Z} = \mathbf{W}^T \mathbf{\Lambda}^{\frac{1}{2}} \mathbf{D}$. By retaining only the principal axes for which the corresponding eigenvalues are relatively large-valued and discarding others, one can utilize PCA to reduce the dimensionality of data, still retaining most of variance in the original data.

2.4 Independent Component Analysis

Independent component analysis (ICA) is another powerful computational method for finding latent factors underlying sets of random variables, measurements or signal. In the data analysis by ICA, it is assumed that the observed variables or signals are generated by some unknown sources, which are assumed to be non-Gaussian² and statistically independent each other (thus called *independent components*) (Hyvärinen and Oja, 2001). Its usage is prevalent in the field of signal processing, including biomedical signals from fMRI and EEG, speech signals, and telecommunication signals.

In the ICA model, a random vector $\mathbf{x} = (x_1, x_2, \dots, x_M)^T$ is expressed as a linear mixtures of independent components $\mathbf{z} = (z_1, z_2, \dots, z_M)^T$:

$$\mathbf{x} = \mathbf{W}\mathbf{z},$$

where $\mathbf{W}$ ($M \times M$) is the mixing matrix³. Here, the task is to find the independent components $\mathbf{s}$ and the mixing matrix $\mathbf{W}$. In general, ICA estimates the demixing matrix $\mathbf{H}$, the (pseudo) inverse of $\mathbf{W}$ rather than $\mathbf{W}$ directly, such that $\mathbf{z} = \mathbf{H}\mathbf{x}$. The estimation process can be done by various approaches, including maximization of nongaussianity (Delfosse and Loubaton, 1995; Hyvärinen, 1998), minimization of mutual information (Comon, 1994; He et al., 2000), maximum likelihood estimation (Bell and Sejnowski, 1995; Amari, 1998), and tensorial methods (Cardoso, 1990; Cardoso and Souloumiac, 1993).

When an observed data is high-dimensional, parameter estimation in ICA may be poor and over-learning may occur. In that case, it is beneficial to reduce the data dimensionality before the estimation of ICA model. A popular approach is to

²This assumption is different from those in factor analysis and principal component analysis where the components are assumed to be Gaussian.

³We assume the case of the linear ICA.

apply PCA first to the data and to represent the data in a lower-dimensional space which is defined by the dominant eigenvectors. With this preprocessing, ICA can also be considered as a method of dimensionality reduction, in that the independent components provides an economical data representation in lower-dimensional space.

2.5 Singular Value Decomposition

Singular value decomposition (SVD) is an important method in statistical computation and in strategies for data compression, based on an approximation of a data matrix with one of lower rank (Horn and Johnson, 1991). Additionally, it is closely related with PCA and is one of the basic approaches to the computation of principal components.

Given an $M \times N$ matrix $\mathbf{D}$, let R be the rank of $\mathbf{D}$ ($R \leq \min(M, N)$). The decomposition of $\mathbf{D}$ by SVD is done in a *dyadic* way by

$$\mathbf{D} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T, \quad (2.2)$$

where $\mathbf{U}^T\mathbf{U} = \mathbf{V}^T\mathbf{V} = \mathbf{I}$ and $\mathbf{\Sigma} = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_R)$, $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_R > 0$. The two matrices $\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_R]$ ($M \times R$) and $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_R]$ ($N \times R$) are the orthogonal matrices of which column vectors are eigenvectors associated with the first R eigenvalues of $\mathbf{D}\mathbf{D}^T$ and $\mathbf{D}^T\mathbf{D}$, respectively. The matrix $\mathbf{\Sigma}$ is an $R \times R$ diagonal matrix of which diagonal elements are the positive square-roots of the eigenvalues of $\mathbf{D}\mathbf{D}^T$ and $\mathbf{D}^T\mathbf{D}$. The decomposition can be formulated in terms of the general setting of dimensionality reduction in Figure 2.5 by an assignment of $\mathbf{W} = \mathbf{U}$ and $\mathbf{Z} = \mathbf{\Sigma}\mathbf{V}^T$ (similarly, $\mathbf{W} = \mathbf{U}\mathbf{\Sigma}$ and $\mathbf{Z} = \mathbf{V}^T$). It is well known that the leading K ($\leq R$) singular triplets, $(\mathbf{u}_k, \sigma_k, \mathbf{v}_k)$ ($1 \leq k \leq K$), provides the best rank- K approximation to the matrix $\mathbf{D}$ in terms of the L_2 matrix norm $\|\mathbf{D} - \mathbf{W}\mathbf{Z}\|_{L_2}$ and the Frobenius matrix norm $\|\mathbf{D} - \mathbf{W}\mathbf{Z}\|_F$ (Golub and Loan, 1996; Berry et al.,

1995), where

$$\|\mathbf{A}\|_{L_2} = \max_{\mathbf{x} \neq \mathbf{0}} \left(\frac{(\mathbf{A}^T \mathbf{x})^T (\mathbf{A}^T \mathbf{x})}{\mathbf{x}^T \mathbf{x}} \right)^{\frac{1}{2}}, \quad (2.3)$$

$$\|\mathbf{A}\|_F = \left(\sum_{i,j} a_{ij}^2 \right)^{\frac{1}{2}}, \quad \mathbf{A} = (a_{ij}). \quad (2.4)$$

Among its broad applications, the SVD is at the heart of the latent semantic analysis (LSA) in the field of information retrieval and text mining. The LSA is a famous method for extracting and representing the contextual-usage meaning of words from a large text corpus (Landauer et al., 1998). Given a *term* ($\mathcal{X}$) $\times$ *document* ($\mathcal{Y}$) matrix $\mathbf{D}$ where each element $v(x_i, y_j)$ contains the co-occurrence information of i -th word and j -document, the matrix is factored into the product of three matrices in a way in Equation (2.2). The matrices $\mathbf{U}$ and $\mathbf{V}$ represents the linearly independent word factors and the document factors, respectively. The dimensionality reduction is accomplished by the truncated-SVD solutions, where all but the K ($< \text{rank}(\mathbf{D})$) largest singular values in $\mathbf{\Sigma}$ are set to zero ($\mathbf{\Sigma}_K$). It is expected that the K factors capture the most prominent association of terms and documents. Then, the association among documents, generally speaking, can be estimated in the K -dimensional space by $\mathbf{D}^T \mathbf{D} \approx (\mathbf{U} \mathbf{\Sigma}_K \mathbf{V}^T)^T (\mathbf{U} \mathbf{\Sigma}_K \mathbf{V}^T) = \mathbf{V} \mathbf{\Sigma}_K^2 \mathbf{V}$. By this, one may expect the semantic matching of documents rather than simple term matching. In a symmetric way, the word association can be given by $\mathbf{D} \mathbf{D}^T \approx (\mathbf{U} \mathbf{\Sigma}_K \mathbf{V}^T) (\mathbf{U} \mathbf{\Sigma}_K \mathbf{V}^T)^T = \mathbf{U} \mathbf{\Sigma}_K^2 \mathbf{U}^T$.

2.6 Non-negative Matrix Factorization:

The nonnegative matrix factorization (NMF) (Lee and Seung, 1999, 2001) has been proposed as a technique for the low rank approximation of *non-negative* matrix data. Given an $M \times N$ non-negative data $\mathbf{D} = \{v_{ij}\}$, NMF decomposes $\mathbf{D}$ into two

non-negative matrices $\mathbf{W}$ ($M \times K$) and $\mathbf{Z}$ ($K \times N$),

$$\mathbf{D} \approx \mathbf{WZ}, \quad (w_{ik} \geq 0, z_{kj} \geq 0, \text{ for } \forall i, j, k),$$

where K is usually much smaller than M and N . The non-negativity constraint for $\mathbf{W}$ and $\mathbf{Z}$ makes it possible to learn the *parts-based* representations, rather than *holistic* encoding, by allowing only additive combination in the reconstruction of original data. In the application to facial images in their introduction of NMF, Lee and Seung (1999) showed that NMF could extract a set of basis images for parts of faces, that is, eyes, noses, lips, and so on, which is not the case for VQ and PCA. Then, many other applications followed, which include text mining (Lee and Seung, 1999; Xu et al., 2003), multimedia data analysis (Kawamoto et al., 2000; Guillaumet et al., 2001, 2002; Cooper and Foote, 2002), biological data analysis (Heger and Holm, 2003; Brunet et al., 2004; Kim and Tidor, 2003).

Two matrices $\mathbf{W}$ and $\mathbf{Z}$ are estimated by iteratively alternating the respective update rules in an EM-like fashion. When the cost function is the square of Euclidean distance, $C = \|\mathbf{D} - \mathbf{WZ}\|_F^2$, the parameter estimation is given by

$$z_{kj} \leftarrow z_{kj} \frac{(\mathbf{W}^T \mathbf{D})_{kj}}{(\mathbf{W}^T \mathbf{WZ})_{kj}}, \quad w_{ik} \leftarrow w_{ik} \frac{(\mathbf{DZ}^T)_{ik}}{(\mathbf{WZZ}^T)_{ik}}.$$

In another way, when it is the case that the generalized Kullback-Leibler (KL) divergence, $C = \sum_{ij} \left(v_{ij} \log \frac{v_{ij}}{(\mathbf{WZ})_{ij}} - v_{ij} + (\mathbf{WZ})_{ij} \right)$, is tried to be minimized, the updates of $\mathbf{W}$ and $\mathbf{Z}$ are given by

$$z_{kj} = z_{kj} \frac{\sum_l w_{lk} v_{lj} / (\mathbf{WZ})_{lj}}{\sum_l w_{lk}}, \quad w_{ik} = w_{ik} \frac{\sum_l w_{kl} v_{il} / (\mathbf{WZ})_{il}}{\sum_l z_{kl}}.$$

Chapter 3

Co-Reference Latent Models for Paired Data Sets

Here, we describe paired data sets and an effective model for the simultaneous analysis of the paired data sets on the basis of the dimensionality reduction of the two data sets, called the *co-reference latent model*.

3.1 Paired Data Sets

Paired data sets are characterized by the fact that data samples are described explicitly by two *distinct* attribute sets. Let $\mathcal{S} = \{s_1, s_2, \dots, s_N\}$ be the set of data samples and $\mathcal{X} = \{X_1, X_2, \dots, X_M\}$ and $\mathcal{Y} = \{Y_1, Y_2, \dots, Y_L\}$ be two variable sets. Here, the numbers of attributes in $\mathcal{X}$ and $\mathcal{Y}$ can be different. A data sample $s_i \in \mathcal{S}$ is given as a joint representation of $s_i^X = (x_1, x_2, \dots, x_M)$ and $s_i^Y = (y_1, y_2, \dots, y_L)$. The two separate representations provide two different views of the same data samples. If a correlated pattern is observed across the representations, it can be said that there exists a corresponding underlying concept to which the two views are commonly related, which is an important issue in the analysis of the paired data set.

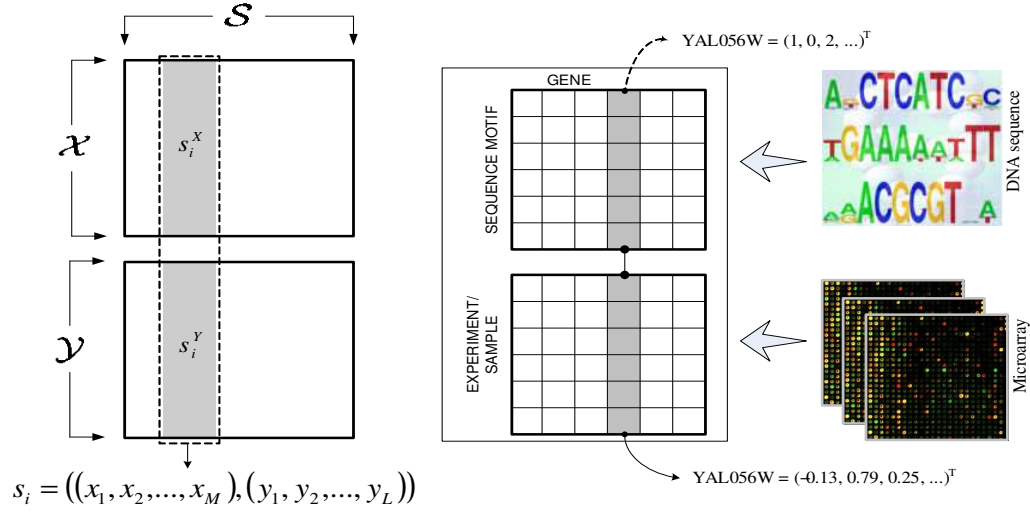


Figure 3.1: A brief view of paired data set. An elementary observation of the data matrix is considered an instantiation of a pair of objects.

Figure 3.1 briefly describes the concept of the two-table case of paired data sets, along with an example in the field of biological data analysis. After a series of microarray experiments, genes in an organism can be represented by their transcriptional activities (expression patterns) across the experiments. Related to the notion of transcription, the genes are also characterized by their associated DNA sequence elements which are presumed to be important in their transcription initiation. Then, every gene is given two separate representations, that is, one by sequence elements and the other by transcription activities. Here, the common concept which is sought between the two views will be concerned with the transcription activity of genes.¹ Among other examples of two-table paired data are text documents in different languages in cross-language retrieval (Littman et al., 1998; Vinokourov et al., 2003) and cancer cell lines for which gene expressions and drug activities are measured (Scherf

¹We discuss this problem in Chapter 4.

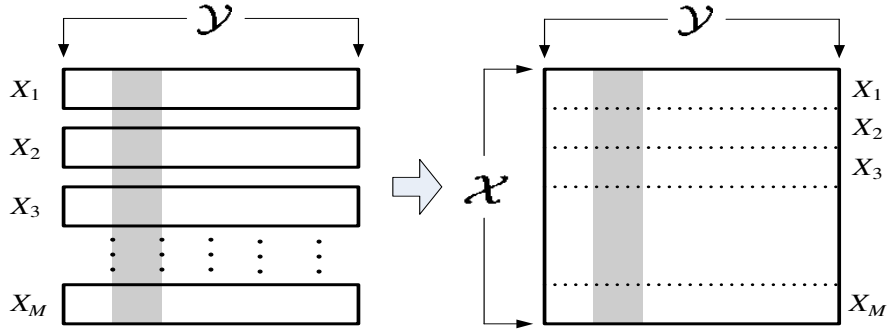


Figure 3.2: Multivariate data set as a juxtaposition of univariate data sets.

et al., 2000; Chang et al., 2002).

A plausible approach to the dependency analysis from the paired data set would be to use dimensionality reduction models for the two data sets respectively, and to pursue explicitly the dependency or co-structure underlying between the two data set while learning the two models. By the use of a separate model for each data set, we can well cope with the case that the variables of the two data sets of different types. Also, the clear modeling of the dependency structure in the latent spaces can be of great help in preventing the co-structure between the two data from fading away, dominated by the self-structure underlying the respective data sets, in the analysis,. This is mainly concerned with the extraction of pairs of statistically salient features associating the two data sets, one for each data set. The *saliency* or *meaningfulness* is the major factor which differentiates various methods in the framework of this approach, as in the case of single table. The following section describe an effective method for the simultaneous analysis of paired data sets in this framework.

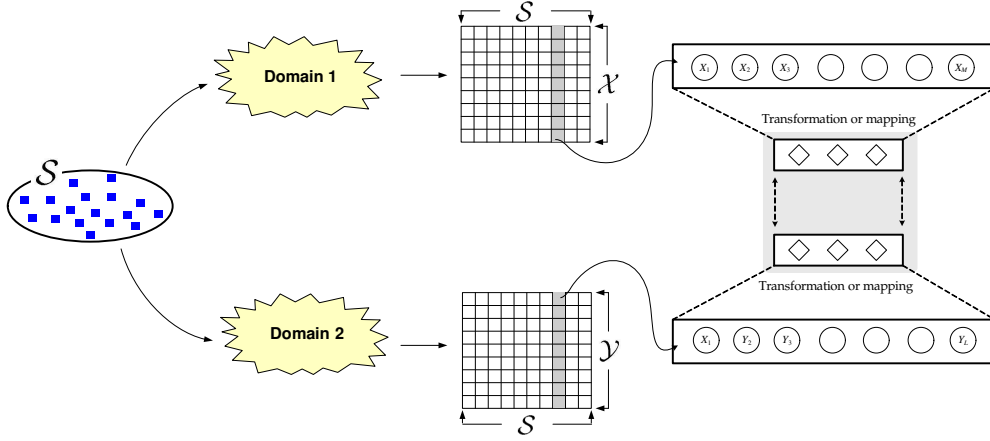


Figure 3.3: An overview of the paired data sets and the analysis by co-reference latent models

3.2 The Co-Reference Latent Model

3.2.1 Model Specification

The co-reference latent model is a latent variable model for associating the two representations by dimensionality reduction and then to find co-structure underlying the two data sets. Figure 3.3 shows an overview of the generation of a paired data set of two tables and the analysis by the co-reference latent model.

Let $\mathbf{D}_{\mathbf{X}}$ ($M \times N$) and $\mathbf{D}_{\mathbf{Y}}$ ($L \times N$) be two separate data sets comprising a paired data set $\mathcal{S} = \{s_1, s_2, \dots, s_N\}$. In a vector form, then, a data sample is represented as $\mathbf{s} = (\mathbf{x}, \mathbf{y})$, where $\mathbf{x}$ and $\mathbf{y}$ are column vectors of $\mathbf{D}_{\mathbf{X}}$ and $\mathbf{D}_{\mathbf{Y}}$ respectively. Under a latent variable model $\mathcal{M}$ which assumes a set of latent variables $\mathcal{Z}$, the probability

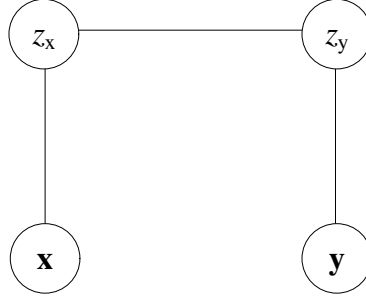


Figure 3.4: The graph-based representation of the co-reference latent model.

of $\mathbf{s}$ can be represented as

$$\begin{aligned}
 p_{\mathcal{M}}(\mathbf{s}) &= p(\mathbf{x}, \mathbf{y}) \\
 &= \int_{\mathbf{z}} p(\mathbf{x}, \mathbf{y}, \mathbf{z}) d\mathbf{z} \\
 &= \int_{\mathbf{z}} p(\mathbf{z}) p(\mathbf{x}, \mathbf{y} | \mathbf{z}) d\mathbf{z}.
 \end{aligned} \tag{3.1}$$

As shown in 3.4, the co-reference latent model assumes respective latent variables for two component data objects $\mathbf{x}$ and $\mathbf{y}$, unlike usual latent variable models for one data set. For the time being, we assume that the number of latent variables of each model is simply one. Then, the settings of latent variables $\mathbf{z}$ is given by $\mathbf{z} = (z_x, z_y)$, where z_x is for $\mathbf{x}$ and z_y is for $\mathbf{y}$. Further, the co-reference latent model conjectures that $\mathbf{x}$ is independent from z_y and $\mathbf{y}$ given z_x . Likewise, $\mathbf{y}$ is independent from z_x and $\mathbf{x}$ given z_y . Then, the probability in Equation (3.1) under a co-reference latent model $\mathcal{M}$ is given by ²

$$\begin{aligned}
 p_{\mathcal{M}}(\mathbf{s}) &= \int_{(z_x, z_y)} p(z_x, z_y) p(\mathbf{x}, \mathbf{y} | z_x, z_y) dz_x dz_y \\
 &= \int_{z_x} \int_{z_y} p(z_x, z_y) p(\mathbf{x} | z_x) p(\mathbf{y} | z_y) dz_x dz_y.
 \end{aligned} \tag{3.2}$$

²Here, we consider the case that z_x and z_y are continuous variables. In a discrete case, the integral can be replaced with the summation ($\sum$).

If data objects $\mathbf{s}_i$ ($1 \leq i \leq N$) are assumed to be independent and identically distributed, the log-likelihood function is

$$\mathcal{L}(\mathbf{S}|\Theta_{\mathcal{M}}) = \ln \prod_{i=1}^N p_{\mathcal{M}}(\mathbf{s}_i) = \sum_{i=1}^N \ln p_{\mathcal{M}}(\mathbf{s}_i). \quad (3.3)$$

Then, the learning task in the maximum likelihood formulation is to maximize $\mathcal{L}(\Theta_{\mathcal{M}})$:

$$\Theta_{\mathcal{M}}^* = \arg \max_{\Theta_{\mathcal{M}}} \mathcal{L}(\mathbf{S}|\Theta_{\mathcal{M}}). \quad (3.4)$$

The joint probability $p(z_x, z_y)$ for two latent variables z_x and z_y in Equation (3.2) can be decomposed into $p(z_x, z_y) = p(z_x)p(z_y|z_x)$. By the employment of the decomposition, the marginal probability $p_{\mathcal{M}}(\mathbf{s})$ can be transformed as

$$\begin{aligned} p_{\mathcal{M}}(\mathbf{s}) &= \int_{z_x} p(z_x)p(\mathbf{x}|z_x) \int_{z_y} p(z_y|z_x)p(\mathbf{y}|z_y) dz_x dz_y \\ &= \int_{z_x} p(z_x)p(\mathbf{x}|z_x)p(\mathbf{y}|z_x) dz_x \\ &= p_{\mathcal{M}}^1(\mathbf{s}), \end{aligned} \quad (3.5)$$

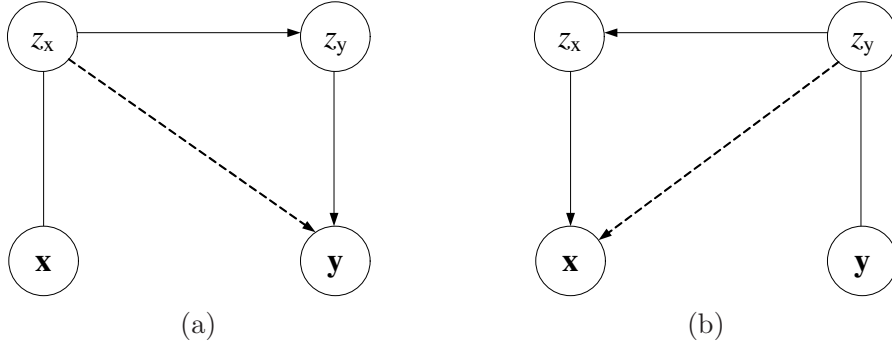


Figure 3.5: A graph-based illustration of two simplification procedures for the co-reference latent model by (a) $p(z_x, z_y) = p(z_x)p(z_y|z_x)$ and then $p(\mathbf{y}|z_x) = \int_{z_y} p(z_y|z_x)p(\mathbf{y}|z_y)$, and (b) $p(z_x, z_y) = p(z_y)p(z_x|z_y)$ and then $p(\mathbf{x}|z_y) = \int_{z_x} p(z_x|z_y)p(\mathbf{x}|z_x)$.

where $p(\mathbf{y}|z_x) = \int_{z_y} p(z_y|z_x)p(\mathbf{y}|z_y)$. We now have a simplified representation for $p_{\mathcal{M}}(\mathbf{s})$, since only one kind of unknown value z_x is concerned with the probability. Again, when we take $p(z_x, z_y) = p(z_y)p(z_x|z_y)$, $p_{\mathcal{M}}(\mathbf{s})$ is

$$\begin{aligned} p_{\mathcal{M}}(\mathbf{s}) &= \int_{z_y} p(z_y)p(\mathbf{y}|z_y) \int_{z_x} p(z_x|z_y)p(\mathbf{x}|z_x) dz_x dz_y \\ &= \int_{z_y} p(z_y)p(\mathbf{y}|z_y)p(\mathbf{x}|z_y) dz_y \\ &= p_{\mathcal{M}}^2(\mathbf{s}), \end{aligned} \tag{3.6}$$

where $p(\mathbf{x}|z_y) = \int_{z_x} p(z_x|z_y)p(\mathbf{x}|z_x)$. The probability $p_{\mathcal{M}}$ is now related with z_y only. Figure 3.5 presents a simple illustration of the two transformations of the co-reference latent model in terms of the graph structure in Figure 3.4. Then the log-likelihood function in Equation (3.4) can be given by

$$\begin{aligned} \mathcal{L}(\mathbf{S}|\Theta_{\mathcal{M}}) &= \frac{1}{2} \sum_{i=1}^N (\ln p_{\mathcal{M}}^1(\mathbf{s}_i) + \ln p_{\mathcal{M}}^2(\mathbf{s}_i)) \\ &= \frac{1}{2} \left[\sum_{i=1}^N \ln p_{\mathcal{M}}^1(\mathbf{s}_i) + \sum_{i=1}^N \ln p_{\mathcal{M}}^2(\mathbf{s}_i) \right] = \frac{1}{2} [\mathcal{L}_1(\mathbf{S}|\Theta_{\mathcal{M}}) + \mathcal{L}_2(\mathbf{S}|\Theta_{\mathcal{M}})]. \end{aligned} \tag{3.7}$$

With an analogy to the “variational free energy” in statistical physics, furthermore, both log-probabilities $\ln p_{\mathcal{M}}^1(\mathbf{s})$ of $\mathcal{L}_1$ and $\ln p_{\mathcal{M}}^2(\mathbf{s})$ in $\mathcal{L}_2$ can be decomposed into sum of log-based terms. Let z be a physical state and $e(\mathbf{s}, z) = -\ln p(\mathbf{s}, z|\Theta)$ be the energy of the state z in a latent variable model parameterized with Θ . Then the log-probability $\ln p(\mathbf{s}|\Theta) = \ln \int_z p(\mathbf{s}, z|\Theta)$ at the local maximum of Θ can be represented in terms of the “average energy” and the entropy of the probability distribution $\tilde{p} = p(z|\mathbf{s}, \Theta)$ (Dayan et al., 1995; Neal and Hinton, 1998; Sahani, 1999), that is,

$$\ln p(\mathbf{s}|\Theta) = -\langle e(\mathbf{s}, z) \rangle_{\tilde{p}} + H(\tilde{p}), \tag{3.8}$$

where the average energy $\langle e(\mathbf{s}, z) \rangle_{\tilde{p}} = \int_z e(\mathbf{s}, z) \ln p(z|\mathbf{s}, \Theta) dz$ and the (differential) entropy $H(\tilde{p}) = - \int_z p(z|\mathbf{s}, \Theta) \ln p(z|\mathbf{s}, \Theta) dz$. In the framework of this representation, if a distribution $\tilde{q}$ is used for the approximation of $\tilde{p}$, the log-probability $\ln p(\mathbf{s}|\Theta)$ can be rewritten in terms of $\tilde{q}$ as follows:

$$\ln p(\mathbf{s}|\Theta) = - \langle e(\mathbf{s}, z) \rangle_{\tilde{q}} + H(\tilde{q}) + KL(\tilde{q}|\tilde{p}), \quad (3.9)$$

where $KL(\tilde{q}|\tilde{p}) = \int_z \tilde{q} \ln \frac{\tilde{q}}{\tilde{p}} dz$ is the Kullback-Leibler (KL) divergence between the two distribution $\tilde{q}$ and $\tilde{p}$.

From the arguments, if we set $\tilde{q} = p(z_x|\mathbf{x}, \Theta_{\mathcal{M}})$ in the formulation of the co-reference latent model in Equation (3.5), which is reasonable since z_x is a latent variable for the data $\mathbf{x}$ in the settings of the co-reference latent model,

$$\begin{aligned} \ln p_{\mathcal{M}}^1(\mathbf{s}) &= \int_{z_x} \tilde{q} \ln [p(z_x)p(\mathbf{x}|z_x)p(\mathbf{y}|z_x)] dz_x + H(\tilde{q}) + KL(\tilde{q}|\tilde{p}) \\ &= f(\mathbf{s}, \tilde{q}, \Theta_{\mathcal{M}}) + KL(\tilde{q}|\tilde{p}) = A(\mathbf{x}), \end{aligned} \quad (3.10)$$

where $\tilde{p} = p(z_x|\mathbf{x}, \mathbf{y}, \Theta_{\mathcal{M}})$ and $f(\mathbf{s}, \tilde{q}, \Theta_{\mathcal{M}})$ is

$$\begin{aligned} f(\mathbf{s}, \tilde{q}, \Theta_{\mathcal{M}}) &= \int_{z_x} \tilde{q} \ln [p(z_x)p(\mathbf{x}|z_x)p(\mathbf{y}|z_x)] dz_x + H(\tilde{q}) \\ &= \int_{z_x} \tilde{q} \ln p(z_x)p(\mathbf{x}|z_x) dz_x + H(\tilde{q}) + \int_{z_x} \tilde{q} \ln p(\mathbf{y}|z_x) dz_x \\ &= \left[- \langle e_{\mathcal{M}_x}(\mathbf{x}, z_x) \rangle_{\tilde{q}} + H(\tilde{q}) \right] + \left[- \langle e_{\mathcal{M}_y}(\mathbf{y}, z_x) \rangle_{\tilde{q}} \right]. \end{aligned} \quad (3.11)$$

Here, $e_{\mathcal{M}_x}(\mathbf{x}, z_x)$ is the energy or *error* from the setting of z_x in a latent variable model $\mathcal{M}_x$ for $\mathbf{x}$, and $e_{\mathcal{M}_y}(\mathbf{y}, z_x)$ is that in a model $\mathcal{M}_y$ for $\mathbf{y}$. In a similar way with $A(\mathbf{x})$, $\ln p_{\mathcal{M}}^2(\mathbf{s})$ can be expressed as

$$\ln p_{\mathcal{M}}^2(\mathbf{s}) = f(\mathbf{s}, \hat{q}, \Theta_{\mathcal{M}}) + KL(\hat{q}|\hat{p}) = A(\mathbf{y}) \quad (3.12)$$

where $\hat{q} = p(z_y|\mathbf{y}, \Theta_{\mathcal{M}})$, $\hat{p} = p(z_y|\mathbf{x}, \mathbf{y}, \Theta_{\mathcal{M}})$, and $f(\mathbf{s}, \hat{q}, \Theta_{\mathcal{M}})$ is

$$\left\{ \left[- \langle e_{\mathcal{M}_y}(\mathbf{y}, z_y) \rangle_{\hat{q}} + H(\hat{q}) \right] + \left[- \langle e_{\mathcal{M}_x}(\mathbf{x}, z_y) \rangle_{\hat{q}} \right] \right\}. \quad (3.13)$$

From Equations (3.10) and (3.12), then the log-likelihood function $\mathcal{L}(\mathbf{S}|\Theta_{\mathcal{M}})$ is given by

$$\begin{aligned}
\mathcal{L}(\mathbf{S}|\Theta_{\mathcal{M}}) &= \frac{1}{2} \left[\sum_{i=1}^N A_i(\mathbf{x}) + \sum_{i=1}^N A_i(\mathbf{y}) \right] \\
&= \frac{1}{2} \left[\sum_{i=1}^N f(\mathbf{s}, \tilde{q}, \Theta_{\mathcal{M}}) + \sum_{i=1}^N f(\mathbf{s}, \hat{q}, \Theta_{\mathcal{M}}) \right] + \frac{1}{2} \sum_{i=1}^N [KL(\tilde{q}|\hat{p}) + KL(\hat{q}|\hat{p})] \\
&\geq \frac{1}{2} \left[\sum_{i=1}^N f(\mathbf{s}, \tilde{q}, \Theta_{\mathcal{M}}) + \sum_{i=1}^N f(\mathbf{s}, \hat{q}, \Theta_{\mathcal{M}}) \right] \\
&= \frac{1}{2} [\mathcal{F}_1(\mathbf{S}, \Theta_{\mathcal{M}_x}, \Theta_{\mathcal{M}_y}) + \mathcal{F}_2(\mathbf{S}, \Theta_{\mathcal{M}_x}, \Theta_{\mathcal{M}_y})], \tag{3.14}
\end{aligned}$$

where $\Theta_{\mathcal{M}} = (\Theta_{\mathcal{M}_x}, \Theta_{\mathcal{M}_y})$. Here, $\Theta_{\mathcal{M}_x}$ and $\Theta_{\mathcal{M}_y}$ are the parameters of $\mathcal{M}_x$ and $\mathcal{M}_y$, respectively. The inequality in the third line is due to the fact that the KL divergence is always non-negative. So, the term in the last line serves as a lower bound of $\mathcal{L}(\mathbf{S}|\Theta_{\mathcal{M}})$ and is employed as the basic objective function $\mathcal{F}$, dropping the constant term, in the learning of co-reference latent model.

$$\begin{aligned}
\Theta_{\mathcal{M}}^* &= (\Theta_{\mathcal{M}_x}^*, \Theta_{\mathcal{M}_y}^*) \\
&= \arg \max_{(\Theta_{\mathcal{M}_x}, \Theta_{\mathcal{M}_y})} \mathcal{F}(\mathbf{S}, \Theta_{\mathcal{M}_x}, \Theta_{\mathcal{M}_y}) \\
&= \arg \max_{(\Theta_{\mathcal{M}_x}, \Theta_{\mathcal{M}_y})} [\mathcal{F}_1(\mathbf{S}, \Theta_{\mathcal{M}_x}, \Theta_{\mathcal{M}_y}) + \mathcal{F}_2(\mathbf{S}, \Theta_{\mathcal{M}_x}, \Theta_{\mathcal{M}_y})] \\
&= \arg \max_{(\Theta_{\mathcal{M}_x}, \Theta_{\mathcal{M}_y})} \left\{ \sum_{i=1}^N \left[-\langle e(\mathbf{x}_i, z_{\mathbf{x}}^i) \rangle_{\tilde{q}_i} + H(\tilde{q}_i) \right] + \sum_{i=1}^N \left[-\langle e(\mathbf{y}_i, z_{\mathbf{y}}^i) \rangle_{\hat{q}_i} \right] \right. \\
&\quad \left. + \sum_{i=1}^N \left[-\langle e(\mathbf{x}_i, z_{\mathbf{y}}^i) \rangle_{\hat{q}_i} + H(\hat{q}_i) \right] + \sum_{i=1}^N \left[-\langle e(\mathbf{y}_i, z_{\mathbf{x}}^i) \rangle_{\tilde{q}_i} \right] \right\} \\
&= \arg \max_{(\Theta_{\mathcal{M}_x}, \Theta_{\mathcal{M}_y})} \{ \phi_{\mathcal{M}_x}(\mathbf{D}_{\mathbf{X}}, \mathbf{z}_{\mathbf{x}}) + \phi_{\mathcal{M}_y}(\mathbf{D}_{\mathbf{Y}}, \mathbf{z}_{\mathbf{y}}) \\
&\quad + \phi_{\mathcal{M}_y}(\mathbf{D}_{\mathbf{Y}}, \mathbf{z}_{\mathbf{y}}) + \phi_{\mathcal{M}_x}(\mathbf{D}_{\mathbf{X}}, \mathbf{z}_{\mathbf{y}}) \} \tag{3.15}
\end{aligned}$$

where $\mathbf{z}_{\mathbf{x}} = (z_{\mathbf{x}}^1, \dots, z_{\mathbf{x}}^N)^T$, $\mathbf{z}_{\mathbf{y}} = (z_{\mathbf{y}}^1, \dots, z_{\mathbf{y}}^N)^T$, $\tilde{q}_i = p(z_{\mathbf{x}}|\mathbf{x}_i, \Theta_{\mathcal{M}_x})$, $\hat{q}_i = p(z_{\mathbf{y}}|\mathbf{y}_i, \Theta_{\mathcal{M}_y})$,

and

$$\phi_{\mathcal{M}_{\mathcal{X}}}(\mathbf{D}_{\mathbf{X}}, \mathbf{z}_{\mathbf{x}}) = -E_{\mathcal{M}_{\mathcal{X}}}(\mathbf{D}_{\mathbf{X}}, \mathbf{z}_{\mathbf{x}}) = -\sum_{i=1}^N \left[\langle e(\mathbf{x}_i, z_{\mathbf{x}}^i) \rangle_{\tilde{q}_i} - H(\tilde{q}_i) \right] \quad (3.16)$$

$$\phi_{\mathcal{M}_{\mathcal{Y}}}(\mathbf{D}_{\mathbf{Y}}, \mathbf{z}_{\mathbf{x}}) = -E_{\mathcal{M}_{\mathcal{Y}}}(\mathbf{D}_{\mathbf{Y}}, \mathbf{z}_{\mathbf{x}}) = -\sum_{i=1}^N \left[\langle e_{\mathcal{M}_{\mathcal{Y}}}(\mathbf{y}_i, z_{\mathbf{x}}^i) \rangle_{\tilde{q}_i} \right] \quad (3.17)$$

$$\phi_{\mathcal{M}_{\mathcal{Y}}}(\mathbf{D}_{\mathbf{Y}}, \mathbf{z}_{\mathbf{y}}) = -E_{\mathcal{M}_{\mathcal{Y}}}(\mathbf{D}_{\mathbf{Y}}, \mathbf{z}_{\mathbf{y}}) = -\sum_{i=1}^N \left[\langle e(\mathbf{y}_i, z_{\mathbf{y}}^i) \rangle_{\hat{q}_i} - H(\hat{q}_i) \right] \quad (3.18)$$

$$\phi_{\mathcal{M}_{\mathcal{X}}}(\mathbf{D}_{\mathbf{X}}, \mathbf{z}_{\mathbf{y}}) = -E_{\mathcal{M}_{\mathcal{X}}}(\mathbf{D}_{\mathbf{X}}, \mathbf{z}_{\mathbf{y}}) = -\sum_{i=1}^N \left[\langle e_{\mathcal{M}_{\mathcal{X}}}(\mathbf{x}_i, z_{\mathbf{y}}^i) \rangle_{\hat{q}_i} \right]. \quad (3.19)$$

Then if we set $\phi_{\mathcal{M}_{\mathcal{Z}}}(\mathbf{z}_{\mathbf{x}}, \mathbf{z}_{\mathbf{y}}) = \phi_{\mathcal{M}_{\mathcal{X}}}(\mathbf{D}_{\mathbf{X}}, \mathbf{z}_{\mathbf{y}}) + \phi_{\mathcal{M}_{\mathcal{Y}}}(\mathbf{D}_{\mathbf{Y}}, \mathbf{z}_{\mathbf{x}})$, the objective function $\mathcal{F}$ says that the co-reference latent model is concerned with the maximization of the three criteria:

- $\phi_{\mathcal{M}_{\mathcal{X}}}(\mathbf{D}_{\mathbf{X}}, \mathbf{z}_{\mathbf{x}})$: the degree of description of $\mathbf{D}_{\mathbf{X}}$ by $\mathcal{M}_{\mathcal{X}}$.
- $\phi_{\mathcal{M}_{\mathcal{Y}}}(\mathbf{D}_{\mathbf{Y}}, \mathbf{z}_{\mathbf{y}})$: the degree of description of $\mathbf{D}_{\mathbf{Y}}$ by $\mathcal{M}_{\mathcal{Y}}$.
- $\phi_{\mathcal{M}_{\mathcal{Z}}}(\mathbf{z}_{\mathbf{x}}, \mathbf{z}_{\mathbf{y}})$: the degree of description of the association patterns underlying $\mathbf{D}_{\mathbf{X}}$ and $\mathbf{D}_{\mathbf{Y}}$, through $\mathbf{z}_{\mathbf{x}}$ and $\mathbf{z}_{\mathbf{y}}$ which are extracted by $\mathcal{M}_{\mathcal{X}}$ and $\mathcal{M}_{\mathcal{Y}}$ respectively.

3.2.2 Learning the Model

Since the objective function $\mathcal{F}$ consists of two sub-objective function $\mathcal{F}_1$ and $\mathcal{F}_2$ as shown in Equation (3.15), the learning process of the co-reference latent model can be basically decomposed into two parts, one for $\mathcal{F}_1$ and the other for $\mathcal{F}_2$.

In the case of $\mathcal{F}_1$, it is concerned with only $z_{\mathbf{x}}$ in terms of latent variable. Furthermore, the distribution $\tilde{q}$ for $z_{\mathbf{x}}$ in $\mathcal{F}_1$ is only dependent on $\Theta_{\mathcal{M}_{\mathcal{X}}}$ and $\mathbf{D}_{\mathbf{X}}$, but not on $\Theta_{\mathcal{M}_{\mathcal{Y}}}$ and $\mathbf{D}_{\mathbf{Y}}$. Given an estimate of the parameter set of the model $\mathcal{M}_{\mathcal{X}}$ as

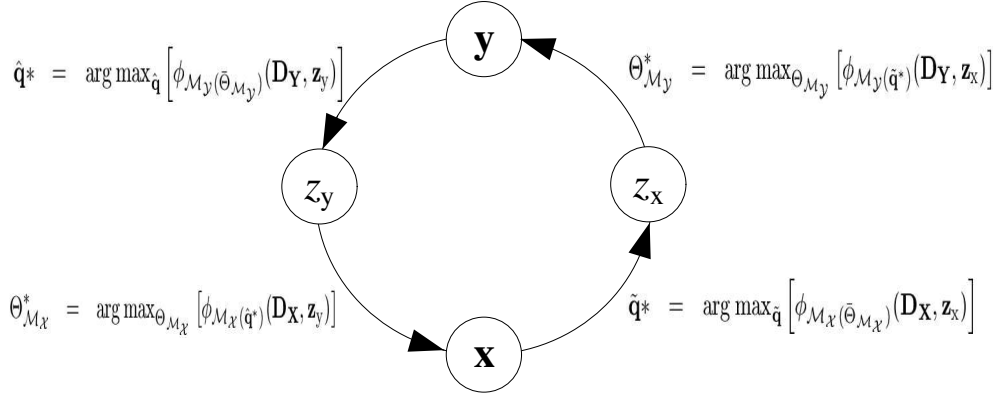


Figure 3.6: The learning scheme of the co-reference latent model

$\bar{\theta}_{\mathcal{M}_x}$, the maximization of $\mathcal{F}_1(\mathbf{S}, \bar{\theta}_{\mathcal{M}_x}, \theta_{\mathcal{M}_y})$ can be separated into two sequential procedures, that is,

$$\tilde{\mathbf{q}}^* = \arg \max_{\tilde{\mathbf{q}}} [\phi_{\mathcal{M}_x}(\bar{\theta}_{\mathcal{M}_x})(\mathbf{D}_X, \mathbf{z}_x)] \quad (3.20)$$

$$\theta_{\mathcal{M}_y}^* = \arg \max_{\theta_{\mathcal{M}_y}} [\phi_{\mathcal{M}_y}(\tilde{\mathbf{q}}^*)(\mathbf{D}_Y, \mathbf{z}_x)], \quad (3.21)$$

where $\tilde{\mathbf{q}} = \{\tilde{q}_i\}_{i=1}^N$.

Then, the maximization of $\mathcal{F}_2$ can be done in a similar way with $\mathcal{F}_1$, since the distribution $\hat{q}$ for z_y is only dependent on $\mathcal{M}_y$ and the estimate of $\theta_{\mathcal{M}_y}$ has been given as in Equation (3.21). Thus, with the setting of $\bar{\theta}_{\mathcal{M}_y} = \theta_{\mathcal{M}_y}^*$, the sequential procedures are

$$\hat{\mathbf{q}}^* = \arg \max_{\hat{\mathbf{q}}} [\phi_{\mathcal{M}_y}(\bar{\theta}_{\mathcal{M}_y})(\mathbf{D}_Y, \mathbf{z}_y)] \quad (3.22)$$

$$\theta_{\mathcal{M}_x}^* = \arg \max_{\theta_{\mathcal{M}_x}} [\phi_{\mathcal{M}_x}(\hat{\mathbf{q}}^*)(\mathbf{D}_X, \mathbf{z}_y)], \quad (3.23)$$

where $\hat{\mathbf{q}} = \{\hat{q}_i\}_{i=1}^N$. Since the parameter $\theta_{\mathcal{M}_x}$ has been updated, the learning process returns again to the maximization of $\mathcal{F}_1$ in Equation (3.20) and Equation (3.21), with the setting of $\bar{\theta}_{\mathcal{M}_x} = \theta_{\mathcal{M}_x}^*$. In conclusion, the learning of the co-reference

latent model can be accomplished by the iterative alternation of the maximization of $\mathcal{F}_1$ and $\mathcal{F}_2$, where one iteration ranges from Equation (3.20) to Equation (3.23). Figure 3.6 presents the summarized illustration of the learning procedure.

In relation with the Expectation-Maximization (EM) algorithm (Dempster et al., 1977), it can be said that Equation (3.20) corresponds to the E-step for $\mathcal{M}_x$ and Equation (3.21) to the M-step for $\mathcal{M}_y$ with an *incorrect* distribution $\tilde{q}$. Likewise, Equation (3.22) corresponds to the E-step for $\mathcal{M}_y$ and Equation (3.23) to the M-step for $\mathcal{M}_x$ with a approximating distribution $\hat{q}$.

3.2.3 Winner-Take-All Approximation

As a special case, we can take the *winner-take-all* or hard assignment approach for z_x and z_y , by which we means that the distributions $\tilde{q}$ and $\hat{q}$ assign zero probabilities to all values except the most plausible values z_x^* and z_y^* , respectively. Then, the optimization step in Equation (3.20) and Equation (3.22) can be stated as

$$\mathbf{z}_x^* = \arg \max_{\mathbf{z}_x} \left[\phi_{\mathcal{M}_x}(\bar{\Theta}_{\mathcal{M}_x})(\mathbf{D}_x, \mathbf{z}_x) \right] \quad (3.24)$$

$$\mathbf{z}_y^* = \arg \max_{\mathbf{z}_y} \left[\phi_{\mathcal{M}_y}(\bar{\Theta}_{\mathcal{M}_y})(\mathbf{D}_y, \mathbf{z}_y) \right], \quad (3.25)$$

where $\mathbf{z}_x = \{z_x^i\}_{i=1}^N$ and $\mathbf{z}_y = \{z_y^i\}_{i=1}^N$.

Given the point estimates $\mathbf{z}_x^*$ and $\mathbf{z}_y^*$, four components from Equation (3.16) to Equation (3.19) can be defined as

$$\phi_{\mathcal{M}_x}(\mathbf{D}_x, \mathbf{z}_x^*) = -E_{\mathcal{M}_x}^{WTA}(\mathbf{D}_x, \mathbf{z}_x^*) = -\sum_{i=1}^N [e_{\mathcal{M}_x}(\mathbf{x}_i, z_x^{i*})] \quad (3.26)$$

$$\phi_{\mathcal{M}_y}(\mathbf{D}_y, \mathbf{z}_y^*) = -E_{\mathcal{M}_y}^{WTA}(\mathbf{D}_y, \mathbf{z}_y^*) = -\sum_{i=1}^N [e_{\mathcal{M}_y}(\mathbf{y}_i, z_y^{i*})] \quad (3.27)$$

$$\phi_{\mathcal{M}_y}(\mathbf{D}_y, \mathbf{z}_y^*) = -E_{\mathcal{M}_y}^{WTA}(\mathbf{D}_y, \mathbf{z}_y^*) = -\sum_{i=1}^N [e_{\mathcal{M}_y}(\mathbf{y}_i, z_y^{i*})] \quad (3.28)$$

$$\phi_{\mathcal{M}_X}(\mathbf{D}_X, \mathbf{z}_y^*) = -E_{\mathcal{M}_X}^{WTA}(\mathbf{D}_X, \mathbf{z}_y^*) = -\sum_{i=1}^N [e_{\mathcal{M}_X}(\mathbf{x}_i, z_y^{i*})], \quad (3.29)$$

since it is always the case that $\tilde{q}(z_x^{i*}) = 1$ and $\hat{q}(z_y^{i*}) = 1$ ($1 \leq i \leq N$).

In the remainder of this dissertation, including the applications in Chapter 4 and Chapter 5, only the winner-take-all framework is considered for the co-reference latent model. Finally, Table 3.1 presents the entire learning procedure of this case in a stepwise manner.

For a specific example, let $\mathcal{M}_X$ and $\mathcal{M}_Y$ be a linear model for the data set $\mathbf{D}_X$ and $\mathbf{D}_Y$, respectively. A sample $\mathbf{x} = (x_1, x_2, \dots, x_M)^T$ of $\mathbf{D}_X$ can be approximated by

$$\mathbf{x} \approx z_x \mathbf{w}_x,$$

where $\mathbf{w}_x$ is the set of linear parameters of $\mathcal{M}_X$. In the linear approximation of a data sample, the adequacy of a model can be usually estimated by the reconstruction error. Thus, if the measure for the reconstruction error is the *Euclidean distance*, the error $e_{\mathcal{M}_X}(\mathbf{x}, z_x)$ for a sample $\mathbf{x}$ can be given by

$$e_{\mathcal{M}_X}(\mathbf{x}, z_x) = \frac{1}{2} \|\mathbf{x} - z_x \mathbf{w}_x\|^2. \quad (3.34)$$

Then $\phi_{\mathcal{M}_X}(\mathbf{D}_X, \mathbf{z}_x)$ is

$$\begin{aligned} \phi_{\mathcal{M}_X}(\mathbf{D}_X, \mathbf{z}_x) &= -\sum_{i=1}^N (e_{\mathcal{M}_X}(\mathbf{x}_i, z_x^i)) \\ &= -\frac{1}{2} \|\mathbf{D}_X - \mathbf{w}_x \mathbf{z}_x\|_F^2, \end{aligned} \quad (3.35)$$

where $\|\mathbf{A}\|_F$ means the Frobenius matrix norm (see Equation (2.4) in Section 2.2) of the matrix $\mathbf{A}$. In the same way, $\phi_{\mathcal{M}_Y}(\mathbf{D}_Y, \mathbf{z}_y)$ is given by

$$\begin{aligned} \phi_{\mathcal{M}_Y}(\mathbf{D}_Y, \mathbf{z}_y) &= -\sum_{i=1}^N (e_{\mathcal{M}_Y}(\mathbf{y}_i, z_y^i)) \\ &= -\frac{1}{2} \|\mathbf{D}_Y - \mathbf{w}_y \mathbf{z}_y\|_F^2, \end{aligned} \quad (3.36)$$

Table 3.1: The stepwise learning procedure of a co-reference latent model in the winner-take-all framework.

STEP 1: Random initialization of the model parameters of $\mathbf{M}_\mathbf{X}$: $\Theta_{\mathcal{M}_\mathbf{X}}^{(0)}$.

Set the iteration count $t \leftarrow 0$.

STEP 2: Estimate $\mathbf{z}_\mathbf{x}$ for $\mathbf{D}_\mathbf{X}$ on the basis of $\Theta_{\mathcal{M}_\mathbf{X}}^{(t)}$.

$$\mathbf{z}_\mathbf{x}^{(t+1)} = \arg \max_{\mathbf{z}_\mathbf{x}} \left[\phi_{\mathcal{M}_\mathbf{X}(\Theta_{\mathcal{M}_\mathbf{X}}^{(t)})}(\mathbf{D}_\mathbf{X}, \mathbf{z}_\mathbf{x}) \right] \quad (3.30)$$

STEP 3: Estimate $\Theta_{\mathcal{M}_\mathbf{Y}}$ for $\mathbf{D}_\mathbf{Y}$ on the basis of $\mathbf{z}_\mathbf{x}^{(t+1)}$.

$$\Theta_{\mathcal{M}_\mathbf{Y}}^{(t+1)} = \arg \max_{\Theta_{\mathcal{M}_\mathbf{Y}}} \left[\phi_{\mathcal{M}_\mathbf{Y}}(\mathbf{D}_\mathbf{Y}, \mathbf{z}_\mathbf{x}^{(t+1)}) \right] \quad (3.31)$$

STEP 4: Estimate $\mathbf{z}_\mathbf{y}$ for $\mathbf{D}_\mathbf{Y}$ on the basis of $\Theta_{\mathcal{M}_\mathbf{Y}}^{(t+1)}$.

$$\mathbf{z}_\mathbf{y}^{(t+1)} = \arg \max_{\mathbf{z}_\mathbf{y}} \left[\phi_{\mathcal{M}_\mathbf{Y}(\Theta_{\mathcal{M}_\mathbf{Y}}^{(t+1)})}(\mathbf{D}_\mathbf{Y}, \mathbf{z}_\mathbf{y}) \right] \quad (3.32)$$

STEP 5: Estimate $\Theta_{\mathcal{M}_\mathbf{X}}$ for $\mathbf{D}_\mathbf{X}$ on the basis of $\mathbf{z}_\mathbf{y}^{(t+1)}$.

$$\Theta_{\mathcal{M}_\mathbf{X}}^{(t+1)} = \arg \max_{\Theta_{\mathcal{M}_\mathbf{X}}} \left[\phi_{\mathcal{M}_\mathbf{X}}(\mathbf{D}_\mathbf{X}, \mathbf{z}_\mathbf{y}^{(t+1)}) \right] \quad (3.33)$$

STEP 6: If the learning has been converged or $t \geq t_{\max}$ then stop. Else, $t \leftarrow t+1$ and go to STEP 2.

under the error measure based on the Euclidean distance. Then, $\phi_{\mathcal{M}_X}(\mathbf{D}_X, \mathbf{z}_y)$ and $\phi_{\mathcal{M}_Y}(\mathbf{D}_Y, \mathbf{z}_x)$ are

$$\begin{aligned}\phi_{\mathcal{M}_X}(\mathbf{D}_X, \mathbf{z}_y) &= -\sum_{i=1}^N (e_{\mathcal{M}_X}(\mathbf{x}_i, z_y^i)) \\ &= -\frac{1}{2} \|\mathbf{D}_X - \mathbf{w}_x \mathbf{z}_y\|_F^2\end{aligned}\quad (3.37)$$

$$\begin{aligned}\phi_{\mathcal{M}_Y}(\mathbf{D}_Y, \mathbf{z}_x) &= -\sum_{i=1}^N (e_{\mathcal{M}_Y}(\mathbf{y}_i, z_x^i)) \\ &= -\frac{1}{2} \|\mathbf{D}_Y - \mathbf{w}_y \mathbf{z}_x\|_F^2.\end{aligned}\quad (3.38)$$

In addition, we constrain $\mathbf{w}_x$ and $\mathbf{w}_y$ to be unit vectors, that is, $\|\mathbf{w}_x\|^2 = \|\mathbf{w}_y\|^2 = 1$.

Initially, we randomly set the values of $\mathbf{w}_x$ of $\mathcal{M}_X$. Then, with the start value of $\mathbf{w}_x^{(t)}$ ($t = 0$), STEP 2 in Table 3.1 is

$$\begin{aligned}\mathbf{z}_x^{(t+1)} &= \arg \max_{\mathbf{z}_x} [\phi_{\mathcal{M}_X}(\mathbf{D}_X, \mathbf{z}_x)] \\ &= \arg \min_{\mathbf{z}_x} \|\mathbf{D}_X - \mathbf{w}_x^{(t)} \mathbf{z}_x\|_F^2.\end{aligned}\quad (3.39)$$

It is well known that the least squares problems have a closed form solution such that

$$\begin{aligned}\mathbf{z}_x^{(t+1)} &= \left(\left(\mathbf{w}_x^{(t)} \right)^T \left(\mathbf{w}_x^{(t)} \right) \right)^{-1} \left(\mathbf{w}_x^{(t)} \right)^T \mathbf{D}_X \\ &= \frac{\left(\mathbf{w}_x^{(t)} \right)^T \mathbf{D}_X}{\left\| \mathbf{w}_x^{(t)} \right\|^2} = \left(\mathbf{w}_x^{(t)} \right)^T \mathbf{D}_X \quad (\|\mathbf{w}_x\|^2 = 1),\end{aligned}\quad (3.40)$$

Based on the estimate of $\mathbf{z}_x^{(t+1)}$, the next step (STEP 3) is

$$\begin{aligned}\mathbf{w}_y^{(t+1)} &= \arg \max_{\mathbf{w}_y} \left[-E_{\mathcal{M}_Y}(\mathbf{D}_Y, \mathbf{z}_x^{(t+1)}) \right] \\ &= \arg \min_{\mathbf{w}_y} (\|\mathbf{D}_Y - \mathbf{w}_y \mathbf{z}_x^{(t+1)}\|_F^2),\end{aligned}\quad (3.41)$$

which produces

$$\begin{aligned}
\mathbf{w}_y^{(t+1)} &= \left(\left(\mathbf{z}_x^{(t+1)} \right) \left(\mathbf{z}_x^{(t+1)} \right)^T \right)^{-1} \mathbf{D}_Y \left(\mathbf{z}_x^{(t+1)} \right)^T \\
&= \frac{\mathbf{D}_Y \left(\mathbf{z}_x^{(t+1)} \right)^T}{\left\| \mathbf{z}_x^{(t+1)} \right\|^2} \\
\mathbf{w}_y^{(t+1)} &= \frac{\mathbf{D}_Y \left(\mathbf{z}_x^{(t+1)} \right)^T}{\left\| \mathbf{D}_Y \left(\mathbf{z}_x^{(t+1)} \right)^T \right\|} \quad (\because \|\mathbf{w}_y\| = 1). \tag{3.42}
\end{aligned}$$

From the sequence of STEP 4 and STEP 5, $\mathbf{z}_y$ and $\mathbf{w}_x$ can be given in the same way with $\mathbf{z}_x^{(t+1)}$ and $\mathbf{w}_y^{(t+1)}$,

$$\begin{aligned}
\mathbf{z}_y^{(t+1)} &= \left(\left(\mathbf{w}_y^{(t+1)} \right)^T \left(\mathbf{w}_y^{(t+1)} \right) \right)^{-1} \left(\mathbf{w}_y^{(t+1)} \right)^T \mathbf{D}_Y \\
&= \frac{\left(\mathbf{w}_y^{(t+1)} \right)^T \mathbf{D}_Y}{\left\| \mathbf{w}_y^{(t+1)} \right\|^2} = \left(\mathbf{w}_y^{(t+1)} \right)^T \mathbf{D}_Y \quad (\|\mathbf{w}_y\| = 1) \tag{3.43}
\end{aligned}$$

$$\begin{aligned}
\mathbf{w}_x^{(t+1)} &= \left(\left(\mathbf{z}_y^{(t+1)} \right) \left(\mathbf{z}_y^{(t+1)} \right)^T \right)^{-1} \mathbf{D}_X \left(\mathbf{z}_y^{(t+1)} \right)^T \\
&= \frac{\mathbf{D}_X \left(\mathbf{z}_y^{(t+1)} \right)^T}{\left\| \mathbf{z}_y^{(t+1)} \right\|^2} \\
\mathbf{w}_x^{(t+1)} &= \frac{\mathbf{D}_X \left(\mathbf{z}_y^{(t+1)} \right)^T}{\left\| \mathbf{D}_X \left(\mathbf{z}_y^{(t+1)} \right)^T \right\|} \quad (\because \|\mathbf{w}_x\| = 1). \tag{3.44}
\end{aligned}$$

Then, the optimization steps from Equation (3.40) to Equation (3.44) are sequentially iterated ($t = 1, 2, \dots$), until convergence.

3.3 Variants of the Model

The Measure for the Adequacy of Data Models

Euclidean Distance Measure

In the example of the linear co-reference latent model in previous section, we can substitute $\mathbf{z}_y$ in Equation (3.44) by Equation (3.43) and $\mathbf{z}_x$ in Equation (3.42) by Equation (3.40). Then we get,

$$\mathbf{w}_x = \frac{\mathbf{D}_X \mathbf{D}_Y^T \mathbf{w}_y}{\|\mathbf{D}_X \mathbf{D}_Y^T \mathbf{w}_y\|}, \quad (3.45)$$

$$\mathbf{w}_y = \frac{\mathbf{D}_Y \mathbf{D}_X^T \mathbf{w}_x}{\|\mathbf{D}_Y \mathbf{D}_X^T \mathbf{w}_x\|}, \quad (3.46)$$

where the iteration number is omitted for convenience. Again, the mutual substitution of $\mathbf{w}_x$ and $\mathbf{w}_y$ in the numerators in Equation (3.45) and Equation (3.46) results in

$$\begin{aligned} (\mathbf{D}_X \mathbf{D}_Y^T \mathbf{D}_Y \mathbf{D}_X^T) \mathbf{w}_x &= \|\mathbf{D}_X \mathbf{D}_Y^T \mathbf{w}_y\| \|\mathbf{D}_Y \mathbf{D}_X^T \mathbf{w}_x\| \mathbf{w}_x \\ &= \lambda \mathbf{w}_x, \end{aligned} \quad (3.47)$$

$$\begin{aligned} (\mathbf{D}_Y \mathbf{D}_X^T \mathbf{D}_X \mathbf{D}_Y^T) \mathbf{w}_y &= \|\mathbf{D}_Y \mathbf{D}_X^T \mathbf{w}_x\| \|\mathbf{D}_X \mathbf{D}_Y^T \mathbf{w}_y\| \mathbf{w}_y \\ &= \lambda \mathbf{w}_y, \end{aligned} \quad (3.48)$$

where $\lambda = \|\mathbf{D}_X \mathbf{D}_Y^T \mathbf{w}_y\| \|\mathbf{D}_Y \mathbf{D}_X^T \mathbf{w}_x\|$. When $\mathbf{D}_X$ and $\mathbf{D}_Y$ have been centered, that is, $\langle \mathbf{x} \rangle = \langle \mathbf{y} \rangle = 0$, it is well known that two unit vectors $\mathbf{w}_x$ and $\mathbf{w}_y$ satisfying Equations (3.47) and (3.48), respectively, define two directions of maximal data covariance between $\mathbf{D}_X$ and $\mathbf{D}_Y$ (Borga, 1998). In fact, the resulting operations for the first pair, the iteration from Equation (3.40) to Equation (3.44), correspond to those of the NIPALS algorithm for the partial least squares (PLS) (Wold, 1975; Wold et al., 1984).

Eventually, the learning algorithm for the linear case with the Euclidean distance measure can be said to produce two latent variables which are maximally covariant, where the squared covariance is given by λ .

$$\begin{aligned}
\lambda &= \|\mathbf{D}_\mathbf{X} \mathbf{D}_\mathbf{Y}^T \mathbf{w}_y\| \|\mathbf{D}_\mathbf{Y} \mathbf{D}_\mathbf{X}^T \mathbf{w}_x\| \\
&= ((\mathbf{w}_x)^T \mathbf{D}_\mathbf{X} \mathbf{D}_\mathbf{Y}^T \mathbf{w}_y) ((\mathbf{w}_y)^T \mathbf{D}_\mathbf{Y} \mathbf{D}_\mathbf{X}^T \mathbf{w}_x) \quad (\text{from (3.45) and (3.46)}) \\
&= [\mathbf{z}_x \mathbf{z}_y^T] [\mathbf{z}_y \mathbf{z}_x^T] \quad (\text{from (3.40) and (3.43)}) \\
&= [\mathbf{z}_x \mathbf{z}_y^T]^2. \tag{3.49}
\end{aligned}$$

Again, λ can be decomposed into three meaningful measures:

$$\begin{aligned}
\lambda &= [\mathbf{z}_x \mathbf{z}_y^T]^2 \\
&= \left[(\|\mathbf{z}_x\| \|\mathbf{z}_y\|) \left(\frac{\mathbf{z}_x \mathbf{z}_y^T}{\|\mathbf{z}_x\| \|\mathbf{z}_y\|} \right) \right]^2 \\
&= \|\mathbf{z}_x\|^2 \|\mathbf{z}_y\|^2 \left(\frac{\mathbf{z}_x \mathbf{z}_y^T}{\|\mathbf{z}_x\| \|\mathbf{z}_y\|} \right)^2 \\
&= \text{var}(z_x) \text{var}(z_y) \text{corr}(z_x, z_y)^2, \tag{3.50}
\end{aligned}$$

where $\text{var}(z_x)$ is the variance of the first latent variable z_x for $\mathbf{D}_\mathbf{X}$ and $\text{var}(z_y)$ is the variance of the corresponding z_y for $\mathbf{D}_\mathbf{Y}$. The value of $\text{corr}(z_x, z_y)$ is the correlation coefficient between z_x and z_y .

When one assumes the number of latent variables to be more than one, the k -th latent feature pair $(\mathbf{w}_k^x, \mathbf{w}_k^y)$ ($\|\mathbf{w}_k^x\|^2 = \|\mathbf{w}_k^y\|^2 = 1$) can be estimated in the same way for $(\mathbf{w}_1^x, \mathbf{w}_1^y)$, except that the process is done for the data sets which are not explained by the predecessors, for example, $\mathbf{D}_\mathbf{X} \leftarrow \mathbf{D}_\mathbf{X} - \sum_{i=1}^{k-1} \mathbf{w}_i^x \mathbf{z}_i^x$ and $\mathbf{D}_\mathbf{Y} \leftarrow \mathbf{D}_\mathbf{Y} - \sum_{i=1}^{k-1} \mathbf{w}_i^y \mathbf{z}_i^y$.

Mahalanobis Distance Error Measure

If the measures in the linear models for both data sets are the *Mahalanobis distance* which is based on correlations between variables, the error for a sample $\mathbf{x}$ by a model $\mathcal{M}_{\mathcal{X}}$ is defined as

$$e_{\mathcal{M}_{\mathcal{X}}}(\mathbf{x}, z_{\mathbf{x}}) = \frac{1}{2}(\mathbf{x} - z_{\mathbf{x}}\mathbf{w}_{\mathbf{x}})^T \Sigma_{\mathcal{X}}^{-1}(\mathbf{x} - z_{\mathbf{x}}\mathbf{w}_{\mathbf{x}}),$$

where $\Sigma_{\mathcal{X}}$ is the sample covariance of $\mathcal{X} = \{X_1, X_2, \dots, X_M\}$, that is, $\Sigma_{\mathcal{X}} = \mathbf{D}_{\mathbf{X}}\mathbf{D}_{\mathbf{X}}^T$ ³. By decomposing $\Sigma_{\mathcal{X}}^{-1}$ into $\Sigma_{\mathcal{X}}^{-\frac{1}{2}}\Sigma_{\mathcal{X}}^{-\frac{1}{2}}$, we can rewrite the error as

$$\begin{aligned} e_{\mathcal{M}_{\mathcal{X}}}(\mathbf{x}, z_{\mathbf{x}}) &= \frac{1}{2}(\Sigma_{\mathcal{X}}^{-\frac{1}{2}}\mathbf{x} - \Sigma_{\mathcal{X}}^{-\frac{1}{2}}z_{\mathbf{x}}\mathbf{w}_{\mathbf{x}})^T(\Sigma_{\mathcal{X}}^{-\frac{1}{2}}\mathbf{x} - \Sigma_{\mathcal{X}}^{-\frac{1}{2}}z_{\mathbf{x}}\mathbf{w}_{\mathbf{x}}) \\ &= \frac{1}{2}(\hat{\mathbf{x}} - z_{\mathbf{x}}\hat{\mathbf{w}}_{\mathbf{x}})^T(\hat{\mathbf{x}} - z_{\mathbf{x}}\hat{\mathbf{w}}_{\mathbf{x}}) \\ &= \frac{1}{2}\|\hat{\mathbf{x}} - z_{\mathbf{x}}\hat{\mathbf{w}}_{\mathbf{x}}\|^2, \end{aligned} \quad (3.51)$$

where $\hat{\mathbf{x}} = \Sigma_{\mathcal{X}}^{-\frac{1}{2}}\mathbf{x}$ and $\hat{\mathbf{w}}_{\mathbf{x}} = \Sigma_{\mathcal{X}}^{-\frac{1}{2}}\mathbf{w}_{\mathbf{x}}$. Then, $\phi_{\mathcal{M}_{\mathcal{X}}}(\mathbf{D}_{\mathbf{X}}, \mathbf{z}_{\mathbf{x}})$ can be defined in a similar way as in Equation (3.35) for the Euclidean distance error measure,

$$\phi_{\mathcal{M}_{\mathcal{X}}}(\mathbf{D}_{\mathbf{X}}, \mathbf{z}_{\mathbf{x}}) = -\frac{1}{2}\|\hat{\mathbf{D}}_{\mathbf{X}} - \hat{\mathbf{w}}_{\mathbf{x}}\mathbf{z}_{\mathbf{x}}\|_F^2, \quad (3.52)$$

where $\hat{\mathbf{D}}_{\mathbf{X}} = [\hat{\mathbf{x}}_1; \hat{\mathbf{x}}_2; \dots; \hat{\mathbf{x}}_N]$. In the same way,

$$\phi_{\mathcal{M}_{\mathcal{Y}}}(\mathbf{D}_{\mathbf{Y}}, \mathbf{z}_{\mathbf{y}}) = -\frac{1}{2}\|\hat{\mathbf{D}}_{\mathbf{Y}} - \hat{\mathbf{w}}_{\mathbf{y}}\mathbf{z}_{\mathbf{y}}\|_F^2, \quad (3.53)$$

$$\phi_{\mathcal{M}_{\mathcal{X}}}(\mathbf{D}_{\mathbf{X}}, \mathbf{z}_{\mathbf{y}}) = -\frac{1}{2}\|\hat{\mathbf{D}}_{\mathbf{X}} - \hat{\mathbf{w}}_{\mathbf{x}}\mathbf{z}_{\mathbf{y}}\|_F^2, \quad (3.54)$$

$$\phi_{\mathcal{M}_{\mathcal{Y}}}(\mathbf{D}_{\mathbf{Y}}, \mathbf{z}_{\mathbf{x}}) = -\frac{1}{2}\|\hat{\mathbf{D}}_{\mathbf{Y}} - \hat{\mathbf{w}}_{\mathbf{y}}\mathbf{z}_{\mathbf{x}}\|_F^2, \quad (3.55)$$

where $\hat{\mathbf{D}}_{\mathbf{Y}} = \Sigma_{\mathcal{Y}}^{-\frac{1}{2}}\mathbf{D}_{\mathbf{Y}}$.

Thus, the learning of a co-reference latent model for this case can be done in the same way for the previous the Euclidean distance measure, on the basis of Equations

³Here, we ignore the scalar scaling factor for the definition of the sample covariance.

from (3.52) to (3.55). Then, the co-reference latent feature pair $(\hat{\mathbf{w}}_x, \hat{\mathbf{w}}_y)$ produced by the learning of the model satisfies

$$\begin{aligned} (\hat{\mathbf{D}}_x \hat{\mathbf{D}}_y^T \hat{\mathbf{D}}_y \hat{\mathbf{D}}_x^T) \hat{\mathbf{w}}_x &= \|\hat{\mathbf{D}}_x \hat{\mathbf{D}}_y^T \hat{\mathbf{w}}_y\| \|\hat{\mathbf{D}}_y \hat{\mathbf{D}}_x^T \hat{\mathbf{w}}_x\| \hat{\mathbf{w}}_x \\ &= \lambda \hat{\mathbf{w}}_x \quad (\|\hat{\mathbf{w}}_x\| = 1), \end{aligned} \quad (3.56)$$

$$\begin{aligned} (\hat{\mathbf{D}}_y \hat{\mathbf{D}}_x^T \hat{\mathbf{D}}_x \hat{\mathbf{D}}_y^T) \hat{\mathbf{w}}_y &= \|\hat{\mathbf{D}}_y \hat{\mathbf{D}}_x^T \hat{\mathbf{w}}_x\| \|\hat{\mathbf{D}}_x \hat{\mathbf{D}}_y^T \hat{\mathbf{w}}_y\| \hat{\mathbf{w}}_y \\ &= \lambda \hat{\mathbf{w}}_y \quad (\|\hat{\mathbf{w}}_y\| = 1), \end{aligned} \quad (3.57)$$

where $\lambda = \|\hat{\mathbf{D}}_x \hat{\mathbf{D}}_y^T \hat{\mathbf{w}}_y\| \|\hat{\mathbf{D}}_y \hat{\mathbf{D}}_x^T \hat{\mathbf{w}}_x\|$. If we multiply $\Sigma_X^{-\frac{1}{2}}$ to the both sides of Equation (3.56) and $\Sigma_Y^{-\frac{1}{2}}$ to those of Equation (3.57), we get

$$(\Sigma_X^{-1} (\mathbf{D}_x \mathbf{D}_y^T) \Sigma_Y^{-1} (\mathbf{D}_y \mathbf{D}_x^T)) \bar{\mathbf{w}}_x = \lambda \bar{\mathbf{w}}_x, \quad (3.58)$$

$$(\Sigma_Y^{-1} (\mathbf{D}_y \mathbf{D}_x^T) \Sigma_X^{-1} (\mathbf{D}_x \mathbf{D}_y^T)) \bar{\mathbf{w}}_y = \lambda \bar{\mathbf{w}}_y, \quad (3.59)$$

where $\bar{\mathbf{w}}_x = \Sigma_X^{-\frac{1}{2}} \hat{\mathbf{w}}_x$ and $\bar{\mathbf{w}}_y = \Sigma_Y^{-\frac{1}{2}} \hat{\mathbf{w}}_y$.

When the two data sets $\mathbf{D}_x$ and $\mathbf{D}_y$ are centered, the two vectors $\bar{\mathbf{w}}_x$ and $\bar{\mathbf{w}}_y$ satisfying Equations (3.58) and (3.59) are known to be those on which sample projections $\mathbf{z}_x$ and $\mathbf{z}_y$ are maximally correlated (Borga, 1998), where the squared correlation is given by λ .

$$\begin{aligned} \lambda &= \|\hat{\mathbf{D}}_x \hat{\mathbf{D}}_y^T \hat{\mathbf{w}}_y\| \|\hat{\mathbf{D}}_y \hat{\mathbf{D}}_x^T \hat{\mathbf{w}}_x\| \\ &= ((\hat{\mathbf{w}}_x)^T \hat{\mathbf{D}}_x \hat{\mathbf{D}}_y^T \hat{\mathbf{w}}_y) ((\hat{\mathbf{w}}_y)^T \hat{\mathbf{D}}_y \hat{\mathbf{D}}_x^T \hat{\mathbf{w}}_x) \\ &= [\mathbf{z}_x \mathbf{z}_y^T] [\mathbf{z}_y \mathbf{z}_x^T] \\ &= [\mathbf{z}_x \mathbf{z}_y^T]^2. \end{aligned} \quad (3.60)$$

Hence, it can be said that a linear co-reference latent model with the Mahalanobis distance measure for both data sets is related to the canonical correlation analysis (CCA) (Hotelling, 1936). The unity conditions for $\mathbf{z}_x$ and $\mathbf{z}_y$ in the definition of the

correlation coefficient is guaranteed by

$$\begin{aligned}
\mathbf{z}_x \mathbf{z}_x^T &= \hat{\mathbf{w}}_x^T \hat{\mathbf{D}}_X \hat{\mathbf{D}}_X^T \hat{\mathbf{w}}_x = \hat{\mathbf{w}}_x^T \left(\Sigma_X^{-\frac{1}{2}} \mathbf{D}_X \mathbf{D}_X^T \Sigma_X^{-\frac{1}{2}} \right) \hat{\mathbf{w}}_x \\
&= \hat{\mathbf{w}}_x^T \left(\Sigma_X^{-\frac{1}{2}} \Sigma_X \Sigma_X^{-\frac{1}{2}} \right) \hat{\mathbf{w}}_x = \hat{\mathbf{w}}_x^T \mathbf{I} \hat{\mathbf{w}}_x = 1, \\
\mathbf{z}_y \mathbf{z}_y^T &= \hat{\mathbf{w}}_y^T \hat{\mathbf{D}}_Y \hat{\mathbf{D}}_Y^T \hat{\mathbf{w}}_y = \hat{\mathbf{w}}_y^T \left(\Sigma_Y^{-\frac{1}{2}} \mathbf{D}_Y \mathbf{D}_Y^T \Sigma_Y^{-\frac{1}{2}} \right) \hat{\mathbf{w}}_y \\
&= \hat{\mathbf{w}}_y^T \left(\Sigma_Y^{-\frac{1}{2}} \Sigma_Y \Sigma_Y^{-\frac{1}{2}} \right) \hat{\mathbf{w}}_y = \hat{\mathbf{w}}_y^T \mathbf{I} \hat{\mathbf{w}}_y = 1.
\end{aligned}$$

As the case for the Euclidean distance error measure, the successive pairs of latent variables can be derived by

$$\hat{\mathbf{D}}_X \leftarrow \hat{\mathbf{D}}_X - \sum_{i=1}^{k-1} \hat{\mathbf{w}}_i^x \mathbf{z}_i^x, \quad \hat{\mathbf{D}}_Y \leftarrow \hat{\mathbf{D}}_Y - \sum_{i=1}^{k-1} \hat{\mathbf{w}}_i^y \mathbf{z}_i^y. \quad (3.61)$$

Finally, for original data sets $\mathbf{D}_X$ and $\mathbf{D}_Y$, the transformation vectors into the latent space are given by $\bar{\mathbf{w}}_k^x$ and $\bar{\mathbf{w}}_k^y$, respectively ($\mathbf{z}_k^x = (\mathbf{D}_X^T \bar{\mathbf{w}}_k^x)^T$, $\mathbf{z}_k^y = (\mathbf{D}_Y^T \bar{\mathbf{w}}_k^y)^T$). For the interpretation of the extracted patterns for $\mathbf{D}_X$ and $\mathbf{D}_Y$, one can look over the original vectors $\mathbf{w}_k^x = \Sigma_X^{\frac{1}{2}} \hat{\mathbf{w}}_k^x$ and $\mathbf{w}_k^y = \Sigma_Y^{\frac{1}{2}} \hat{\mathbf{w}}_k^y$.

Figure 3.7 shows a network-based representation of the presented two linear co-reference latent models. According to the target criterion and applications, many other approaches can be derived in the same framework. They are differentiated in term of data transformation for the two data sets $\mathbf{D}_X$ and $\mathbf{D}_Y$. Data may be centered or not and rescaled or not. Further, a different transformation may be employed for each data set. The choice is data- and application-specific.

One Data Set with a Self-Derived Data

In some cases, an analysis of one data set can be done by the learning of a co-reference latent model for itself and an auxiliary data derived from the data, thus two different data, unlike the previous cases for one-table data. For example, Borga

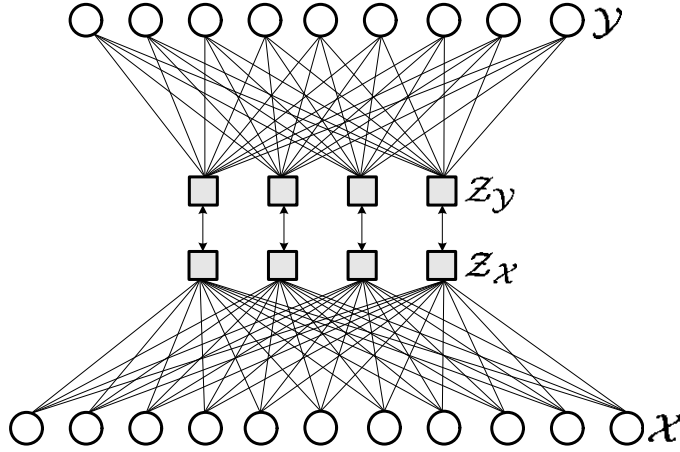


Figure 3.7: A graph-based representation of the linear co-reference latent models.

and Knutsson (2001) presented an efficient method for solving the problem of blind source separation under the framework of the canonical correlation analysis (CCA)⁴, a linear co-reference latent model with the Mahalanobis distance measure. Given a time series data, they first generate its counterpart data set of temporal correlation, from the original data set. Then CCA was applied to the two data sets, by which the components of maximum auto-correlation is pursued.

To summarize, the learning of the co-reference model is achieved by an iterative alternation of two learning steps for $\mathcal{M}_{\mathcal{X}}$ and $\mathcal{M}_{\mathcal{Y}}$, while optimizing an integrative function which is basically composed of three criteria. Two ($\phi_{\mathcal{M}_{\mathcal{X}}}(\mathbf{D}_{\mathbf{X}}, \mathbf{z}_{\mathbf{x}})$ and $\phi_{\mathcal{M}_{\mathcal{Y}}}(\mathbf{D}_{\mathbf{Y}}, \mathbf{z}_{\mathbf{y}})$) of the three are for measuring the degree of description of the respective models for $\mathbf{D}_{\mathbf{X}}$ and $\mathbf{D}_{\mathbf{Y}}$, and the other ($\phi_{\mathcal{M}_{\mathcal{Z}}}(\mathbf{z}_{\mathbf{x}}, \mathbf{z}_{\mathbf{y}})$) is for evaluating the relation among the extracted latent variables.

⁴For a paired data set of $\mathbf{D}_{\mathbf{X}}$ and $\mathbf{D}_{\mathbf{Y}}$, CCA is known to be closely related to the maximization of mutual information for Gaussian distributed latent variables, one for $\mathbf{D}_{\mathbf{X}}$ and the other for $\mathbf{D}_{\mathbf{Y}}$.

Chapter 4

Dependency Analysis of DNA Sequence Motifs and Gene Expression Patterns

4.1 Introduction

Recent developments in the technology for DNA microarrays have made it possible for us to acquire a parallel view of the expression pattern of tens of thousands of genes under various environmental conditions or cellular processes. An important issue in computational biology in the analysis of the massive gene expression data is to understand the mechanism underlying the transcriptional regulation of genes, possibly often by incorporating other biological information sources.

The regulation of transcription initiation is at the core of the control of gene expression and transcription factors (TF) play a major role in the regulation process. Transcription factors affect the transcription of a set of genes by first binding to specific DNA sequence elements in regulatory regions upstream of the genes. The regulatory sequence elements are usually rather short sequence fragments and act as binding sites for the transcription factors. The identification of the sequence regulatory elements can provide an important clue to the understanding of gene

regulation at the transcription level. The problem is often tackled by relating the transcriptional behaviors of the genes with promoter sequence data.

One of the well-established computational approaches to the identification of regulatory sequence elements is based on the notion that the co-expressed genes are also co-regulated and thus may share common regulatory motifs. The identification process proceeds in a serialized fashion. Genes are first grouped into clusters by their expression similarity, then a motif search algorithm is run for identification of over-represented sequence elements in the upstream regions of the member genes of each cluster (Spellman et al., 1998; Roth et al., 1998; Tavazoie et al., 1999).

Segal et al. (2003) proposed another cluster-based approach which is based on probabilistic graphical models, where they integrate both gene expression data and upstream sequence data in a unified way, unlike the previous decoupled approach. The grouping of genes and the search for regulatory sequence motif are intertwined and are iteratively refined during the learning procedure by an EM algorithm.

Another useful approach is to associate gene expression values directly with the abundance of short sequence motifs in the sequences upstream of genes, without clustering of genes as a prior step (Bussemaker et al., 2001; Keles et al., 2002). Most works of this type formulate the problem in the regression framework where the input variables are predefined sequence features and the output is the expression levels at various conditions. Bussemaker et al. (2001) used a linear regression model where single genome-wide set of expression ratios are regressed on regulatory elements. Keles et al. (2002) proposed a similar approach where they incorporated location information as well as the raw motif counts in the calculation of motif scores and included additional terms for capturing interactions of the sequence elements. Conlon et al. (2003) also presented oligomer motif-expression regression analysis, but unlike the previous two approaches, they first identified significant sequence

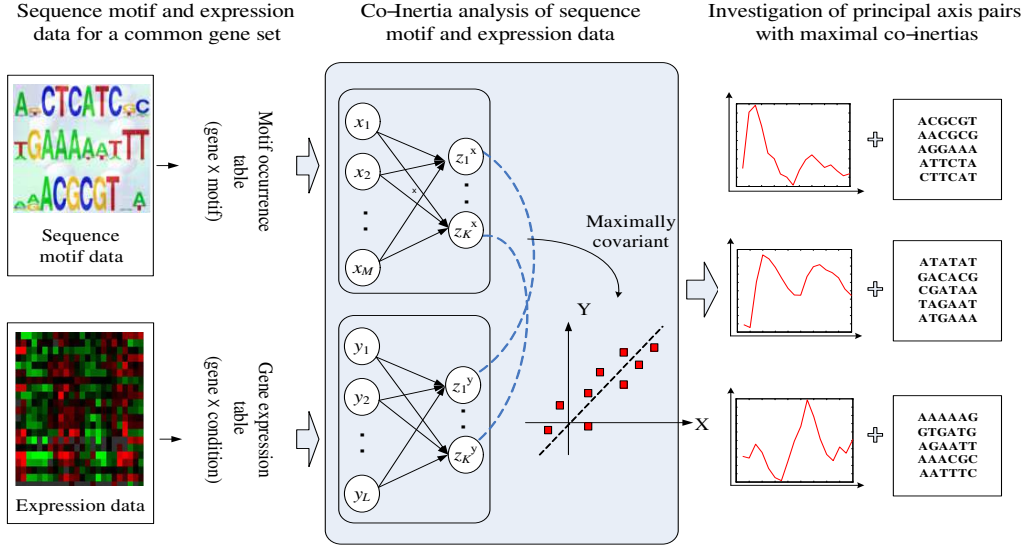


Figure 4.1: The outline of the combined analysis of sequence motif profiles and gene expression profiles using a co-reference latent model. For a set of genes, first, two tables which contain sequence motif profiles and gene expression patterns of the genes are constructed. Then, the two tables are fed into the co-reference latent model and several pairs of latent factors which are maximally covariant are extracted. We finally investigate the sequence motifs which have high contributions to each extracted feature. Also, the high-significant sequence motifs are displayed in a low-dimensional plot with the characteristic expression.

motifs by the application of a sequence pattern finding algorithm and used only the significant motifs in their accompanying linear regression model. Phuong et al. (2004) employed a tree-based regression method. They handled the expression data over multiple experiments simultaneously, rather than separately.

In this work, we present a linear co-reference latent model-based approach to the problem of identification of regulatory sequence elements using sequence motif data and expression data. Figure 4.1 shows the overall analysis procedure. Given a set of genes, we first construct two separate data matrices of the genes. One is the sequence motif data where each gene is represented by a profile of sequence motifs which

locate in the upstream region of the gene. The other matrix is established in such a way that a gene is described by the expression profile over various experimental conditions. On the basis of the learning of the linear co-reference latent model for the two data sets of genes, we extract a series of latent feature pairs such that the gene representations by the latent features from the sequence motif profile data and those by the latent features from the expression profile data are maximally covariant. Significant regulatory elements are identified by investigating their contributions to the definition of each latent feature for the sequence motif data. The contribution of the putative regulatory elements to the specific transcription patterns are illustrated in a understandable way by plotting them in the low-dimensional space defined by the latent spaces for DNA sequence motif data and expression data.

Our method is similar to the regression-based approach in that the sequence motif and expression patterns are directly linked during the dependency analysis of them. However, it is far from a simple regression between a sequence motif and an experimental condition, in that the linking of the two patterns in our co-reference latent model is accomplished in latent spaces derived from the respective data set, by which one can acquire a global view of the relation underlying the two different properties and a low-dimensional interpretation of the characteristic association pattern.

4.2 The Co-Reference Latent Model for Sequence Motif Data and Gene Expression Data

Let $\mathbf{V}$ be an $M \times N$ sequence motif data matrix of N genes and M DNA sequence motifs, where v_{ij} is set to the number of the i -th motif in the upstream regions of the j -th gene. Thus $\mathbf{V}$ can be considered a co-occurrence data of genes and sequence motifs. For this data, we introduce a latent variable model $\mathcal{M}_{\mathcal{X}}$ which

is based on the framework of the correspondence analysis (Greenacre, 1993). The correspondence analysis (CA) is an effective multivariate statistical analysis method of the associations in a single data table of frequencies. In bioinformatics fields, the CA-based technique was previously explored for the low-dimensional graphical analysis of gene expression data (Fellenberg et al., 2001) and the discrimination of microarray samples (Culhane et al., 2002).

Let n_{i+} and n_{+j} be the sum of i -th row elements and the sum of j -th column elements, respectively, and n_{++} be the total sum of all element of the matrix $\mathbf{V}$:

$$\begin{aligned} n_{i+} &= \sum_j v_{ij}, \\ n_{+j} &= \sum_i v_{ij}, \\ n_{++} &= \sum_{i,j} v_{ij} \\ &= \sum_i n_{i+} = \sum_j n_{+j}. \end{aligned}$$

From the definitions, the mass of i -th row is defined as $r_i = n_{i+}/n_{++}$ and the mass of j -th column is $c_j = n_{+j}/n_{++}$. The final sequence motif data $\mathbf{D}_{\mathbf{X}}$ to be analyzed by the co-reference latent model was acquired by

$$\mathbf{D}_{\mathbf{X}} = \left[x_{ij} \mid x_{ij} = \frac{f_{ij} - r_i c_j}{r_i c_j} \right],$$

where $f_{ij} = n_{ij}/n_{++}$. Then, the degree of description of $\mathcal{M}_{\mathcal{X}}$ for $\mathbf{D}_{\mathbf{X}}$ was set to

$$\begin{aligned} \phi_{\mathcal{M}_{\mathcal{X}}}(\mathbf{D}_{\mathbf{X}}, \mathbf{Z}_{\mathbf{x}}) &= -E_{\mathcal{M}_{\mathcal{X}}}^{WTA}(\mathbf{D}_{\mathbf{X}}, \mathbf{Z}_{\mathbf{x}}) \\ &= -\frac{1}{2} \|\mathbf{M}_{\mathbf{x}}^{\frac{1}{2}}(\mathbf{D}_{\mathbf{X}} - \mathbf{W}_{\mathbf{x}}\mathbf{Z}_{\mathbf{x}})\|_F^2, \end{aligned} \quad (4.1)$$

where $\mathbf{W}_{\mathbf{x}} = [\mathbf{w}_1^{\mathbf{x}}, \dots, \mathbf{w}_K^{\mathbf{x}}]$ and $\mathbf{Z}_{\mathbf{x}} = [\mathbf{z}_1^{\mathbf{x}}; \dots; \mathbf{z}_K^{\mathbf{x}}]$ with the number of latent variables being K . $\mathbf{M}_{\mathbf{x}}$ is a diagonal matrix of which the i -th element is $\sqrt{r_i}$ ($1 \leq i \leq M$).¹

¹In the original application of CA to a single data set, the analysis is done on the matrix $\mathbf{M}_{\mathbf{x}}^{\frac{1}{2}}\mathbf{D}_{\mathbf{X}}\mathbf{M}_{\mathbf{s}}^{\frac{1}{2}}$, where $\mathbf{M}_{\mathbf{s}}^{\frac{1}{2}}$ is the diagonal matrix of c_j ($1 \leq j \leq N$) and serves as a weight table of

In terms of the meaning, a column vector $\mathbf{w}_k^x$ of $\mathbf{W}_x$ represents the k -th salient sequence motif profile features for $\mathbf{D}_x$ and the corresponding row vector $\mathbf{z}_k^x$ of $\mathbf{Z}_x$ is the projections of genes onto the latent space defined by $\mathbf{w}_k^x$ in the metric of $\mathbf{M}_x$.

In the case of the gene expression data set, let $\mathbf{T}$ be an $L \times N$ expression data matrix of N genes across L experimental conditions, where t_{ij} is the expression measurement of the j -th gene in the i -th experimental condition. The final data matrix $\mathbf{D}_Y$ to be analyzed was constructed by centering $\mathbf{T}$ such that

$$\mathbf{D}_Y = \left[y_{ij} \mid y_{ij} = t_{ij} - \frac{1}{N} \sum_j t_{ij} \right].$$

Then, we assumed a linear model $\mathcal{M}_Y$ for $\mathbf{D}_Y$ and the metric for reconstruction error of $\mathbf{D}_Y$ was set to the Euclidean distance, that is,

$$\begin{aligned} \phi_{\mathcal{M}_Y}(\mathbf{D}_Y, \mathbf{Z}_Y) &= -E_{\mathcal{M}_Y}^{WTA}(\mathbf{D}_Y, \mathbf{Z}_Y) \\ &= -\frac{1}{2} \|\mathbf{D}_Y - \mathbf{W}_Y \mathbf{Z}_Y\|_F^2, \end{aligned} \quad (4.2)$$

where $\mathbf{W}_Y = [\mathbf{w}_1^y, \dots, \mathbf{w}_K^y]$ and $\mathbf{Z}_Y = [\mathbf{z}_1^y; \dots; \mathbf{z}_K^y]$. In terms of the meaning, a column vector $\mathbf{w}_k^y$ of $\mathbf{W}_Y$ defines the k -th latent feature and the corresponding vector $\mathbf{z}_k^y$ contains the projection values of all genes onto the latent space defined by $\mathbf{w}_k^y$. Finally, the degree of association between two latent factors from $\mathcal{M}_X$ for $\mathbf{D}_X$ and $\mathcal{M}_Y$ for $\mathbf{D}_Y$ is

$$\phi_{\mathcal{M}_Z}(\mathbf{Z}_x, \mathbf{Z}_y) = (\phi_{\mathcal{M}_X}(\mathbf{D}_X, \mathbf{Z}_y) + \phi_{\mathcal{M}_Y}(\mathbf{D}_Y, \mathbf{Z}_x)), \quad (4.3)$$

where $\phi_{\mathcal{M}_X}$ and $\phi_{\mathcal{M}_Y}$ are those of Equation (4.1) and Equation (4.2) respectively, except that $\mathbf{Z}_x$ in Equation (4.1) is replaced with $\mathbf{Z}_y$ and $\mathbf{Z}_y$ in Equation (4.2) is substituted with $\mathbf{Z}_x$.

samples. Since it is rather not adequate to scale each sample differently for two data sets in our co-reference latent model, however, we simply dropped the term $\mathbf{M}_s$ in the definition of $\phi_{\mathcal{M}_X}(\mathbf{D}_x, \mathbf{Z}_x)$. This leads to the same weights for all samples.

Since the co-reference latent model here assumes a linear latent variable model for DNA sequence motif data and gene expression data, the learning can proceed in a way as presented for the linear cases in Section 3.2.2 and Section 3.3. Thus, the final solution by the learning procedure gives successive pairs of $\mathbf{w}_k^x$ and $\mathbf{w}_k^y$ of which each satisfies

$$\mathbf{M}_x^{\frac{1}{2}} \mathbf{D}_k^x (\mathbf{D}_k^y)^T \mathbf{D}_k^y (\mathbf{D}_k^x)^T \mathbf{M}_x^{\frac{1}{2}} = \lambda_k \mathbf{M}_x^{\frac{1}{2}} \mathbf{w}_k^x, \quad (4.4)$$

$$\mathbf{D}_k^y (\mathbf{D}_k^x)^T \mathbf{M}_x \mathbf{D}_k^x (\mathbf{D}_k^y)^T = \lambda_k \mathbf{w}_k^y, \quad (4.5)$$

where $\mathbf{D}_k^x = \mathbf{D}_x - \sum_{i=1}^{k-1} \mathbf{w}_i^x \mathbf{z}_i^x$ and $\mathbf{D}_k^y = \mathbf{D}_y - \sum_{i=1}^{k-1} \mathbf{w}_i^y \mathbf{z}_i^y$. The value λ_k is given as $[(\mathbf{z}_k^x)(\mathbf{z}_k^y)^T]^2$. In the aspect that one is allowed to employ a separate linear model with a different type of measure for the degree of model-based data description for each data set, the linear co-reference latent model well corresponds to the *co-inertia analysis* (Doledec and Chessel, 1994; Dray et al., 2003) which is an inter-table ordination method. Therefore, we hereafter call the latent features $\mathbf{w}_k^x$ and $\mathbf{w}_k^y$ as the *k-th co-reference axes* for a paired data set, similarly to the nomenclature in the co-inertia analysis. Likewise, the value λ_k is called as the *k-th co-reference score* for the corresponding co-reference axis pair.

4.3 Experimental Results

4.3.1 Data Sets

Sequence Motif Data Set

For the construction of sequence motif profile data set, we adopted a DNA sequence motif collection established by Pilpel et al. (2001). The repository contains 356 putative or known regulatory sequence motifs of the yeast *Saccharomyces cerevisiae*, where 37 known motifs are included. Pilpel et al. (2001) derived the motifs by the

application of the AlignACE program (Hughes et al., 2000) to the upstream regions of genes in the MIPS (Mewes et al., 2000) functional categories, where some of the known motifs were obtained from the literature and SCPD database (Zhu and Zhang, 1999). Then, the ScanACE program (Hughes et al., 2000) was used to identify all the genes containing each motif in their upstream regions. The data set with detailed description is available at <http://genetics.med.harvard.edu/~tpilpel/MotComb.html>.

Microarray Data Set

Two temporal microarray data sets of the yeast *Saccharomyces cerevisiae* were analyzed: One is the yeast sporulation data set (Chu et al., 1998) and the other is the data of expression measurement during the cell cycle of the yeast (Spellman et al., 1998). The sporulation data set explores the temporal program of gene expression during meiosis and spore formation of *S. cerevisiae* over seven time points, 0h, 0.5h, 2h, 5h, 7h, 9h, and 11.5h. Over 6,000 genes, we first removed genes that do not show a 2.5-fold change in either direction, induction or repression, for at least one time point. Then we retained only those genes, 1865 genes in total, of which upstream regions contain at least one DNA sequence motif among 356 elements in the above sequence motif collection.

The cell cycle data set by Spellman et al. (1998) contains the time series of relative expression measurements from cell cultures synchronized by three independent methods. Among the data sets, we use the α -factor expression data of 18 time points, starting from release from arrest in M/G1 phase. We first picked out genes of which expressions vary significantly across time points, where the difference between maximum and minimum values of log2 expression ratio was used as the measure of significant variation and threshold value was set to 1.0. Then, we retained only those genes of which missing values are not more than three and their promoter regions

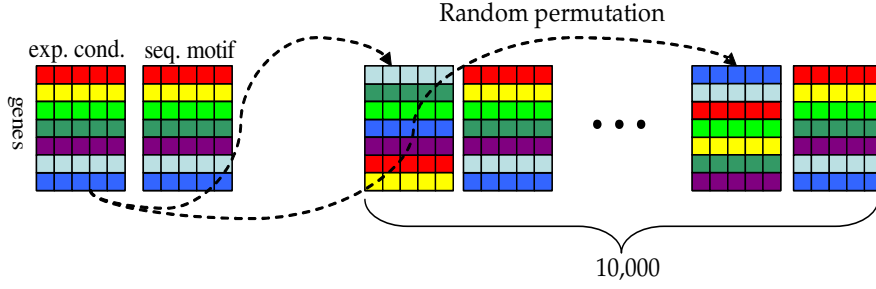


Figure 4.2: The process of random matching of sequence motif profiles and expression profiles of the same set of genes.

are assigned to at least one DNA sequence motif in the collection of Pilpel et al. (2001), which resulted in 1877 genes in total. Missing values of the retained genes were imputed using the k -NN (k-nearest neighbors) method (Troyanskaya et al., 2001) with $k = 10$ and the Euclidean distance measure.

To summarize, the paired data set of the yeast sporulation consists of a (356×1865) DNA sequence motif profile data matrix and a (7×1865) gene expression data matrix. And, the yeast cell-cycle paired data set contains a (356×1877) sequence motif profile data matrix and an (18×1877) gene expression data matrix.

4.3.2 Co-structure between Sequence Motif and Gene Expression

We first investigated whether there exists a meaningful level of co-structure between the sequence motif profiles and the expression profiles of genes. A data permutation-based test was employed for the significance test of the co-structure. The permutation of a paired data set was accomplished by the random matching of sequence motif profiles and expression profiles of genes (Figure 4.2). Then, the co-reference latent model described in Section 4.2 was learned for the permuted data in the same way as for the original paired data set.

Figure 4.3 shows the test results for the sporulation data set and the cell-cycle data set. For each original data set, we generated 10,000 permuted data sets in total. The histogram in Figure 4.3(a) is the frequency distribution of *total co-reference scores* between the sequence motif data and gene expression data for all randomly matched data sets from the sporulation data set. The total co-reference score is defined as

$$\Lambda = \sum_{k=1}^{\min(M,L)} \lambda_k, \quad (4.6)$$

where M is the number of candidate DNA sequence motifs and L is the number of experimental conditions in the microarray data. λ_k is the k -th co-reference score. Against those values, we also depict the observed total co-reference score for the original data pair. No random matching, among 10,000 trials, shows the higher total co-reference score than the observed co-reference score. Thus, we can say that the co-structure between the regulatory sequence motif profiles and gene expression profiles of this data is highly significant, far from occurring by chance. A similar tendency can be observed in Figure 4.3(b) for the yeast cell-cycle data set, which supports the high significance of the co-structure underlying the data set.

4.3.3 Analysis of Co-Reference Latent Factors

With the above observation that there underlies a highly significant co-structure between the DNA sequence motif data and gene expression data, we then analyze the extracted co-reference axes for each data set and identify the putative regulatory sequence motifs along with their corresponding gene expression patterns.

Sporulation data

Prior to the learning of the co-reference latent model, the candidate sequence motifs which do not appear in the upstream regions of more than 15 genes were excluded.

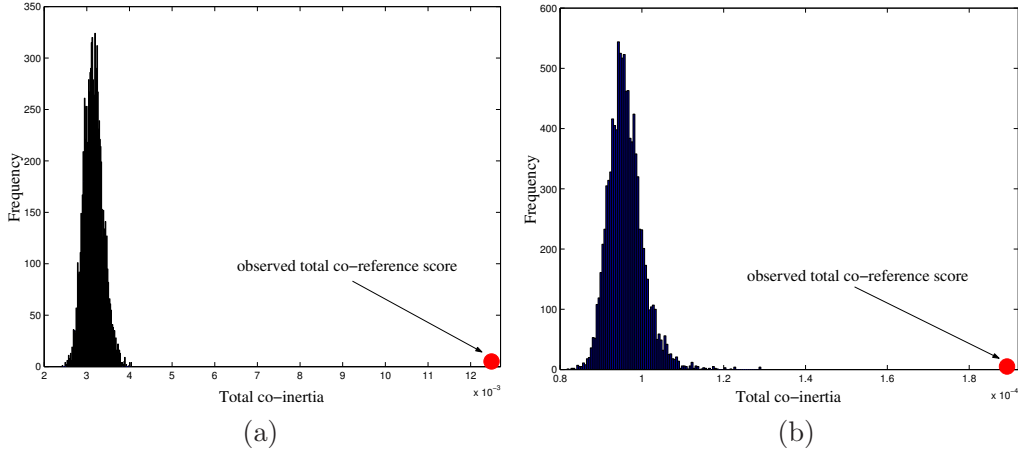


Figure 4.3: Frequency distribution of total co-reference scores by the learning of co-reference latent models for 10,000 randomly matched data of sequence motif data and expression data of (a) the sporulation data set and (b) the cell-cycle data set. Values at the filled circles are those for the original pair of data sets, which shows the high significance of the respective co-structure underlying the two data sets.

We also filtered out those sequence motifs which do not show a meaningful correlation with gene expression data at any time point. For a candidate motif m_i , let its linear regression coefficient for the j -th time point ($1 \leq j \leq 7$) of the paired gene expression data set be a_{ij} . If none of the z -scores of the coefficients a_{ij} among the entire time points are significantly nonzero at the level of p , the motif m_i is considered to be not significantly correlated with gene expression and thus is excluded from the set of candidate sequence motifs. We set the threshold level to $p = 0.05$. After the process, the number of candidate motifs resulted in 176.

Figure 4.4 shows the result of the learning of the linear co-reference latent model with graphical display. In the bar plot which shows the co-reference scores (λ_k , $1 \leq k \leq 7$) for each pair of co-reference axes, we can see that the first two factors are most dominant, where they explain more than 95% of the total co-reference scores between the sequence motif and the expression data. The diagonal boxes plot the

expression factors extracted by the analysis. The two expression patterns are very similar to those presented by Raychaudhuri et al. (2000) where PCA was applied to the analysis of only the gene expression data in yeast sporulation. The first expression factor captures the positive expression levels on average across all time points and the second factor shows the increasing tendency of expression from the second time point. The upper right box plots each time point on two-dimensional space of the two expression factors.

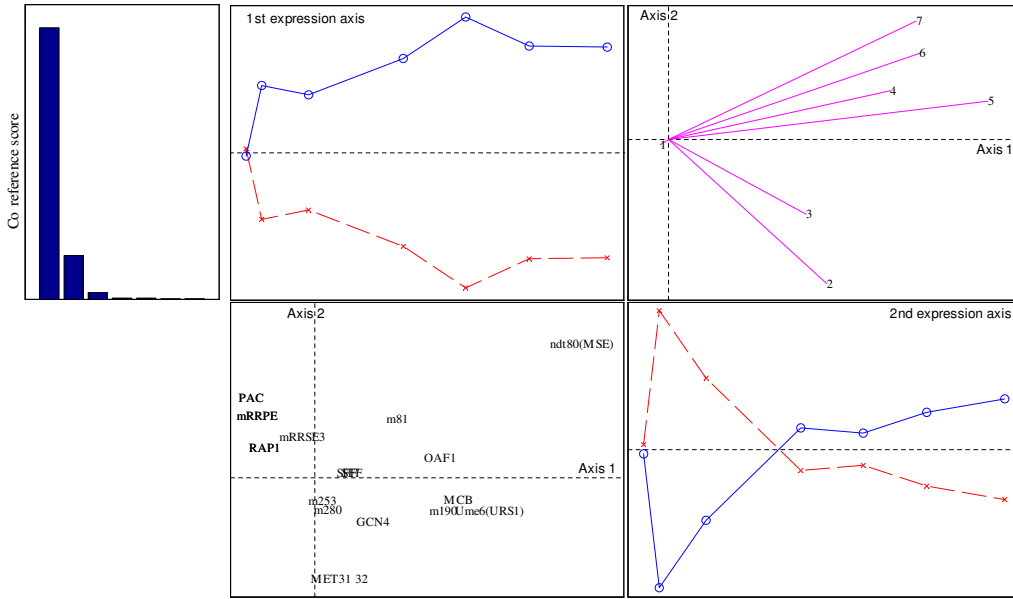


Figure 4.4: A low-dimensional display of the result for the yeast sporulation data set. The left-most plot of the top row shows the co-reference scores of the corresponding pair of co-reference axes. Diagonal boxes show the expression axis extracted by the learning of co-reference latent model, where the solid lines are the original axes and the dashed lines are the axes of opposite direction. The right-top box is the experiment condition plot which shows the coordinates of all experimental time points and the numbers represent the order of time points. The left-bottom is the motif plot depicting the coordinates of sequence motifs which is significant in defining co-reference axis for sequence motif data.

Table 4.1: Motifs with high absolute contribution for the first two co-reference factors in the analysis of the sporulation data set. Top 10 motifs are shown for each factor. The description of putative motifs of which names start with ‘m’ are given in Table 4.2.

1st		2nd	
ndt80(MSE)	0.2688 (+)	ndt80(MSE)	0.2205 (+)
m190	0.0543 (+)	PAC	0.1843 (+)
Ume6(URS1)	0.0484 (+)	mRRPE	0.1814 (+)
SFF	0.0461 (+)	mRRSE3	0.0501 (+)
SFF'	0.0403 (+)	m81	0.0110 (+)
MCB	0.0380 (+)	RAP1	0.0105 (+)
OAF1	0.0203 (+)	m280	0.0087 (-)
RAP1	0.0195 (-)	GCN4	0.0098 (-)
PAC	0.0666 (-)	m253	0.0144 (-)
mRRPE	0.1177 (-)	MET31-32	0.0229 (-)

Sequence motifs are also plotted on the two-dimensional space (the lower left box in Figure 4.4) of the first two motif factors corresponding to the two expression factors. A sequence motif m_i on the k -th co-reference axis $\mathbf{w}_k^x$ is located at the coordinate c_{ki}^x which is given by

$$c_{ki}^x = \sqrt{\lambda_k} w_{ki}^x, \quad (4.7)$$

where λ_k is the k -th co-reference score which is associated with $\mathbf{w}_k^x$ and $\mathbf{w}_k^y$. Similarly, the coordinate of the j -th experimental condition in gene expression data on the corresponding expression axis is

$$c_{kj}^y = \sqrt{\lambda_k} w_{kj}^y. \quad (4.8)$$

Only ten most significant motifs for each co-reference motif factor are shown and Table 4.1 presents such ten sequence motifs for each factor. The significance of a motif m_i in the k -th co-reference motif factor was determined by its *absolute*

contribution values (Hardle and Simar, 2003) to the factor. The value is defined as

$$C_a^x(i, k) = \frac{\mathbf{M}_x(i, i) \times (c_{ki}^x)^2}{\lambda_k}, \quad (4.9)$$

where $\mathbf{M}_x(i, i)$ is the i -th diagonal element of the matrix $\mathbf{M}_x$. As can be seen in Table 4.1, ndt80(MSE) was identified as the most significant motif for the first and the second co-reference factors. In the motif plot in Figure 4.4, the middle sporulation element (MSE) is located in the right-top quadrant which is the region of the positive combination of both the two factors. While the first expression axis captures the inductive expression over time points, the second axis bears the highly negative values for the 2nd and 3rd time points (30 min and 2 hours, respectively), which is also shown in the experimental condition plot in 2-dimensional space. Therefore, the induction trend between 0 and 2 hours in the first expression axis will be canceled out by the combination with the second expression factor, and thus those motifs such as MSE with high positive values for both factors are expected to correspond to the expression behaviors which highly induces between the 3rd and the 4th time points (2 and 5 hours) and thereafter. In fact, ndt80 is known to be the primary transcriptional activator of the middle sporulation genes at the end of meiotic prophase, and binds to a DNA sequence element, that is, the motif MSE (Chu et al., 1998; Pierce et al., 2003). In the original microarray data (Chu et al., 1998), the time of the exit from meiotic prophase corresponds to between 2 and 5 hours after the transfer to sporulation medium. The result of our analysis is well reflecting this relation between the motif and the expression behavior.

Two motifs, Ume6(URS1) and m190, are located in the right-bottom quadrant of the motif plot in Figure 4.4. They have positive coefficients for the first co-reference factor, but negative values for the second factor, unlike MSE. Ume6 is an important transcription factor in regulating metabolic and meiotic gene expression in yeast. The upstream regulatory sequence URS1 is the Ume6-binding site and is

Table 4.2: The description of putative motifs in Table 4.1 and Table 4.3, which are from (Pilpel et al., 2001).

Motif	Full names of the putative motifs
m76	m_organization_of_cell_wall_orfnum2SD_n6
m81	m_organization_of_centrosome_orfnum2SD_n5
m109	m_nutritional_response_pathway_orfnum2SD_n7
m123	m_other_cation_transporters_orfnum2SD_n14
m178	m_homeostasis_of_other_ions_orfnum2SD_n30
m180	m_organization_of_cell_wall_orfnum2SD_n14
m190	m_meiosis_orfnum2SD_n3
m207	m_pheromone_response_generation_orfnum2SD_n10
m208	m_other_signal-transduction_activities_orfnum2SD_n8
m235	m_sugar_and_carbohydrate_transporters_orfnum2SD_n14
m253	m_glyoxylate_cycle_orfnum2SD_n8
m258	m_other_energy_generation_activities_orfnum2SD_n17
m280	m_nitrogen_and_sulphur_metabolism_orfnum2SD_n16
m281	m_other_energy_generation_activities_orfnum2SD_n4
m294	m_other_cell_growth_cell_division_and_dna_synthesis_activities_orfnum2SD_n10.scn
m296	m_biogenesis_of_chromosome_structure_orfnum2SD_n9
mOCSE15	m_organization_of_chromosome_structure

found in the upstream region of many of the early genes in sporulation of yeast (Chu et al., 1998; Williams et al., 2002). Two time points, 30 min and 2 hours, correspond to meiotic prophase in sporulation and are located in the lower-right quadrant of the experimental condition plot as for URS1 in the motif plot. In the second expression axis, we can see that the opposite direction-profile of the second expression axis shows the high induction of transcription at the two time points. Then, the combination of the two expression profiles (the first axis and the second axis in opposite direction) is expected to produce a transcription pattern which is characterized by the early induction and the sustained expression thereafter. Thus, we can say that our result well characterize the expression behavior of early genes which have URS1 motif. The sequence element m190 is the motif derived from the

MIPS *meiosis* functional category. The DNA sequence of m190 is very similar to that of URS1 and the two motifs were identified in many common promoters in the work of Pilpel et al. (2001). Two sets of genes that contain URS1 and m190 motifs, respectively, have shown similar mean expression profiles (data not shown). In fact, they are closely located in the sequence motif plot.

Yeast cell cycle data

As for the sporulation data set, we first picked out the sequence motifs of which gene frequencies are more than 15 and regression coefficients with the transcription levels are significant at the level of 0.05 in at least one time point. The resulting number of candidate motifs was 246.

Figure 4.5 shows the result of the learning of co-reference latent model for this data set. The top-leftmost bar plot shows the co-reference score for each pair of axes. We selected the first four co-reference factors which together explain 84% of the total co-reference score for the data set. The diagonal plots show the four expression factors respectively. Each of the upper right boxes from the diagonals locates the 18 experimental conditions on the two-dimensional space which is defined by the corresponding two expression axes among the four. The bottom-left boxes are those which display the sequence motifs on the space defined by two different motif axes. Only the first ten motifs of the highest contributions (Equation (4.9)) to the construction of each factor are shown, and they are also listed, with their absolute contribution values, in Table 4.3.

Among the most significant are PAC, mRRPE, MCB, ECB, MCM1, STE12. Sequence elements PAC and mRRPE are the two highest contributors to the first co-reference factor for DNA sequence motifs. They have been also identified in many previous studies for various data set (Tavazoie et al., 1999; Bussemaker et al., 2001;

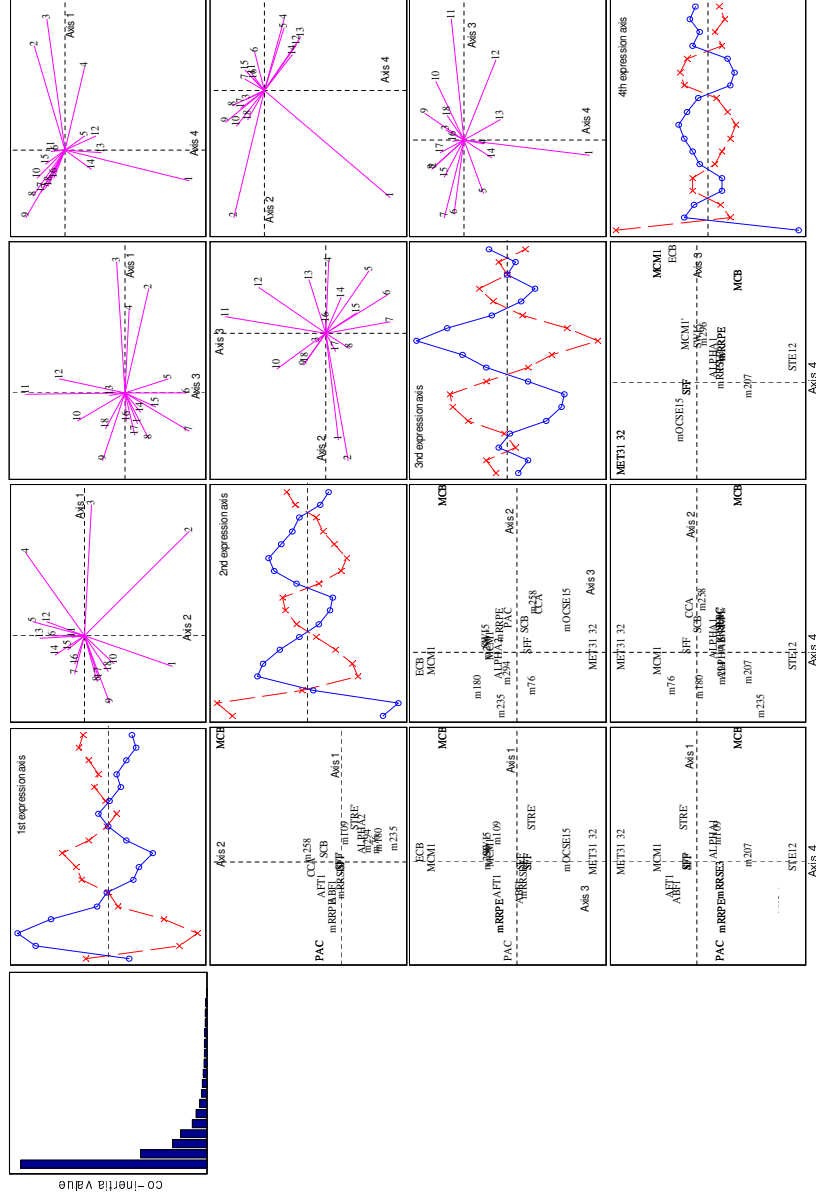


Figure 4.5: A low-dimensional display of the result for the yeast cell-cycle data set. The top-leftmost bar plot shows the co-reference scores. The diagonals show expression patterns (—o—) for the first four co-reference factors, with their reverses (—x—). The box in the i th row and j th column ($i > j$) plots the high-contributing sequence motifs of the i -th and j -th motif axes on the 2-dimensional space defined by the two axes. The corresponding box in the j -th row and i -th column shows all experimental conditions in the plot defined by the j -th and i -th expression axes.

Table 4.3: Motifs with high absolute contribution for the first four co-reference factors in the analysis of the cell-cycle data set. Top 10 motifs are shown for each factor. The numbers in every second column are the absolute contribution values and the signs in parentheses are those of the principal coordinates of the respective motifs. The description of putative motifs of which names start with ‘m’ are given in Table 4.2.

1st		2nd		3rd		4th	
MCB	0.1587 (+)	MCB	0.3519 (+)	ECB	0.1533 (+)	MCM1	0.0376 (+)
STRE'	0.0066 (+)	SCB	0.0289 (+)	MCM1	0.1251 (+)	SFF	0.0214 (+)
m109	0.0058 (+)	CCA	0.0246 (+)	MCB	0.0605 (+)	MET31-32	0.0182 (+)
SFF	0.0073 (-)	PAC	0.0235 (+)	MCM1'	0.0602 (+)	m207	0.0192 (-)
AFT1	0.0085 (-)	m258	0.0212 (+)	SWI5	0.0433 (+)	mRRSE3	0.0211 (-)
SFF'	0.0101 (-)	m180	0.0161 (-)	m296	0.0173 (+)	PAC	0.0227 (-)
ABF1	0.0264 (-)	m235	0.0167 (-)	mRRPE	0.0127 (+)	MCB	0.0290 (-)
mRRSE3	0.0374 (-)	ALPHA2	0.0180 (-)	MET31-32	0.0130 (-)	ALPHA1	0.0341 (-)
mRRPE	0.2311 (-)	m294	0.0221 (-)	SFF	0.0171 (-)	mRRPE	0.0377 (-)
PAC	0.2793 (-)	m76	0.0221 (-)	mOCSE15	0.0204 (-)	STE12	0.1275 (-)

Conlon et al., 2003; Phuong et al., 2004). Their principal coordinates on the axis are negative-valued and thus can be said to be associated with the opposite direction of the corresponding expression axis which is largely characterized by its trend of conspicuous down-regulation in the early stage of the expression experiment. The MCB element, one of the well-known sequence motifs, contributes most highly to the definition of the 2nd co-reference factor for DNA sequence motifs. The corresponding 2nd co-reference factor for gene expression data is characterized its high-induction at the time points of 4th-5th and 11th-12th. This is well consistent with the observation that MCB serves as the binding site for the transcription factor MBF that mediates the induced expression at the G1/S transition during the cell cycle (Spellman et al., 1998). The element shows relatively high contribution values also for the 1st motif factor.

In the lattice showing the sequence motifs and time points in 2-dimensional space in Figure 4.5, the box of the (row, column) index being (2, 1) shows the highly contributing motifs to the 1st and 2nd motif axes and that of the index (1, 2) is the corresponding plot of experimental time points. Two most significant motifs PAC and mRRPE with negative signs of the 1st axis are located on the top-left quadrant of the motif plot. And the right opposite side in the corresponding plot of experimental conditions, the bottom-right quadrant, locates the 2nd and 3rd time points, which characterizes the repression of gene transcription in those stages. The motif element mRRSE3 is also in the same quadrant with the two sequence elements. In fact, it has been previously observed that they significantly co-occur in the upstream regions of the genes involved in rRNA-related activities and have an synergistic effect in the regulation of those genes (Pilpel et al., 2001; Sudarsanam et al., 2002). The cell cycle motif MCB is located far from the origin in the top-right quadrant of the plot and the time points in the same direction further illustrates the important role of MCB in the positive regulation of genes which contain the motif in their promoters, at the G1/S phase of the cell cycle. The SCB element, the second contributor for the 2nd axis, is also known to be related with the transcription induction at the G1/S phase. Accordingly, it is located in the same quadrant with MCB, but its significance is shown to be rather moderate, compared to MCB. The STRE' is located in the right-bottom quadrant which is associated with the transcription induction at early time points, 2nd and 3rd.

ECB and MCM1 are two most significant elements for the 3rd motif factor. ECB shows high sequence similarity with MCM1 and was observed to contain Mcm1p binding site (Spellman et al., 1998; Sudarsanam et al., 2002). They are known to promote the transcription of genes at M/G1 boundary in the cell cycle. Among the moderate contributors, MCM1' is a variant of MCM1 and SWI5 is often found in

the upstream of genes of which expression peak at the M/G1 boundary (Spellman et al., 1998). These are supported by the transcription pattern of the 3rd expression factor. They are on the nearly same direction in the motif plots with which the 3rd axes are involved, that is, those of (3, 1), (3, 2), and (4, 3) indices in Figure 4.5. Also the direction well coincides with the late M/G1 boundary of the cell cycle in their corresponding plots for experimental conditions.

STE12 is the most significant element in the last factor and contributes to the reverse of the corresponding 4th expression pattern since its projection on the axis is negative-valued. The high induction in the earliest time point may be explained by the fact that genes in the data set were synchronized through the release from mating pheromone and Ste12p is an important factor in the regulation of the pheromone-induced transcription (Conlon et al., 2003). MCM1 and SFF take the positive values in the construction of the motif factor with moderate contributions, which means their association to the original 4th expression patterns. The cooperation between the two motifs for the genes which are induced in the G2/M phase, such as CLB2 and SWI5, was previously suggested (Spellman et al., 1998; Pilpel et al., 2001).

Other plausible motifs include STRE' and MET31-32. STRE' is a variant of the STRE motif and the induction effect at the early points, shown in the motif and condition plots of the indices of (2, 1) and (1, 2) in Figure 4.5, may be explained as a stress response to the centrifugation and handling required for the release from cell-cycle arrest (Conlon et al., 2003). MET31-32 was previously identified in the upstream regions of genes which show the peak expression at S/G2 boundary (Spellman et al., 1998). It is among the highest contributor for the 3rd and 4th co-reference factors. In the motif plot and the condition plot defined by the two factors, those indexed by (4, 3) and (3, 4) respectively, the location of MET31-32 is well reflecting the observation.

Computational complexity: As can be seen Equations (4.1) and (4.2), the models for DNA sequence motif data and gene expression data in the co-reference latent model are both linear. Basically, one iteration in the learning of a pair of co-reference axes requires the successive execution of the operations in Equations from (3.44) to (3.43) in Section 3.3. Since each operation is the process of multiplication of a data matrix and a vector, the time complexity of the learning of a pair of co-reference axes is $\mathcal{O}(RN(M+L))$. Here, R is the number of iteration for the extraction of the single latent factors. The deflation procedure for the construction of the next axis pair requires also $\mathcal{O}(RN(M+L))$. So, the total time complexity for K pairs of co-reference axes is $\mathcal{O}(RKN(M+L))$. The additional space complexity is $\mathcal{O}(K(N+M+L))$, which is required for $\mathbf{z}_k^x$, $\mathbf{z}_k^y$, $\mathbf{w}_k^x$, and $\mathbf{w}_k^y$ ($1 \leq k \leq K$).

Comparison with regression-based analysis method: An intuitive method for the dependency analysis of gene expression patterns and DNA sequence elements in the presence of an paired data set of the two different profiles of the same set

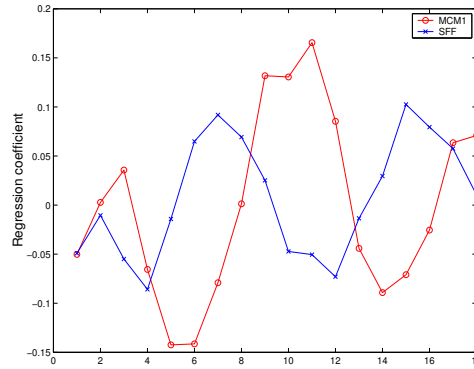


Figure 4.6: The regression coefficients of MCM1 and SFF across 18 time points of the yeast cell cycle data set.

of genes would be to perform the regression analysis or to estimate the correlation coefficients between all pairs of entities (Bussemaker et al., 2001; Keles et al., 2002). Though it is simple and have been shown to be successful in some applications, the use of expression data of only one experimental condition can be vulnerable to the adverse effect of experimental noise and missing value (Bussemaker et al., 2001; Phuong et al., 2004). The approach may also have a difficulty in dependency analysis when a DNA sequence regulatory element are concerned with gene expression patterns of more than one. Figure 4.6 shows an example for the yeast cell cycle data set, where the independent linear regression coefficients across all experimental conditions (18 time points) are plotted for two sequence motifs MCM1 and SFF. In a global view, the coefficient plots present the mutually antagonistic effects of MCM1 and SFF on expression behavior of genes (shown in the third latent factor in Table 4.3) (Bussemaker et al., 2001). As previously stated, however, the two sequence motifs are also frequently found together upstream of genes induced in G2/M phase, which supports their synergistic effect in transcriptional regulation of genes (the fourth latent factor in our co-reference latent model-based analysis). It shows that a simple regression-based approach may not be adequate to identify this local type of dependency between sequence motifs and gene expression profiles. Finally, the linear regression approach may require an additional post-processing step, such as clustering, for the investigation of motif groups which have similar effects on gene expression patterns across multiple experimental conditions (Conlon et al., 2003), while the co-reference latent model-based approach can naturally accomplish the task, directly from the result of learning.

Comparison with the juxtaposition-based method: Given a paired data set, one popular integrated analysis method based on dimensionality reduction is

to first merge the two data sets and then extract prominent latent factors from the resulting single juxtaposed data matrix (Figure 4.7). This approach has been previously employed for the automatic construction of multilingual semantic space in the context of cross-language retrieval, where an SVD-based dimensionality reduction technique is utilized (Rehder et al., 1997; Littman et al., 1998). We also tried this method and compared the result with that of the co-reference latent model.

Figure 4.8 shows the four extracted expression patterns by the application of SVD to the juxtaposed yeast cell cycle data set. Except for the first, the latent factors from the second to the fourth have in common the highest expression level in the second time point. Furthermore, the third and the fourth patterns are rather redundant over all experimental conditions, which is not the case in the result by the co-reference latent model. Along with the expression patterns, the sequence motifs identified as significant (shown in Table 4.4) are somewhat redundant across the extracted latent factors. They are also missing several important DNA sequence regulatory elements presented in Table 4.3, MCB, STRE', STE12, MET31-32, for

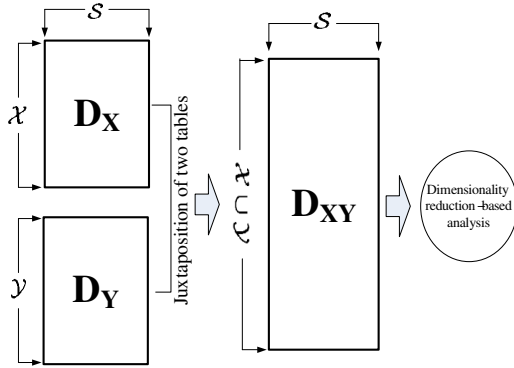


Figure 4.7: Extraction of latent factors from the juxtaposed data matrix of a paired data set.

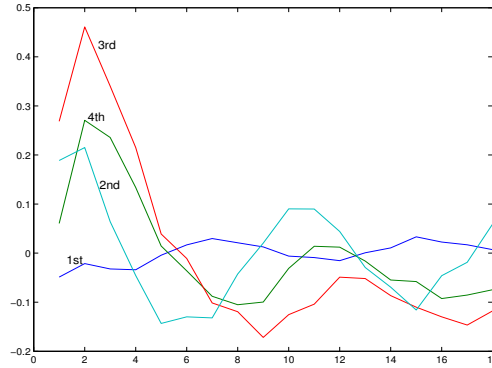


Figure 4.8: The extracted expression patterns by the analysis of the juxtaposed motif profile data and gene expression data.

Table 4.4: DNA sequence motifs with high absolute contribution in the analysis of the cell-cycle data set by the analysis of juxtaposed motif profile data and gene expression data.

1st		2nd		3rd		4th	
SFF'	(+)	MCM1'	(+)	SFF	(+)	MCM1'	(+)
SFF	(+)	ALPHA1	(+)	SFF'	(+)	ECB	(+)
MCM1'	(+)	m252	(+)	mRRSE3	(-)	MCM1	(+)
m310	(+)	CSRE	(+)	m262	(-)	m253	(-)
ALPHA1	(+)	m262	(+)	SWI5	(-)	m112	(-)
SWI5	(+)	mMERE4	(+)	ECB	(-)	mMERE4	(-)
SCB	(+)	SWI5	(+)	MCM1	(-)	CSRE	(-)
ATRepeat	(+)	ATRepeat	(+)	PAC	(-)	m252	(-)
MCM1	(+)	PAC	(-)	mRRPE	(-)	m262	(-)
m289	(+)	mRRPE	(-)	MCM1'	(-)	ATRepeat	(-)

example.

The rather ineffectiveness of the juxtaposition-based dimensionality reduction in this case may be due to the difference in data-inherent scale, such as the number of variables and the variation of the variables, of the respective data sets constituting the paired data set. The disparity can give rise to the trouble that the co-structure between paired data sets fades away in the analysis of the merged data, dominated by the self-structure underlying the respective data sets.

4.4 Summary

In this work, we presented an effective approach to the identification of sequence regulatory elements by the learning of co-reference latent model for sequence motif data and gene expression measurement data. The analysis proceeds by successively extracting pairs of motif and expression axes which are assumed to be highly contributing to the association of the motif profile and expression profile data.

In the application to two data sets of the yeast *S. cerevisiae*, the sporulation data and the cell-cycle data, we first showed that there exist highly significant co-structure between the sequence motif profile and expression pattern for both data set. Then, we could identify several well-known sequence regulatory motifs and also a few putative regulatory elements. The significant sequence motifs were investigated by listing those which highly contributes in the definition of each motif co-reference latent factors. Among the sequence elements, several motifs, including MCB, PAC, mRRPE, and MET31-32, were common for both data set.

As the SVD— or PCA—based analysis of gene expression data provides a summarized view on the transcription behaviors of genes underlying the data (Alter et al., 2000; Raychaudhuri et al., 2000), our associated analysis of motif profiles and expression profiles by co-reference latent models could provide a global summarization of the co-structure underlying the two different but related data. In particular, the coupled display of sequence motifs and experimental conditions in low-dimensional plots was shown to be effective in the intuitive grasping of the characteristic transcriptional role of a regulatory sequence element at specific conditions.

The transcription in eukaryotes is often effected through the combinatorial regulation by multiple transcription factors. Though our analysis does not explicitly incorporate this fact into the model, two motifs which are located on the near position or similar direction in low-dimensional motif plots may be a candidate for further investigation of their combinatorial regulation. Such example pairs include PAC-mRRPE, mRRPE-mRRSE3, MCB-SCB, and MCM1-SFF in the yeast cell-cycle data set.

Chapter 5

Gene-Drug Dependency Mining: Paired Analysis of Gene Expression and Drug Activity Data

5.1 Introduction

Enormous amount of biological data obtained from the high-throughput experimental techniques offers a new opportunity to molecular biology and medicine. For example, combined analysis of the microarray and massive drug-activity data sets can give an insight into various correlations among gene expressions and drug activities in the malignant cell (Scherf et al., 2000; Butte et al., 2000).

Here, we harness Bayesian networks to such analysis. Bayesian networks have mainly been used for the inference of gene networks from microarray data (Friedman et al., 2000; Pe’er et al., 2001; Hartemink et al., 2002; Imoto et al., 2003a,b). It is also a promising model for the integrative dependency analysis from heterogeneous data sets, which is the concern of this dissertation. When adopting Bayesian networks for the analysis of huge biological data, however, the curse of dimensionality

problem usually arises, leading to the inapplicability of a general structure learning algorithm and the intractability of inference in the dependency model. To address this problem, we utilize the basic framework of the co-reference latent model for dependency analysis.

At the core of the dependency analysis by co-reference latent model is the dependency modeling in a low-dimensional latent space, rather than in a high-dimensional input space. A latent feature in the reduced space is usually defined by a set of variables which describe a coherent aspect of the data set. In the application in Chapter 4, for example, DNA sequence motifs with high contribution values of the same sign for a co-reference axis may co-occur frequently or cause similar expression patterns of the genes of which upstream regions contain them. Thus, it can be anticipated that the dependency analysis with the latent features enables a global understanding of the dependency structure between two data sets, while a specific analysis is still possible by investigating the connected latent features.

Under the observation, we take the approach of Bayesian network learning with the reduced number of features. Dimensionality reduction is done by the clustering of genes and drugs. Each resulting cluster maintains a gene prototype or drug prototype, and thus serves as a latent variable or feature for data samples. Then, the prototypes are used as a component (node) of the Bayesian network. As a result, it is expected that an established learning and probabilistic inference algorithm for Bayesian networks will be applicable with a moderate cost.

A similar approach was adopted for genetic network analysis using the graphical Gaussian modeling (GGM) technique in the work of Toh and Horimoto (Toh and Horimoto, 2002). Their goal of data dimensionality reduction was to eliminate linear dependencies in the correlation coefficient matrix. Recently, Segal *et al.* (Segal et al., 2003) suggested a method for revealing gene regulation from gene expression profiles.

They also adopted a clustering technique for identifying a set of co-regulated genes (modules). In contrast, the main purpose of data dimensionality reduction of our study is to make it feasible to analyze the massive biological data sets using Bayesian network learning. Moreover, we apply probabilistic inference for quantifying various relationships among genes, drugs, and cancer subtypes. Previous works did not exploit this feature of Bayesian networks for data analysis.

Our method is validated on the NCI60 data set (<http://discover.nci.nih.gov>) which provides the combined data set of gene expression profiles and drug activity patterns on 60 human cancer cell lines. In the context of the paired data sets, there also exists an auxiliary variable, that is, the type of cancer. By the analysis of the paired data set, we aim at the identification of various probabilistic dependencies among genes, anti-cancer drugs, and the type of cancer. In this aspect, the Bayesian network-based dependency model can be advantageous in that we can also get an insight into the association of genes and drugs in relation with a cancer type, in addition to that over all data samples.

5.2 The NCI60 Data Set

The NCI60 data set (Scherf et al., 2000) consists of 60 human cancer cell lines from 9 kinds of cancers: colorectal, renal, ovarian, breast, prostate, lung, and central nervous system origin cancers, as well as leukemias and melanomas. On each cell line, the gene expression pattern was measured by a cDNA microarray of 9,703 genes including ESTs (expressed sequence tags). Separately from this, 1,400 chemical compounds were tested on the same 60 cell lines. The drug (chemical compound) activity on each cell line was measured by the growth inhibition activities (GI_{50}) assessed from changes in total cellular protein after 48 hours of drug treatment using sulphorhodamine B assay.

In advance of the analysis, we eliminated some genes and drugs for more reliable results. From 9,703 genes in total, we picked out 1,376 genes which have 4 or fewer missing values and showing strong variation among 60 cell lines (more than 3 measurements have red-green intensity ratios > 2.6 or < 0.38), as in the work of Scherf et al. (2000). Furthermore, ESTs with no name were excluded from the analysis. From 1,400 chemical compounds in the drug activity data, 118 anti-cancer drugs with known mechanisms were chosen for the analysis. Among them, drugs with more than 3 missing values were also thrown out. Consequently, the data set in our analysis consists of 60 samples with 890 features: 805 gene expression levels in $\log_2$ ratios, 84 drug activities in $\log_{10}(1/\text{GI}_{50})$ values, and one additional variable for the cancer type.

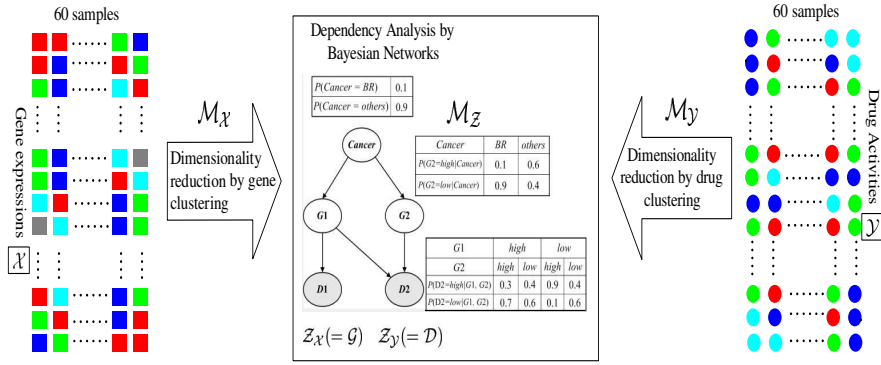


Figure 5.1: The analysis procedure for the NCI60 data set. The original data set is condensed through the dimensionality reduction procedure by clustering ($\mathcal{M}_X$ and $\mathcal{M}_Y$). Then, the Bayesian network ($\mathcal{M}_Z$) is learned for the dependency analysis from the reduced data set.

5.3 The Co-Reference Latent Model for Gene-Drug Dependency Analysis

Figure 5.1 outlines the entire analysis scheme for the NCI60 data set. First, genes and drugs are independently clustered on the basis of their patterns across 60 cell lines of the NCI60 data set. Then, we construct a prototype data set by concatenating the two kinds of prototype entities (genes and drugs), where each cell line is represented by the combined entities. Finally, Bayesian networks are learned for the dependency analysis from the prototype data set. In the framework of the co-reference latent model, the two clustering serves as the models for gene expression data ($\mathcal{M}_{\mathcal{X}}$) and drug activity data ($\mathcal{M}_{\mathcal{Y}}$) respectively. A Bayesian network serves as an independent dependency model ($\mathcal{M}_{\mathcal{Z}}$). One different thing from the case in previous chapter is that the model $\mathcal{M}_{\mathcal{Z}}$ for dependency analysis is explicitly specified, with its own structure and parameter sets. Another is that the learning proceeds in a one-by-one fashion, in that $\mathcal{M}_{\mathcal{X}}$ and $\mathcal{M}_{\mathcal{Y}}$ are first completely learned and then $\mathcal{M}_{\mathcal{Z}}$ is estimated on the resulting latent space. We begin with a brief description of Bayesian networks.

5.3.1 Bayesian Networks as a Dependency Model

A Bayesian network compactly represents the joint probability distribution over a set of M random variables, $\mathbf{X} = \{X_1, X_2, \dots, X_M\}$:

$$P(\mathbf{X}) = \prod_{i=1}^M P(X_i | \mathbf{Pa}_{\mathcal{N}}(X_i)), \quad (5.1)$$

where $\mathbf{Pa}_{\mathcal{N}}(X_i)$ denotes the set of parents of X_i in the Bayesian network structure $\mathcal{N}$, which encodes the conditional independencies among the variables in $\mathbf{X}$, assuming a directed-acyclic graph (DAG) structure whose nodes are one-to-one correspondent

to the variables. The conditional probability distribution $P(X_i|\mathbf{Pa}_{\mathcal{N}}(X_i))$ is called the local probability distribution of X_i .

A Bayesian network for gene-drug dependency analysis can be learned using a greedy search algorithm (Friedman and Goldszmidt, 1999). The greedy search is guided by a scoring metric such as the BD (Bayesian Dirichlet) score (Heckerman et al., 1995):

$$\begin{aligned} BD(\mathcal{N}; \mathcal{S}) &::= P(\mathcal{N}, \mathcal{S}) = P(\mathcal{N})P(\mathcal{S}|\mathcal{N}) \\ &= P(\mathcal{N}) \int P(\mathcal{S}|\boldsymbol{\Theta}, \mathcal{N})P(\boldsymbol{\Theta}|\mathcal{N})d\boldsymbol{\Theta}, \end{aligned} \quad (5.2)$$

where $\mathcal{N}$ is a structure and $\mathcal{S}$ is the training data. $\boldsymbol{\Theta}$ denotes the set of the parameters for all the local probability distributions. With several assumptions such as *i.i.d.* complete data and Dirichlet prior, the integral in Equation (5.2) can be calculated in closed form as

$$\int P(\mathcal{S}|\boldsymbol{\Theta}, \mathcal{N})P(\boldsymbol{\Theta}|\mathcal{N})d\boldsymbol{\Theta} = \prod_{i=1}^M \prod_{j=1}^{q_i} \frac{\Gamma(\alpha_{ij})}{\Gamma(\alpha_{ij} + N_{ij})} \prod_{k=1}^{r_i} \frac{\Gamma(\alpha_{ijk} + N_{ijk})}{\Gamma(\alpha_{ijk})}, \quad (5.3)$$

Figure 5.1 where q_i is the number of configurations of parents of X_i , and r_i is the number of values of X_i . N_{ijk} is the number of cases in $\mathcal{S}$, where X_i assumes its k -th value when its parents assume their j -th configuration. α_{ijk} denotes the parameters of Dirichlet prior. $N_{ij} = \sum_k N_{ijk}$ and $\alpha_{ij} = \sum_k \alpha_{ijk}$. $\Gamma(\cdot)$ denotes the Gamma function.

5.3.2 Dimensionality Reduction for Feature Abstraction

The scheme described in the previous section is not appropriate for the analysis of the massive biological data sets. One reason involves the inapplicability of the greedy search algorithm. The massive biological data sets usually have more than hundreds of data features. Because the space of the possible Bayesian network structures

grows super-exponentially with the number of variables, a greedy search requires an extremely large amount of time. To make matters worse, the obtained solutions are likely to be very far from the global optimum. The search space of Bayesian network structures is known to be very complicated and multimodal, having many local maxima (Friedman and Koller, 2003).

Another reason is related with the infeasibility of probabilistic inferences. The memory requirement or the computation time of general probabilistic inference algorithms is usually exponential in the number of variables of the Bayesian network. Hence, the probabilistic inference to quantify the dependencies is impractical for Bayesian networks consisting of hundreds of variables. In order to mitigate these problems, we reduce the number of features in the massive biological data sets using a feature abstraction technique.

Formally, feature abstraction reduces an M -dimensional feature space $\mathcal{X} = (X_1, \dots, X_M)$ into a K -dimensional prototype feature space $\mathcal{Z} = (Z_1, \dots, Z_K)$, where $K \ll M$. In this work, the meaning of the process is confined to aggregating individual features (genes or drugs) which show similar patterns in their properties, that is, transcriptional expression (genes) or activity (drugs) across samples. A prototype feature Z_k is defined by a partition subset $\mathbf{B}_k$ of the original feature set $\mathbf{X}$, where $\mathbf{B}_k \subset \mathbf{X}$ and $\mathbf{B}_k \cap \mathbf{B}_l = \emptyset$ ($l \neq k$, $1 \leq l, k \leq K$).

For the feature abstraction for dimensionality reduction, we tried two different clustering algorithms: k -means clustering with the Pearson correlation coefficient and the sequential Information Bottleneck (sIB) method (Slonim et al., 2002). The k -means algorithm is a partitioning clustering algorithm, where the total within-cluster variance is tried to be minimized. The distance between a gene (drug) x and a cluster center (average value of the cluster) z was set to

$$d(x, z) = 1 - r(x, z), \quad (5.4)$$

where $r(x, z)$ is the Pearson correlation coefficient between x and z . The Pearson correlation measure was chosen to group genes (drugs) which show similar relative, rather than absolute, variation pattern across patient samples. In terms of the metric between x and z , our clustering is closely related with the spherical k -means algorithm (Dhillon and Modha, 2001).

The sIB algorithm formulates the clustering problem in an information-theoretic view and runs like the sequential k -means algorithm. Let X and S be the random variables representing the set of genes (drugs) $x \in \mathcal{X}$ and the set of tissue samples $s \in \mathcal{S}$, respectively. The cost of merging a gene (drug) x into a cluster z is given by

$$d(x, z) = (p(x) + p(z)) \cdot JS(p(s|x), p(s|z)), \quad (5.5)$$

where $JS(\cdot, \cdot)$ is the Jensen-Shannon (JS) divergence (Lin, 1991).

The number of clusters (prototype features) was set by a method based on random permutation of data. Here, correlation coefficients are first calculated for all of the permuted gene (drug) pairs, and a cut-off value θ is set in such a way that only upper $\alpha\%$ of the values are higher than θ . Then, genes (drugs) in the original data are hierarchically clustered using an agglomerative method with average-linkage until all inter-cluster similarities are lower than θ . Finally, the number of clusters is set to the number of top-level subtrees in the dendrogram.¹

Let 805 genes and 84 drugs be clustered into K_1 and K_2 groups, respectively, such that the number of clusters is $K = K_1 + K_2$ in total. From the results, a condensed data set with the reduced dimensionality ($K + 1$: the number ‘1’ is added for the cancer type) is generated by *feature averaging* or *feature sampling*, which we

¹This method is similar to the approaches in the works of Herrero *et al.* (Herrero et al., 2001) and Lukashin and Fuchs (Lukashin and Fuchs, 2001) in that the number of clusters is determined using a random permutation-based test, but is different in that the cut-off value is applied at the level of clusters not of individual elements.

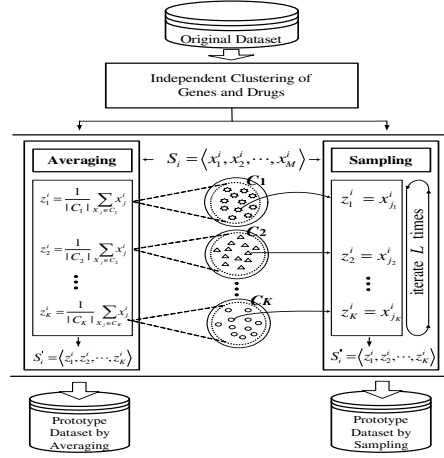


Figure 5.2: Outline of the construction of prototype data sets by feature abstraction for dimensionality reduction. Genes and drugs are clustered independently, then prototype data sets are generated by feature averaging and feature sampling techniques, respectively.

call the *prototype data set* hereafter. Figure 5.2 summarizes the prototype data set construction process.

In the feature averaging approach, the original sample S_i is transformed to S'_i of which the k -th feature value z_k^i ($1 \leq k \leq K$) is given by the arithmetic mean of the expression (activity) values of genes (drugs) of the k -th cluster. In the feature sampling approach, the k -th feature value z_k^i of S'_i is set to the value x_j^i of an individual gene (drug) X_j randomly selected from the cluster C_k . The sampling process is repeated L times for each S_i , by which L new prototype samples are generated from S_i .

5.3.3 Learning the Model

As the local probability distribution model in learning Bayesian networks from the prototype data set, we use the multinomial model which has mainly been adopted in

the gene expression data analysis (Friedman et al., 2000; Pe’er et al., 2001; Hartemink et al., 2002; Chang et al., 2002).² All the feature values in the prototype data set were discretized for the multinomial model. We divided the feature values into three levels (*low*, *normal*, and *high*) based on the mean (arithmetic average) and standard deviation of each feature. Here, *normal* just denotes the value around the mean value across all the samples. The discretization boundaries were calculated as $\mu - b \cdot \sigma$ and $\mu + b \cdot \sigma$, where μ is the mean value, σ is the standard deviation, and b is the constant which determines the breadth of the value *normal*. We tried several b values and set b as 0.60 in the experiments.

In the definition of the BD score in Equation (5.2), $P(\mathcal{S})$ denotes the prior probability of the structure $\mathcal{S}$. We set $P(\mathcal{N}) \propto 2^{-\sum_i (\log M + \log(|\mathbf{P}_{\mathbf{a}_{\mathcal{N}}(X_i)}|^M))}$ for penalizing the complex structures. The parameters for Dirichlet prior α_{ijk} in Equation (5.3) is set to the uninformative value 1.0. A greedy search algorithm (Friedman and Goldszmidt, 1999) was adopted for the structural learning and the following constraints on the network structure were imposed to reduce the search space: 1) gene prototypes and drug prototypes can not be the parents of the cancer node, 2) drug prototypes can not be the parents of gene prototypes. The first constraint is based on the belief that the cancer type affects the gene expression patterns and the drug activities. The second is due to the nature of NCI60 data set, where expression measurements of genes have been made on untreated cell lines, but not on those after the treatment by anti-cancer drugs, hence edges from a drug to genes are rather senseless. In addition, the maximum in-degree of the network is confined to two for compromising the extremely small sample size. Finally, to escape the local maxima which usually arise when relying on the greedy search, the greedy search with random restarts was run 20 times and the one with the highest BD score was

²Friedman *et al.* (Friedman et al., 2000) experimented also with the linear Gaussian model and Imoto *et al.* (Imoto et al., 2003a,b) used the nonparametric regression model.

selected.

5.4 Results

We first investigate the influence of feature abstraction level (the number of prototype features) on the quality of the learned model. This task is done by comparing the results of our method with pairwise correlation analysis of the original data set.

Distinct prototype data sets were generated with four confidence levels, $\alpha = 5, 10, 15, 20$, in the random permutation-based test as described in the previous section about feature abstraction by clustering. The corresponding (# of gene prototypes, # of drug prototypes) were (55, 11), (34, 9), (24, 6), and (16, 5), where the cut-off value pairs were $(\theta_{gene}, \theta_{drug}) = (0.214, 0.218), (0.168, 0.169), (0.136, 0.134), (0.110, 0.110)$, respectively. In each case, two kinds of prototype data sets were constructed by averaging and sampling. In the latter approach, 100 subsamples were generated for each cell line (that is, $L = 100$ in Figure 5.2). Consequently, 16 prototype data sets, thereby 16 Bayesian networks, were constructed according to the number of clusters (4 cases), clustering method (2 cases), and the method for making prototype values (2 cases).

From the original data set, the correlation coefficients for the following feature pairs were calculated. Among 323,610 gene-gene pairs, the top 5,000 negatively-correlated ones were tested. For 67,620 gene-drug pairs, the top 50 negatively-correlated pairs and the top 50 positively-correlated pairs were tested.³

For the comparison, we defined an information-theoretic measure based on the concept of conditional probability. Assume that gene A in the gene prototype $G1$ and gene B in the gene prototype $G2$ are highly negatively-correlated in the original

³In the case of the gene-gene (drug-drug) pair, positively correlated pairs are likely to be clustered into one gene (drug) prototype. Hence, we only consider negatively correlated gene-gene (drug-drug) pairs here.

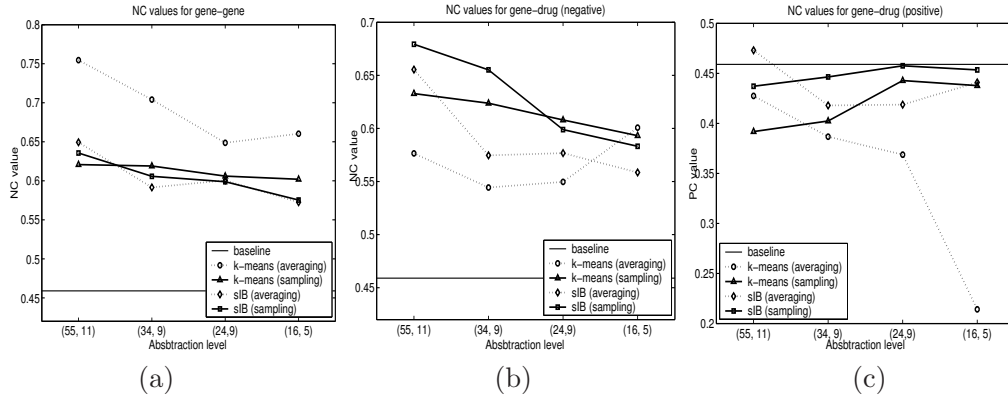


Figure 5.3: Comparison of the Bayesian network-based dependency analysis over various levels of feature abstraction with the plain correlation analysis on the prominent pairwise correlations in the original NCI60 data set. The graphs show the mean of $NC(\cdot||\cdot)$ and $PC(\cdot||\cdot)$ for (a) gene-gene, (b) and (c) gene-drug pairs.

data set. Then, the measure for the negative correlation between $G1$ and $G2$ in the Bayesian network, $NC(G1||G2)$ is defined as follows:

$$\begin{aligned}
 NC(G1||G2) ::= & 0.5 \cdot JS(P(G1 = low), P(G1|G2 = low)) \\
 & + 0.5 \cdot JS(P(G1 = high), P(G1|G2 = high)) \\
 & + JS(P(G1|G2 = low), P(G1|G2 = high)), \quad (5.6)
 \end{aligned}$$

where $JS(\cdot, \cdot)$ is the JS divergence. The measure for the positive correlation, $PC(G1||G2)$ is defined as the same as $NC(G1||G2)$ except that the sign of the third term, $JS(P(G1|G2 = low), P(G1|G2 = high))$ is minus. The baseline for the two measures is when $G1$ and $G2$ are independent from each other, as well as $P(G1|G2 = low)$ and $P(G1|G2 = high)$ are $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ respectively. The baseline value amounts to about 0.459 using the logarithm of base 2. All the values above (below) this baseline are considered the sign of negative (positive) correlation in the Bayesian network.

Figure 5.3 shows the comparison results. In the figure, it can be seen that the

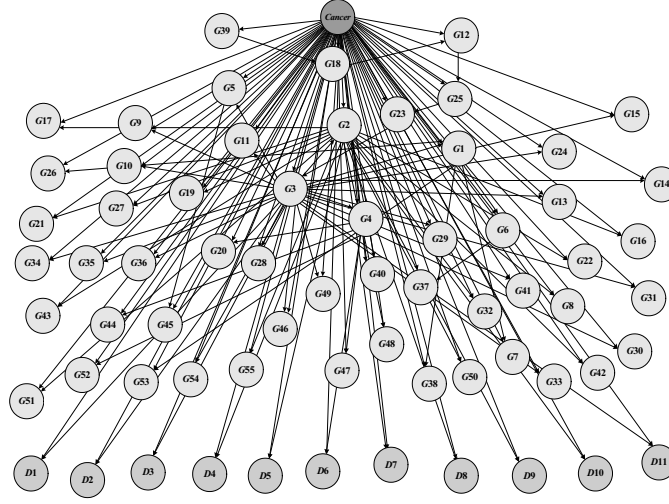


Figure 5.4: The Bayesian network learned from the prototype data set consisting of 55 gene prototypes, 11 drug prototypes, and the cancer type. This prototype data set was built by k -means clustering with 5% confidence level and the sampling approach for making prototype feature values.

quality of the learned Bayesian network is affected by the level of feature abstraction. As the number of prototype features increases, the positive and negative correlations are represented well in general. In the sequel, we present the analysis results using the model built by k -means clustering of 5% confidence level, ($\#$ of gene prototypes, $\#$ of drug prototypes) = (55, 11), with the sampling approach for prototype data generation, since it shows acceptable agreement with four kinds of pairwise correlations.

The Bayesian network is shown in Figure 5.4. From this figure, we could gain insight on the dependency structure over gene prototypes, drug prototypes, and the cancer type, i.e. a condensed representation of the relationships among the features of the NCI60 data set. Also, arbitrary conditional probabilities of interest can be calculated through the probabilistic inference, which is intractable on the Bayesian

network consisting of hundreds of variables due to the exponential time/space complexity in the size of the network (Cooper, 1990).

Our feature abstraction process requires an additional computation of time complexity $\mathcal{O}(T^3 + RKT)$, where the first is for the cluster initialization by a hierarchical clustering and the second is for the cluster refinement by k -means algorithm. Here, T is the numbers of genes (drugs), K is the determined number of gene (drug) clusters, and R is the number iteration in k -means algorithm. When considering the time complexity of the learning/inference in Bayesian networks, however, the additional cost may be trivial. As a concrete example, Figure 5.5 illustrates a trend of the learning time over a variety of numbers of genes and drugs.

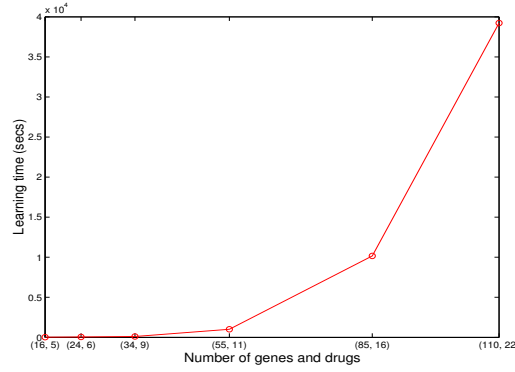


Figure 5.5: An illustration of the variation of the learning time of Bayesian networks according to the number of attributes (genes and drugs). Each value is the CPU time on the system with an AMD Athlon 1900+ 1.60 GHz processor. A greedy search algorithm is used for the learning. The first four cases include the time for clustering of genes and drugs also.

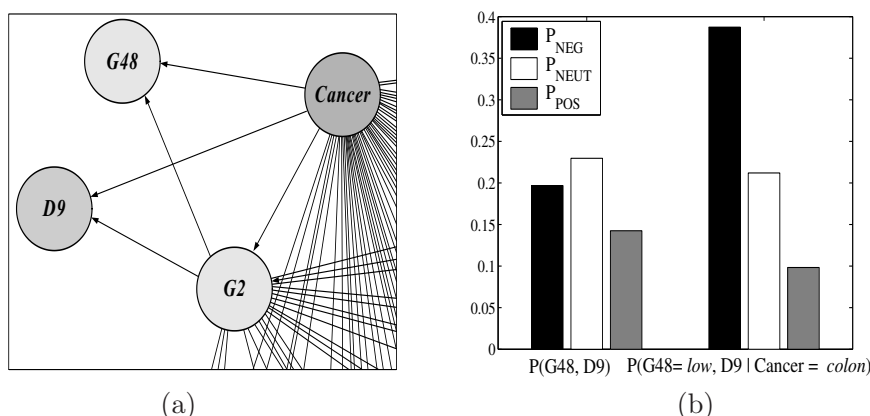


Figure 5.6: (a) Part of the Bayesian network around *G48* (containing *DPYD*) – *D9* (containing 5-FU) and (b) the probabilistic inference in the network. Each bar group in (b) depicts three kinds of probabilities: ‘negative’, ‘neutral’, and ‘positive’ cases from left to right.

5.4.1 Dependency Analysis by Probabilistic Inference

Beyond Pairwise Correlations

Here, it is shown that more intricate relationships involving several features can be detected by our method. We first investigate two plausible relationships among gene expressions, drug activities, and the kind of cancer, obtained from Scherf *et al.*’s work (Scherf *et al.*, 2000).

One is the relationship among the gene *DPYD* (SID W 278125, dihydropyrimidine dehydrogenase [5’:N94809, 3’:N63511]), the drug 5-FU (5-fluorouracil), and the kind of cancer. The drug 5-FU is clinically used to treat colorectal and breast cancers. The expression level of gene *DPYD* and the activity of drug 5-FU are highly negatively-correlated. Figure 5.6a shows the part of the Bayesian network around the gene prototype *G48* and the drug prototype *D9* to which *DPYD* and 5-FU belong, respectively.

To quantify the relationship among gene (G) expressions, drug (D) activities, and the kind of cancer ($Cancer$), we infer the probabilities $P(G, D)$ and $P(G, D|Cancer)$ from the Bayesian network. In specific, the probabilities for the following three cases according to the drug activities when the gene expression level is *low* or *high* are calculated: $P_{NEG}(G, D) = P(G = low, D = high) + P(G = high, D = low)$, $P_{NEUT}(G, D) = P(G = low, D = normal) + P(G = high, D = normal)$, $P_{POS}(G, D) = P(G = low, D = low) + P(G = high, D = high)$, where $P_{NEG}(G, D)$ and $P_{POS}(G, D)$ are the joint probabilities of G and D with negative and positive relations respectively, and $P_{NEUT}(G, D)$ is for the neutral case where drug activity is *normal* irrespective of the expression level of G . $P(G, D|Cancer)$ is calculated in the same way as for $P(G, D)$, except that each probability is conditioned on the specific kind of cancer.

Figure 5.6(b) shows the results of probabilistic inference on $G48$ and $D9$. Here, the negative tendency between $G48$ and $D9$ is observed well. Furthermore, this negative tendency becomes more definite when the kind of cancer is colorectal. For the cell lines of colorectal cancer where $G48$ is *low*, especially, the probability that $D9$ is *high* ($P(G48 = low, D9 = high|Cancer = colon) = 0.387$) is prominent compared to the probability that $D9$ is *low* ($P(G48 = low, D9 = low|Cancer = colon) = 0.098$). In the original data set, in practice, all of the colon-derived cell lines show low expression level of $DPYD$ and are highly susceptible to 5-FU. The probabilities for $G48 = high$ were not included in the result, since the probability $P(G48 = high|Cancer = colon)$ inferred from the Bayesian network is so small (0.023) that it is statistically insignificant (actually, no high expression of $DPYD$ are observed in colorectal lines of the original data set).

The other relationship is among the expression of gene $ASNS$ (asparagine synthetase Chr.7 [510206, (IW), 5':AA053213, 3':AA053461]), the activity of anti-

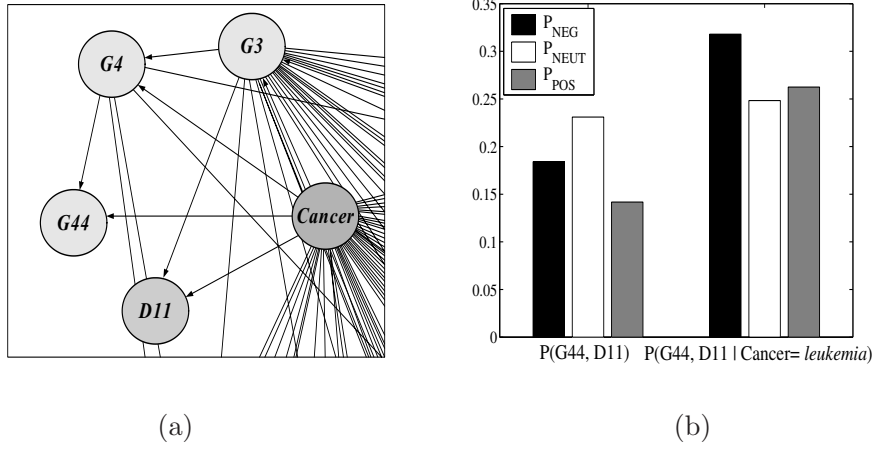


Figure 5.7: (a) Part of the Bayesian network around $G44$ (containing *ASNS*) – $D11$ (containing L-asparaginase) and (b) the probabilistic inference in the network.

cancer drug L-asparaginase, and the kind of cancer. L-asparaginase depletes the amino acid asparagine external to cells. Tumor cells, especially lymphatic tumor cells, need a great amount of asparagine for their growth. There exists a moderately high negative-correlation between the expression level of *ASNS* within tumor cells and the activity of L-asparaginase. Figure 5.7 shows the part of the Bayesian network around the gene prototype $G44$ and the drug prototype $D11$ that contain *ASNS* and L-asparaginase, respectively, and the inference results in the network. In Figure 5.7(b), it can be seen that $P_{NEG}(G44, D11)$ is higher than $P_{POS}(G44, D11)$, where $P_{NEG}(G44, D11) = 0.184$ and $P_{POS}(G44, D11) = 0.142$. For the leukemia cell lines, $P_{NEG}(G44, D11 | \text{Cancer} = \text{leukemia}) = 0.318$ and $P_{POS}(G44, D11 | \text{Cancer} = \text{leukemia}) = 0.263$. The results accord with the negative tendency between the expression level of *ASNS* and the activity of L-asparaginase.

Exploratory Analysis using Bayesian Networks

In this section, we illustrate several salient features discovered by the Bayesian network described in the previous section and present the results of the literature survey related to them.

Relation between $G17$ and $D10$: The dependency structure for $G17$ and $D10$ is depicted in Figure 5.8a. These two components are highly negatively-correlated ($NC(G17||D10)$ is 0.720). However, the negative correlation seems to be mediated by other components, such as the kind of cancer and other genes in the Bayesian network. Eighteen genes in all belong to the prototype $G17$, including *ITGB1* (integrin, beta 1 (fibronectin receptor, beta polypeptide, antigen CD29 includes MDF2, MSK12) Chr.10 [486375, (IW), 5':AA044145, 3':AA044261]), *uPA* (SID 486215, urokinase-type plasminogen activator [5':, 3':AA040727]), *IL-8* (interleukin 8 Chr.4 [328692, (DW), 5':W40283, 3':W45324]), *IL-6* (IL6 Interleukin 6 (B cell stimulatory factor 2) Chr.7 [310406, (RW), 5':W31016, 3':N98591]), and so on. The drug prototype $D10$ contains six drugs, where four members (paclitaxel, colchicine, trityl-cysteine, and vinblastine-sulphate) are tubulin-active antimitotic agents and the remaining two drugs are geldanamycin and bisantrene.

Paclitaxel (taxol) is a novel anti-cancer compound with activity in varieties of carcinomas of breast, ovary, prostate, and so on (Duan et al., 1999). The clinical agent is known to act as an anti-cancer agent by inducing microtubule stabilization or apoptosis of tumor cells. Two genes in $G17$, *uPA* and *ITGB1*, can affect the activation of paclitaxel in breast cancer cells, respectively. *uPA* catalyzes the conversion of plasminogen to plasmin. Plasmin induces the formation of multicellular spheroids of breast cancer cells, by which tumor cells get to acquire an increased resistance to chemotherapeutic drugs, including paclitaxel (Chun, 1997). In the case of *ITGB1*, its signaling has been identified as an important survival pathway in drug-induced

apoptosis in breast cancer cells (Aoudjit and Vuori, 2001). Especially, the ligation of *ITGB1* by extracellular matrix ligands can considerably inhibit apoptosis induced by paclitaxel. From these, it is expected that the chemotherapeutic effect of paclitaxel might be restrained by high expression of *ITGB1* or *uPA*. In fact, *ITGB1* and *paclitaxel* are ranked at the 1st and the 2nd in the Pearson correlation similarity with the center profiles of *G17* and *D10*, respectively. The *uPA* - *paclitaxel* pair is within the lowest-valued top ten among all member pairs, in terms of the JS divergence between its empirical probabilistic dependency profile and the inference result for *G17* and *D10* in the Bayesian network. The empirical pairwise dependency of a gene-drug pair is obtained directly from data, by calculating conditional probabilities based on their discretized patterns.

For another cases, IL-6 is a growth factor for several tumors and was suggested to be an inhibitor of apoptosis which is associated with cytotoxic chemotherapy (Duan et al., 1999). IL-8 has previously shown to play an important role in the progression of human tumors and to enhance tumor metastasis (Singh et al., 1994; Shi et al., 2001). Especially, it was also identified as a key factor involved in angiogenesis and invasion in human breast cancer (Lin et al., 2004), and was observed to be relatively overexpressed in some cell lines which are resistant to taxol (Duan et al., 1999). Table 5.1 lists the member genes and member drugs of *G17* and *D10*.

Relation between *G2* and *D4*: Figure 5.8(b) shows the network structure around *G2* and *D4*. These two nodes are directly linked in the network structure and also affected by the node of cancer type. The $NC(G2||D4)$ is 0.685 and the two nodes are mediated by the node of the kind of cancer in the network structure. *G2* and *D4* contain the gene *BMI1* (murine leukemia viral (bmi-1) oncogene homolog Chr.10 [418004, (REW), 5':W90704, 3':W90705]) and the drug azacytidine, respectively. Azacytidine is used as a chemotherapeutic agent for several cancer types in-

Table 5.1: The list of member genes and drugs of $G17$ and $D10$

$G17$	$D10$
Urokinase-type plasminogen activator [5':, 3':AA040727]	Paclitaxel
ITGB1 Integrin, beta 1 (fibronectin receptor, beta polypeptide, antigen CD29 includes MDF2, MSK12) Chr.10 [486375, (IW), 5':AA044145, 3':AA044261]	Geldanamycin
IL8 Interleukin 8 Chr.4 [328692, (DW), 5':W40283, 3':W45324]	Trityl-cysteine
IL6 Interleukin 6 (B cell stimulatory factor 2) Chr.7 [310406, (RW), 5':W31016, 3':N98591]	Vinblastine-sulfate
Human mitochondrial 1,25-dihydroxyvitamin D3 24-hydroxylase mRNA, complete cds [5':H82378, 3':H82272]	Colchicine
TISSUE FACTOR PATHWAY INHIBITOR PRECURSOR [5':AA004583, 3':AA004712]	Bisantrene
TISSUE FACTOR PATHWAY INHIBITOR 2 PRECURSOR Chr.7 [361247, (DEW), 5':AA017445, 3':AA017446]	
Human chromosome 17q21 mRNA clone 1046:1-1 Chr. [130476, (DE), 5':R22363, 3':R21832]	
GRO2 oncogene Chr.4 [80971, (DIW), 5':T70148, 3':T70079]	
Crystallin zeta (quinone reductase) [5':AA035179, 3':AA035180]	
Human cytochrome P450-IIB (hIIB3) mRNA, complete cds [5':T71165, 3':T72862]	
MALATE OXIDOREDUCTASE [5':H13370, 3':H06037]	
H.sapiens mRNA for TRAMP protein Chr.8 [149355, (IEW), 5':H01598, 3':H01495]	
H.sapiens mRNA for TRAMP protein Chr.8 [489060, (EW), 5':AA056956, 3':AA057165]	
Homo sapiens caveolin-2 mRNA, complete cds [5':, 3':AA036724]	
Homo sapiens regulator of G-protein signalling 12 (RGS12) mRNA, complete cds Chr.4 [343073, (IRW), 5':W67133, 3':W67134]	
ALDH7 Aldehyde dehydrogenase 7 Chr.11 [308598, (IW), 5':W24901, 3':N95790]	
ATPase, Na+/K+ transporting, beta 1 polypeptide [5':AA004976, 3':AA004863]	

cluding acute leukemia. Incidentally, when the node of cancer type is set to leukemia (LE), the negative correlation is much more prominent ($P_{NEG}(G2, D4|LE) = 0.829$, $P_{POS}(G2, D4|LE) = 0.002$, and $NC(G2||D4) = 1.954$). The structure and functioning mechanism of the anti-cancer drug azacytidine (member of $D4$) are very similar to 5-aza-2'-deoxycytidine. And 5-aza-2'-deoxycytidine is known to promote the tu-

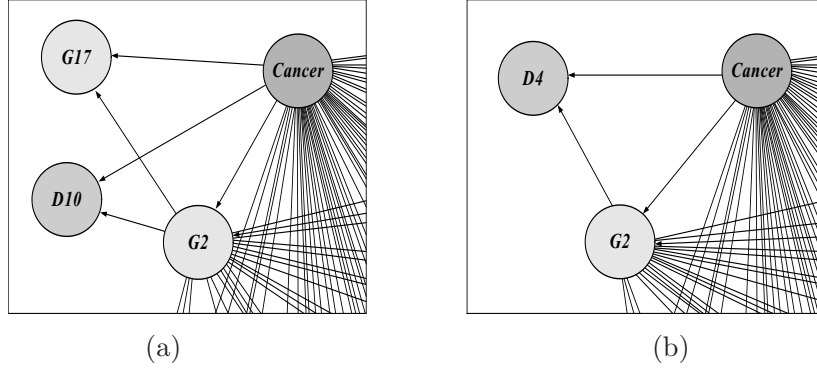


Figure 5.8: The network structures around (a) *Cancer*, *G17*, and *D10* and (b) *Cancer*, *G2*, and *D4*. They show prominent negative correlations in the Bayesian network measured by the $NC(\cdot||\cdot)$ metric.

mor suppressing protein p14/ARF (Magdinier and Wolffe, 2001), whereas the gene *BMI1* (member of *G2*) is known to negatively regulate p14/ARF (Lindstroem et al., 2001). In the cancer cell line which shows high expression level of *BMI1*, it is anticipated that the functioning mechanism of azacytidine might not work because of the possibility of the interruption by *BMI1*, suppressor of p14/ARF. This hypothesis supports the observed negative correlation between the expression level of *G2* and the activity of *D4*.

5.4.2 Comparison with a Linear Dependency Model

For the purpose of comparison, we also test a linear dependency model for the prototype data set. This corresponds to a double reduction approach for original NCI60 data set. For both gene expression data and drug activity data, the error measure is set to the Euclidean distance.

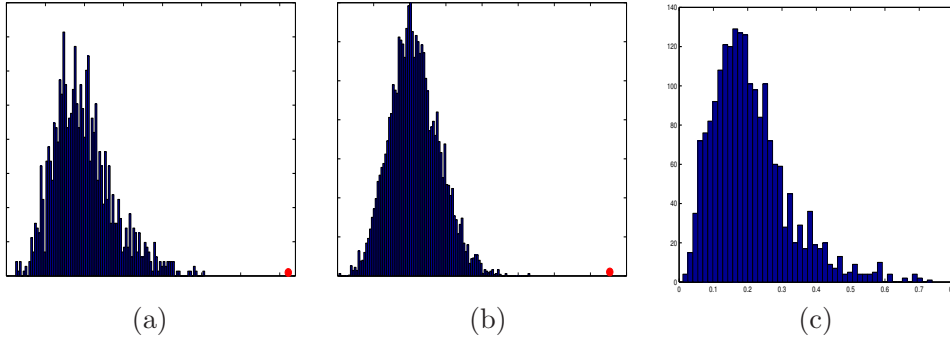


Figure 5.9: Significance test for the co-structure between gene expression data and drug activity data. The left two histograms show the distribution of total co-reference scores for 10,000 randomly matched data (a) with no feature abstraction and (b) with $(\text{gene}, \text{drug}) = (55, 11)$. The filled circles represent the total co-reference scores for original data sets prior to random permutation. (c) The histogram of the significance values for 2,000 prototype data sets produced by random clustering of $(\text{gene}, \text{drug}) = (55, 11)$ (see the text).

Significance of the Co-structure

First, we investigate the significance of the co-structure underlying the condensed data set. As in Section 4.3.2, this was done by comparing the co-reference score for the original paired data set and those for randomly matched data sets. Figure 5.10(a) and Figure 5.10(b) presents the comparison results for the NCI60 data set with no feature abstraction (805 genes and 84 drugs) and the condensed data set. For both cases, none of the 10,000 permuted data shows the higher co-reference score than the original paired data set. Thus, it can be said that the prototype data set is still well retaining the co-structure observed in the original NCI60 data set. The adequacy of the prototype data set was further confirmed by the comparison with the result for data sets from random clustering of genes and drugs. Figure 5.10(c) shows the result. Among 2,000 trials of random clustering, we could observe no case which is as significant as the data set by the k -means clustering.

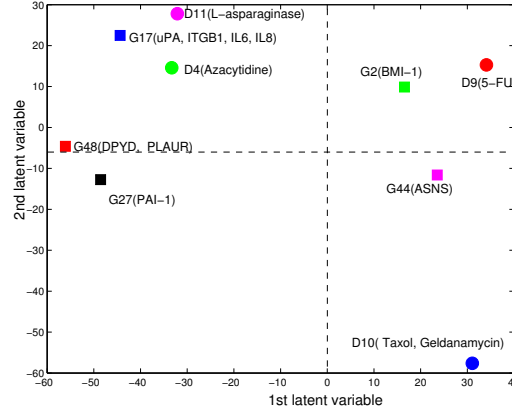


Figure 5.10: The display of several genes and drugs in the 2-dimensional space defined by the first and the second latent factors from a linear co-reference latent model.

Analysis Result

Figure 5.10(a) shows 4 gene prototypes and 4 drug prototypes which were identified by the Bayesian network, along with another gene prototypes $G27$, in the 2-dimensional space defined by the first two latent factors. For $G48$ (DPYD) and $G17$ (uPA , $ITGB1$, $IL-6$, $IL-8$), it can be seen that their negative tendency with $D9$ (5-FU) and $D10$ (taxol), respectively, are being well reflected. Interestingly, $G48$ and $G17$ are those with the first and the third high contribution values with negative sign in the first latent factor for gene expression data, respectively. Also, their corresponding $D9$ and $D10$ takes the top two highest positive values in the first latent factor for drug activity data. In particular, $D10$ is the highest contribution-valued element in the second latent factor for drug.

The gene prototype $G27$ takes the high contribution value in the definition of the first latent variables for gene expression data (ranked at 2nd), along with $G48$ and $G17$. The gene $PAI-1$ (Plasminogen activator inhibitor, type I Chr.7

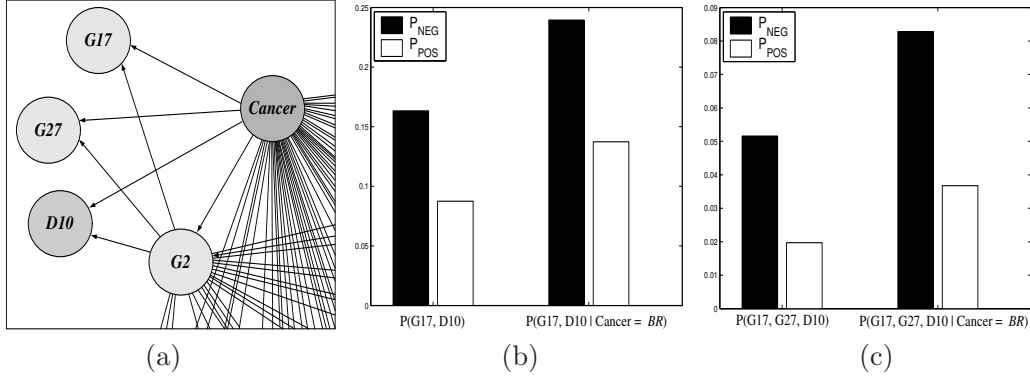


Figure 5.11: (a) The part of the Bayesian network around $G17$, $G27$, and $D10$. The probabilistic inference results are also presented for (b) $G17 - D10$ and (c) $(G17, G27) - D10$.

[487394, (DIW), 5':AA046572, 3':AA046659]) is the member of $G27$ and it is an important component of the plasminogen activator system, together with $uPA(G17)$ and $PLAUR$ (Plasminogen activator, urokinase receptor Chr.19 [309553, (DI), 5':, 3':N94408], $G48$). The plasminogen activator system has been identified as an important factor in tumor growth and metastasis (Andreasen et al., 1997; Kwaan et al., 2000). On the basis of this observation, we performed an probabilistic inference in the previous Bayesian network for the relation between $(G17, G27)$ and $D10$. Figure 5.11 shows the network structure around the three elements and the inference result. When the two gene prototypes were set to *low* and *high* together (Figure 5.11(c)), the more distinctive negative tendency could be inferred than that for $D10$ and $G17$ only (Figure 5.11(b)). By the two results, a possible role of the elevated level of $PAI-1$ in the resistance to some chemotherapy may be anticipated.

Figure 5.12(a) shows the representation of 60 tumor samples on the first co-reference latent factor. For a tumor sample s_i , the representation based on gene expression profiles is plotted by its original value z_i^x and that based on drug activity

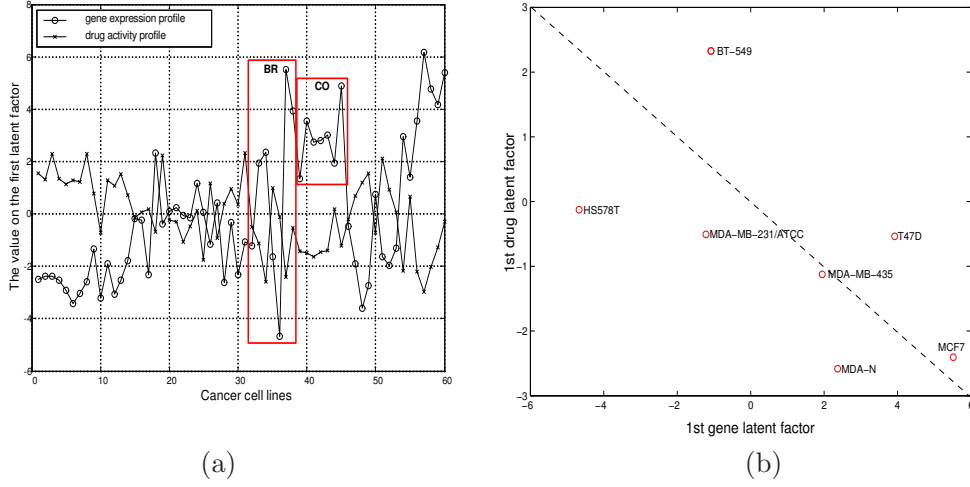


Figure 5.12: The representation of tumor samples on the first latent factor of the linear co-reference latent model: (a) all 60 cancer cell lines of the NCI60 data set and (b) the breast tumor samples only.

profile is plotted by the negative value of z_i^y . One can see the negatively-correlated tendency in the plot. In particular, all the colon cancer cell lines (CO) have the positive values on the gene expression latent factor and the negative values on the drug activity latent factor, except only one case. Thus, it can be said that the observation of the relation between $DPYD(G48)$ and $5-FU(D10)$ in the original NCI60 data set is well reflected. One can also observe that the breast cancer cell lines (BR) shows the high variance in terms of the projected values. Figure 5.12(b) plots the breast tumor samples based on gene expression profiles against those based on drug activity patterns. In a global view, the two representations are negatively correlated, which further supports the relation between $(G17, G27)$ and $D10$ stated in the above.

Two other relations, one for between $ASNS(G44)$ and L-asparaginase($D11$) and the other for between $BMI1(G2)$ and azacytidine($D4$), also correspond to those by Bayesian networks. However, their contribution values for the first and the second

latent factors were rather moderate, compared with the previous three cases.

To summarize, both the Bayesian network-based dependency model and the linear dependency model could extract several common salient relations among genes and drugs. While the identification process is done by the investigation of network structure and the probabilistic inference in the former model, the task in the latter model can be done by looking over the contribution values of genes and drugs in latent factors and the low-dimensional display. The latter model have a merit in that it can represent the relation between genes and drugs in an intuitive manner by the display of low-dimensional space. The former can be advantageous in that when an auxiliary variable, the cancer type here, besides the paired data set of gene expression and drug activity, it is able to represent the association of genes and drugs in relation with a cancer type. This is among important issues in our analysis of NCI60 data set for the dependency analysis between genes and drugs. The relation between $G2$ and $D4$ in leukemia is an example.

Despite the difference, the two approaches can be used in a synergistic way, exemplified by the identification of a possible complex association among $G17$, $G27$, and $D10$ during the comparison of the two results. In addition, when the number of samples are much smaller than the number of variables, which is often the case in bioinformatics as well as the data analyzed here, the combinatorial analysis will be more useful in the assessment of the plausibility of an extracted relationship as a candidate for further research.

5.5 Summary

We presented an efficient analysis method using Bayesian networks that incorporates a dimensionality reduction technique to alleviate the curse of dimensionality problem, which thus makes it applicable to the analysis of massive biological data

sets. The Bayesian network can capture the entire dependency structure immanent in the data set. This property provides an effective means for encoding the valuable information contained in the large-scale data sets in a probabilistic way. Further, the dependency among several features can be quantified by probabilistic inference. The probabilistic inference is used for calculating the conditional probability of interest, assisting the decision making.

In our problem setting, any gene or drug prototype can be regarded as a dependent or an independent variable. In this sense, our method produces a huge number of dependency structures of similar shapes. To identify a small set of eligible ones among them, we exploited a measure for the negative correlation between gene and drug prototypes. The dependency structures with high negative-correlation ($NC(\cdot||\cdot)$) were examined by literature survey. Through this, we were able to confirm several dependency structures among gene expressions, drug activities, and the kind of cancer. From the comparison of the result by a linear dependency model, furthermore, a potential complex relationship incorporating more than one gene could be demonstrated. Our findings were obtained in a purely data-driven way, and this demonstrates the potential of our method as an efficient data mining tool for the revelation of the relations between gene expression patterns and drug activities in cancer cells.

Since the learning of the dependency model is done on the prototype genes and drugs, there may occur some information loss in the dependency analysis. For example, the expression of the gene *ASNS* and the activity of drug L-asparaginase was previously shown to be more highly negatively correlated in leukemia cancer cell lines than other kinds of tumor samples. Though our analysis result also presents the negative tendency between the two across all samples, the higher significance of the relation for leukemia samples was rather obscure (Figure 5.7). The limitation

may be alleviated by an additional confined analysis for member genes and drugs of the corresponding groups assumed to be mutually associated (Chang et al., 2002), which can be regarded as a hierarchical or module-based approach.

There are several directions for future work. One direction is to combine other information sources as prior knowledge. The protein-protein or DNA-protein interaction DB can be exploited for the more precise structural learning. Additionally, other sorts of data such as polymorphisms in key enzymes or gene expression patterns from cell lines treated with some drugs (e.g., expression profile data of responses to drugs, as in the work of Imoto *et al.* (Imoto et al., 2003b), for cancer cell lines) could provide more biologically useful information.

Chapter 6

Conclusions

This dissertation has been concerned with the explorative analysis of paired data sets in the framework of dimensionality reduction. By the paired data set, we mean the case where each observation is represented by two variable sets. Explorative analysis tasks are generally related to the descriptive mining where one mainly seeks to present a concise and summarized description of the observed data. For a paired data set, a key element in the descriptive task is the association pattern between two component data sets comprising the pair. For a large-scale data set, however, the association can be obscured and thus it may require a tedious and time-consuming work for one to understand the data. We approached this problem by the application of dimensionality reduction, especially in the problem domains of dependency analysis of paired biological data sets. Here, the dimensionality reduction is utilized to extract salient features underlying the *surface association* between two components of paired data sets.

We presented the co-referent latent model for an efficient and effective identification of such latent features and thus salient association patterns underlying between two data sets which are given as being paired. The co-reference latent model is a gen-

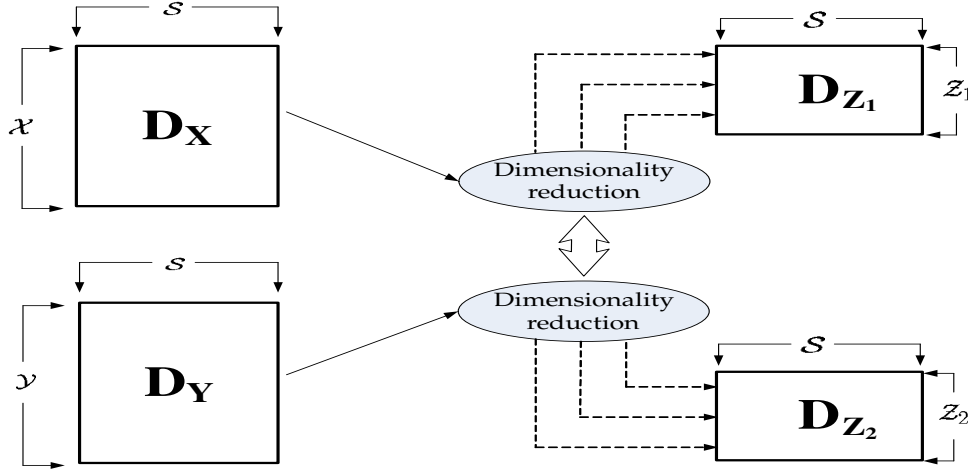


Figure 6.1: Dependency analysis of paired data sets by co-reference latent models

eral model for the analysis of two data sets in the framework of data/dimensionality reduction. Given a paired data set $(\mathbf{D}_X, \mathbf{D}_Y)$, the model constructs two separate component models, $\mathcal{M}_X$ for $\mathbf{D}_X$ and $\mathcal{M}_Y$ for $\mathbf{D}_Y$. The association patterns between $\mathbf{D}_X$ and $\mathbf{D}_Y$ can be efficiently grasped by correlating the latent variables which are produced by $\mathcal{M}_X$ and $\mathcal{M}_Y$, under the constraint that they should be of the same meaning or concept (probability or linear projection, for example). Figure 6.1 shows the schematic view of the co-reference latent model in this framework.

If the feature pairs which are extracted by a co-reference latent model are found to be well reflecting the regularity underlying data generation or association patterns among variables, we have the advantage of the acquisition of effective data summary with only a small number of latent descriptors. We showed that this is the very case in two practical applications to biological data sets, one for dependency analysis of DNA sequence elements and gene expression patterns, and the other for gene-drug dependency mining from a microarray data and drug activity data.

Table 6.1: The summary of two application problems in the constitutional aspect of paired data sets.

	$X \in \mathcal{X}$	$Y \in \mathcal{Y}$	$s \in \mathcal{S}$
Chapter 4	occurrence of a sequence motif	expression at an experimental condition	gene
Chapter 5	expression of a gene	activity of a drug	cancer cell line

6.1 Summary of Main Results

Table 6.1 summarizes the two applications in terms of the paired components in their data. An important issue in the analysis of massive gene expression data in computational biology is to understand the mechanism underlying the transcriptional regulation of genes. It is known that at the core of gene expression control is the binding of transcription factors to specific DNA sequences upstream of genes. The first application deals with the subject of identifying significant DNA regulatory sequence elements and linking them to specific expression patterns with which they are expected to be involved. We tackled the problem on the basis of a co-reference latent model with its two component models being linear. The data set of DNA sequence element was modeled in the framework of the correspondence analysis since its elements are count and non-negative. The gene expression data set was modeled using the error measure based on ordinary Euclidean distance. In the computational experiments for two data sets of the yeast *S. cerevisiae*, we could identify several well-known regulatory sequence elements and also a few putative elements. More importantly, our analysis method could provide a coupled display of sequence elements and experimental conditions in low-dimensional plots, which was possible by virtue of the simultaneous dimensionality reduction. The display in the reduced space was shown to be very effective in the intuitive grasping of the transcriptional role of a

regulatory sequence element in relation with specific experimental conditions.

The second application in Chapter 5 focused on the use of dimensionality reduction as a tool for facilitating the learning and inference of salient association patterns from a paired data, rather than the analysis of the result by dimensionality reduction itself. Data samples are cancer cell lines and each sample is represented by two separate measurements, one is the gene expression profile and the other is drug activity profile. Here, the task was to gain an insight into various correlations among gene expressions and drug activities in the malignant cell lines. Under the general framework of co-reference latent model, the dimensionality of gene expression data and drug activity data were separately reduced. Unlike the model used in Chapter 4, however, the dependency structure between genes and drugs was analyzed by an explicit model with its own structure and parameters, that is, Bayesian networks. The dimensionality reduction was accomplished by the clustering of genes and drugs. In the experiment for NCI60 data set, the effectiveness of the approach was shown in terms of two aspects. After the dimensionality reduction, we could still identify most of the salient pairwise correlations in the original data set. In addition, several relationships inferred from the learned dependency model were confirmed to be biologically meaningful.

We also tested a linear co-reference latent model for the condensed data set and could relatively well extract the prominent associations between genes and drugs which were identified by Bayesian network. By the combined analysis of the two results, furthermore, an additional plausible relation including two genes could be identified. When the number of data samples is small, the combinatorial analysis would be useful in assessing the feasibility of a suggested dependency for further study.

6.2 Future Work

Future research direction includes the followings. One important thing is on the determination of appropriate number of latent factors in the learning of co-reference latent models, that is, the dimension of the reduced space. In a probabilistic view, the problem can be cast as a model selection problem. Then, some standard criteria from statistics or information theory such as BIC or MDL could be used. Cross-validation is another candidate.

Another issue in a methodological aspect is concerned with the application in Chapter 5. In the abstraction-based gene-drug dependency analysis, the grouping of genes and drugs was separated from the dependency learning with probabilistic graphical models. However, if the two phases are accomplished in an integrative way as the application in Chapter 4 for the dependency analysis of DNA sequence elements and gene expression, a more plausible model could be expected.

Recent advances in sensor and high throughput data acquisition is increasingly facilitating the measurement of diverse aspects of a set of entities. In this circumstance, we need to work on extending the analysis framework in this dissertation to more than two data sets. Such integrated analysis of the heterogeneous knowledge sources is expected to be highly valuable in that it can provide one with a content-rich information and thus broaden his/her understanding of the data while bridging the gap among the respective data. A circular learning scheme would be a candidate for the co-analysis of such more than two data sets, where co-reference latent models are applied in succession on a sequence of data pairs.

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초 록

실세계의 많은 데이터들은 결합쌍 형태로 주어지며, 이는 동일 데이터 샘플 집합에 대해 두 가지 서로 다른 영역 또는 측면의 자질이나 변수 집합에 의한 표현이 주어지는 경우를 의미한다. 데이터 마이닝이나 기계학습 분야에서 중요한 이슈 중의 하나는 이러한 결합쌍 데이터로부터 특징적 의존관계 또는 연관관계를 동정(identification)하는 것이다. 하지만, 고용량/고차원의 결합쌍 데이터의 경우에는 데이터의 복잡성 때문에 이러한 연관관계의 직접적 파악이 곤란해지는 문제가 있다.

본 논문은 이러한 문제점을 해결하기 위하여 차원축소 기반의 결합쌍 데이터의 동시 분석을 위한 효과적 모델을 제시한다. 제안되는 모델의 핵심은 결합쌍을 구성하는 각 데이터에 대해 차원 축소를 위한 별도의 은닉변수 모델을 도입하고, EM 알고리즘 형태의 학습과정을 통해 각 은닉자질(latent factor)들을 교차 반복적으로 상호 참조함으로써 데이터 쌍간의 연관관계나 공통구조를 명시적으로 파악하고자 하는 것이다. 모델의 또 다른 특징은 데이터를 구성하는 자질들간의 복잡한 연관관계에 내재된 규칙적 또는 특징적 패턴을 저차원 공간상에 간략한 형태로 사상(mapping)시킨다는데 있다. 따라서, 제안되는 “상호참조 은닉모델”은 결합쌍 데이터에 대한 요약적 설명(description)을 제시할 수 있을 뿐 아니라, 은닉자질 기반의 가시화를 통해 데이터 자질쌍간 연관관계에 대한 직관적 이해를 증진시킬 수 있다.

두 가지의 실세계 바이오 데이터 쌍 분석 문제에 대한 응용을 통해 제시된 모델의 유용성을 검증하였다. DNA 서열 요소 및 유전자 발현 패턴의 결합쌍 데이터에 대해 선형적(linear) 상호 참조 은닉 모델을 적용함으로써, 특정 발현 패턴에 대해 생물학적으로 기존에 알려진 중요한 전사조절서열요소와 더불어 가능성 있는 몇 가지 서열 요소들을 동정할 수 있었다. 특히, 수백 개의 후보 서열 요소에 대해 훨씬 적은 수의 은닉자질들으로써 발현패턴과의 관계를 요약적으로 제시하였으며 저차원

상의 가시화(visualization)를 통해 DNA서열요소들의 특정발현조건에 대한 연관성을 손쉽게 파악할 수 있었다. 유전자 발현과 약물 반응간의 의존관계 분석에 대한 응용에서는, 군집화(clustering)에 의한 데이터 차원 축소를 통해 효율적 학습 비용으로 간결한 형태의 의존관계 모델 구축이 가능하였다. 더욱이, 구축된 모델 상에서의 추론 복잡도가 경감됨으로써, 확률 기반의 정량적 분석에 의해 유전자 발현 및 약물 반응간의 의미있는 상관 관계들을 추출해 낼 수 있었다.

주요어: 상호참조 은닉모델, 결합쌍 데이터, 차원 축소, 의존관계 분석, 바이오 데이터

학 번: 97419-816

감사의 글

먼저, 저에게 학문의 자세를 가르쳐 주시고 관심어린 격려와 때로는 질책으로써 여기까지 저를 이끌어주신, 지도 교수님이신 장병탁 교수님께 가슴 깊이 감사드립니다. 장병탁 교수님께서 저에게 기계학습 분야를 접하게 해 주셨을 뿐 아니라, 언제나 많은 지원과 함께 좋은 기회를 주셨습니다. 그리고, 석사 시절 은사님이시자 박사 과정 초기까지 저를 지도해 주시고 지금도 애정어린 충고를 아끼지 않으시는 김영택 교수님께 감사드립니다. 또한, 바쁘신 중에도 저의 부족한 논문에 많은 조언과 도움을 주신 유석인 교수님, 김형주 교수님, 조성배 교수님께 머리 숙여 감사의 인사를 드립니다.

컴퓨터 공학부 자연언어 처리 연구실에서의 기계번역 연구 시절, 많은 도움을 주신 양승현, 이재원, 김나리, 김성동, 김유섭 박사님께도 감사의 말씀을 드립니다. 연구실 바로 옆 선배로서 저를 잘 이끌어 주신 성배 형, 제가 처음 대학원 생활을 시작할 때 부터 많은 관심과 도움을 주고 연구실 뿐 아니라 일상생활에서도 함께 많은 것을 공유했던 친형같은 중우 형, 저의 모자란 부분을 기꺼이 채워준, 모든 일에 열심인 좋은 후배 황규백, 오장민에게 정말 고맙다는 말을 하고 싶습니다. 그리고, 프로젝트와 연구를 수행하면서 많은 시간을 연구실에서 함께 보냈던 바이오지능 연구실의 모든 구성원들에게도 고마운 마음을 전합니다.

지금의 제가 있을 수 있는 것은 사랑과 신뢰로써 저를 받쳐주는 가족들이 있기 때문입니다. 저 때문에 늘 걱정하시라는 어머니께서는 이 논문이 자그마한 선물이 되길 바라며 당신께서 보여주시는 사랑에 늘 감사한다는 말씀을 드립니다. 그리고, 여전히 철없는 이 동생을 어른스럽게 잘 이끌어주는 형과 별로 해준 것이 없는 오빠지만 잘 따라주는 동생 은정에게도 감사합니다. 저를 사위로 맞아주시고 친자식처럼 사랑으로 대하고 보살펴 주시는 장모님께 존경과 감사의 마음을 드리며, 이 논문이 그 동안 저에게 보여주신 사랑에 조금이나마 보답이 될 수 있었으면 합니다. 매형으로써 별로 해준 것도 없는데, 언제나 웃음으로써 따뜻하게 대해주는 처남 영민에

계도 고맙다는 말을 하고 싶습니다. 무엇보다도, 힘들 때마다 곁에서 저에게 큰 힘이 되어주고 모든 기쁨과 슬픔을 함께 하는, 부족한 저를 사랑과 인내로써 변함없이 믿고 지지해주는 저의 가장 소중한 친구이자 평생의 동반자인 아내 경은에게 무한한 감사와 함께 언제나 사랑한다는 말을 전합니다.

마지막으로, 이 아들의 결혼식도 제대로 못보시고 돌아가신, 지금은 하늘 나라에 계신 아버지께 이 논문을 바칩니다.